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Umweltmesstechnik AG

Dust Concentration Measuring Device

PFM 06 ED

Operation Manual

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1 General

1.1 Information for operation manual

This operation manual contains the required information for intended use of the described product. It is firm part of the scope of supply, also if the possibility of separated order respectively delivery had been planned due to logistic reasons. By reasons of clarity it does not contain all details for all types of the described product and it cannot consider each possible case in operation with the product.

Read the operation manual completely and attentively. Observe the safety instructions and operation directions in this operation manual as well as the labelling at the device and the packaging. For later use keep the operation manual at a safe place.

If you should need further information or if problems should occur which are not described elaborately in this operation manual, please refer to Dr. Födisch Umweltmesstechnik AG (contact details: see cover inside).

For operation with the optional additional devices please read the technical documentations of the suppliers. The contents are at responsibility of the respective manufacturers.

1.2 Intended use

The product described in this operation manual has been developed, manufactured, tested and documented in observation of the corresponding safety standards and has left the factory in safety-related correct and tested condition.

By observing the actions and safety instructions described for configuring, assembly, intended use and upkeep, there is no danger coming up by the product itself in normal case. The correct and safe operation presupposes furthermore the proper transport, professional storage, placement and assembly as well as careful operating and upkeep.

The dust concentration measuring device PFM 06 ED is used for continuous extractive measurement of dust contents in wet and sticky exhaust gases.

The PFM 06 ED represents a device of group 1 and class A according to DIN EN 61326-1. Thereby it is intended for the use in industrial range. The connection to low voltage supply networks used for residential buildings can result in conduction-bound or radiated interference of devices in rare cases.

To keep the correct condition of the device and to achieve a proper and safe operation it must solely be used in the way described by the manufacturer. Any kind of differing use as described in this operation manual is regarded as non-intended use and can result in personal or material damage.

Non-intended use results in termination of guarantee.

1.3 Warranty

The Dr. Födisch Umweltmesstechnik AG advises that the content of this operation manual is not part of a prior or present arrangement, commitment or legal relationship or that does not change these. All liabilities result from the respective sales contract which also contains the complete and solely legal warranty regulations. These contractual warranty terms are neither extended nor limited by the contents in this operation manual.

Rebuilding and modification at the product is not permitted. Any intervention into the device as well as any kind of non-intended use results in termination of guarantee. The manufacturer assumes no liability at all.

1.4 Standards and regulations

As far as possible, the harmonised European standards have been applied to specification and production of this device. If no harmonised European standards have been applied, the standards and regulations for the Federal Republic of Germany apply.

1.5 Declaration of conformity

The dust concentration measuring device PFM 06 ED has a CE label. Therewith we declare that the device in its conception and design as well as in the execution put into circulation by us corresponds to the fundamental safety and health requirements.

NOTICE
On request the document of conformity declaration is placed at the disposal by Dr. Födisch Umweltmesstechnik AG (contact details: see cover inside).

2 Safety

The device must solely be operated in correct condition and in strict observation of the safety instructions.

Working at the device must solely be executed by qualified specialised personnel (see section 2.2 “Requirements for personnel”, page 11). Personal protective equipment according to the current legal accident prevention regulations must be worn.

Please read the safety instructions for the optional additional devices in the technical documentations of the suppliers. The contents are at responsibility of the respective manufacturers.

2.1 Safety instructions

Safety instructions serve the prevention of hazards for life and health of users or upkeep personnel respectively for avoiding material damage. In this operation manual they are emphasised by the here defined signal terms. Furthermore, special safety instructions can be characterised by additional symbols.

2.1.1 General safety instructions

DANGER
Notes with signal word “DANGER” indicate possible hazards which cause personal damage in terms of death or most serious injury in case of non-observing the safety precautions.

WARNING
Notes with signal word “WARNING” indicate possible hazards which cause personal damage in terms of simple up to serious injury in case of non-observing the safety precautions.

CAUTION
Notes with signal word “CAUTION” indicate possible hazards which cause material damage in case of non-observing the safety precautions.

NOTICE
Notes with this indication describe helpful information and tips for operation with the product and serve the avoidance of failure.

2.1.2 Specific safety instructions

DANGER
Hazardous voltage!
Parts of the device can be energised with hazardous voltage.
Danger of electric shock.
Any work at the device must solely be executed by qualified personnel.

DANGER
Explosion hazard!
Personal damage in immediate surround as well as material damage at the device and its vicinity can be caused.
The device must not be operated in potentially explosive atmosphere.

DANGER
Poisonous substances!
Poisonous gases can cause serious health damage or death.
Irritating of eyes, skin or respiratory system organs can be caused.
The exhaust gases from the device must be led back to the exhaust duct or outdoor. Furthermore, the correct operation of the blowers as well as their safe accommodation must be ensured.
Make all necessary safety arrangements (e.g. personal protective equipment) to assure riskless handling in the environment of the exhaust duct.

WARNING
Corrosive substances!
Irritating or corrosive gases or substances can cause chemical burn of body tissue and serious eye injury in case of contact.
In case of contact with skin or eyes the affected spots must be cleaned immediately!
Objects which have been contacted with irritating or corrosive gases or substances must be cleaned accurately.

WARNING
Hot surface!
Several device parts can develop high temperatures.
Burn of skin can be caused!
For protection against possible injury protective gloves must be worn.

WARNING
Laser beam!
Serious eye injury up to blindness can be caused.
Do not look into the laser beam. Never direct the laser to persons. Avoid reflexions of the laser beam.
For protection against possible injury eye protection must be worn.

2.1.3 Personal protective equipment

WEAR PROTECTIVE CLOTHING
For protection against injury at any kind of hazard protective clothing must be worn.

WEAR PROTECTIVE FOOTWEAR
For protection against possible injury protective footwear (e.g. safety shoes) must be worn.

WEAR PROTECTIVE GLOVES
For protection against possible injury at touching of device components protective gloves must be worn.

WEAR HEAD PROTECTION
For protection against possible injury by falling objects or bouncing hazard a head protection must be worn.

WEAR RESPIRATORY PROTECTION
For protection against possible suffocation hazard or injury of respiratory system organs by poisonous gases a respiratory protection must be worn.

WEAR EYE PROTECTION
For protection against possible eye irritation by corrosive gases or substances eye protection must be worn.

WEAR HEARING PROTECTION
For protection against possible hearing impairment by high noise level hearing protection must be worn.

2.2 Requirements for personnel

This operation manual is directed to technically qualified personnel which have been specially instructed or which possesses appropriate knowledge in the field of measuring, control and feedback control technology, called automation technology further on.

Qualified personnel are persons who

- are either familiar as configuring personnel with the safety concepts of automation technology
- or are instructed as operating personnel in operation with automation technology equipment and are acquainted with the contents of these instructions referring to operation
- or have been instructed as commissioning and/or service personnel to be qualified for repair of such automation technology equipment respectively who are authorised to energise, ground and tag circuits and devices/systems according to the standards of safety engineering.

The knowledge and the technically correct realisation of the safety instructions and operating directions described in this operation manual are the requirement for hazard-free assembly and commissioning as well as for safety at operation and upkeep. The specialised personnel must be familiar with the general risks and hazards and know and observe the respective safety precautions.

Unqualified interventions into the device or non-observance of the operation manual or of the affixed labels on the product can result in personal or material damage.

Any work at the device must solely be executed by qualified specialised personnel in observation of the corresponding regulations (central association of electrical engineering and industry).

2.3 Electrical power supply

DANGER
Hazardous voltage! Danger of electric shock.
Also when the device is switched off, there can be high voltage inside it.
Any work at the device must solely be executed by qualified personnel. The following requirements must be observed.

- The power supply has to be installed and secured according to the corresponding legal safety regulations and prescriptions.
- The device must only be connected to the supply voltage designated on the type plate.
- A protective separation between primary and secondary circuit is generally ensured. Low voltage which is connected must also be generated by protective separation.
- Grounding contact:
  - The device must always be grounded. It has only to be operated at a power supply with grounding contact.
  - The grounding conductors in the device or the feeder must not be separated or removed.
  - The protection effect must not be abolished by an extension without grounding conductor. Every kind of interruption of the grounding conductor inside or outside of the device is hazardous and not permitted.
  - In the case of insufficient grounding or damaged grounding conductor the device must be shut down and secured against unauthorised or inadvertent activation.
- Fuses:
  - In the case of a required fuse change, only fuses must be used which are according to the old fuses in type and capacity.
  - Auxiliary fuses must not be used.
  - Fuse holders must not be shorted.
- Cables must be laid that an accident risk by tripping or getting caught is excluded.
- Covers:
  - The device must not be operated when covers or other parts have been removed, because current carrying parts are divested of covering in operation.
  - If not explicitly requested, work inside the device must not be executed.
  - Before opening the device it must be de-energised by exerting the pre-fuse.
  - If work at the opened device is necessary (adjustment, maintenance etc.), this work must only be executed by appropriate qualified personnel which is familiar with the hazard points and which has knowledge of avoiding hazards by proper safety precaution.

Electrical safety

If the electrical safety of the device is not given anymore, the device must be shut down and secured against unauthorised or inadvertent activation.

The electrical safety of the device is not given any more if it:

- has visible external damage
- does not work correctly anymore
- has been stored under impermissible or inappropriate conditions for any length of time
- has been encountered impermissible strain during transport

2.4 Components

The device as well as the single components must only be operated as original variant. In case of exchanging single elements only the original parts of the manufacturer must be used. Components are configured device-specific and hence they are not interchangeable between different devices.

2.5 Electronic components

Electrostatic discharges can cause damage at the electronic components. The following precautions must be taken:

- Electronic components must be kept away from statically charged surfaces (PVC plastics, plastic bags etc.).
- Wear a special ESD wrist band or use a grounded, antistatic working surface.

2.6 Configuration settings

Changes of configuration can endanger the safety and function of the device. Configuration settings must solely be executed by an authorised service technician or by factory personnel of the manufacturer.

2.7 Blowers

The blowers only serve the conveyance of clean air. They must never be operated with open suck-in or blow-out opening. Possible foreign material or impurities must be filtered before entering the blowers.

DANGER
Poisonous substances!
In the case of insufficient ventilation or outage of a blower poisonous gases can escape through the respective blower and cause most serious injury or death.
In the case of operation of the device in closed rooms the blowers must be placed in a separate sector with sufficient ventilation. Defective blowers must be exchanged immediately.

WARNING
Hot surface!
During operation the blower housing heats up.
Burn hazard!
For protection against possible injury protective gloves must be worn.
If the temperature increases above + 50 °C, the blower must be secured by the licensee against direct contact.

WARNING
Heavy suck-in and blow-out force!
At the suck-in opening objects, clothing or hair can be sucked. Sucked objects can be expelled from the blow-out opening with high speed.
Injury can be caused.
Do not grasp into the suck-in/blow-out opening.
The suck-in opening must be covered by a grid with a fine filter cartridge. The blow-out opening must be covered by a protective grid according to DIN EN 294.
During operation a respective distance to the openings must be observed.

The sounds emitted from the blowers reach up to approx. 80 dB(A), noise level LA.

NOTICE
Dr. Födisch Umweltmesstechnik AG recommends precise measurements by the licensee at location. Possibly, an acoustic insulation is necessary.

In case of necessity of acoustic insulation, this has to be performed by the licensee so that the legally allowed maximum values at working places in the environment of the blower are not exceeded.

Stationary heating installation (option)

DANGER
Hazardous voltage!
When using a stationary heating installation, there is also a voltage at the stationary heating installation if the motor protection switches -S01 and -S02 are switched off.
Danger of electric shock.
Any work at the device must solely be executed by qualified personnel.

3 Function

The dust concentration measuring device PFM 06 ED is a highly sensitive system. It is used for continuous extractive measurement of dust contents in wet and sticky exhaust gases.

For dust concentration measurement the measuring gas is sampled from the process by a temperature-controlled probe and conveyed to a measuring chamber which contains an optical measuring unit. The sucked off measuring gas is continuously diluted and dried with hot and dust-free ambient air. The active principle of dust measurement is based on the optical scattered light measurement. Therefore a laser lance unit is arranged in a cylindrical measuring chamber and streamed with the conditioned measuring air. In the electronics of the control unit the signal of the optical unit is converted to an equivalent dust signal.

Via the display of the operating device the current measuring value is displayed. All necessary parameters can be reviewed and set by the menu. The operating is carried out via the keypad.

The functional scheme of PFM 06 ED is shown in Fig. 1. Alternatively, the PFM 06 ED can be operated with compressed-air respectively nitrogen (see Fig. 2, page 16).

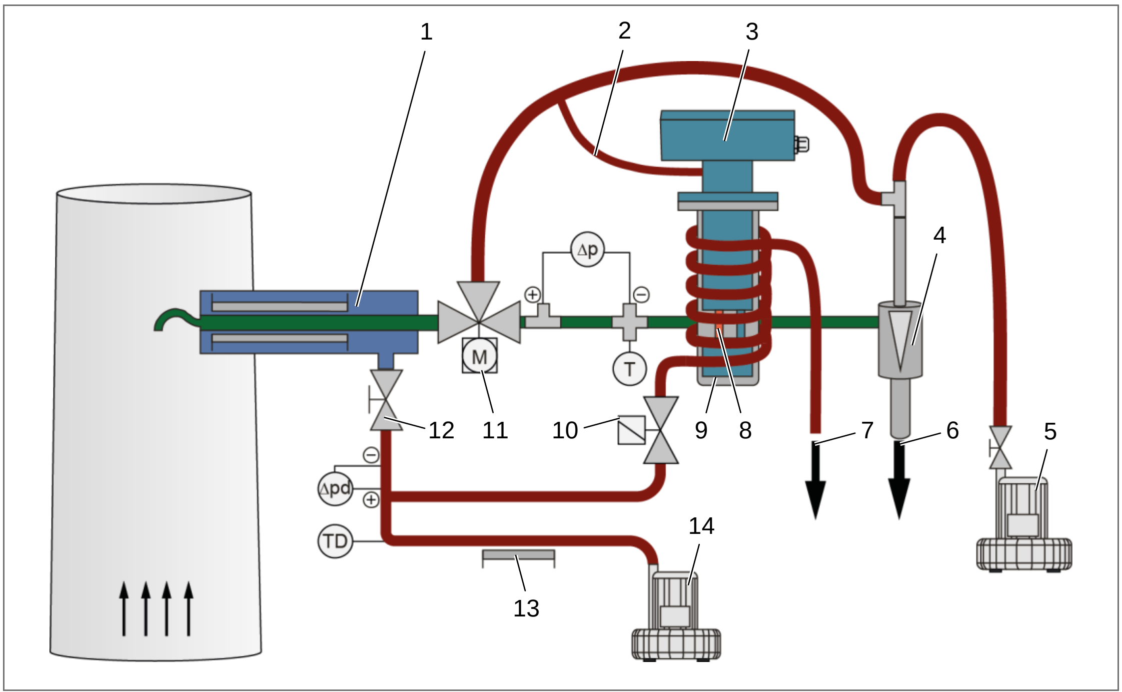

Fig. 1: Functional scheme with two blowers

1 Probe tube with heating
2 Purge air of probe
3 Probe electronics
4 Injector
5 Blower for injector air
6 Exhaust air
7 Heating air of measuring chamber
8 Laser beam
9 Measuring chamber
10 Magnetic valve
11 3/2-way ball valve
12 Valve of dilution air supply
13 Heating for dilution air
14 Blower for dilution air

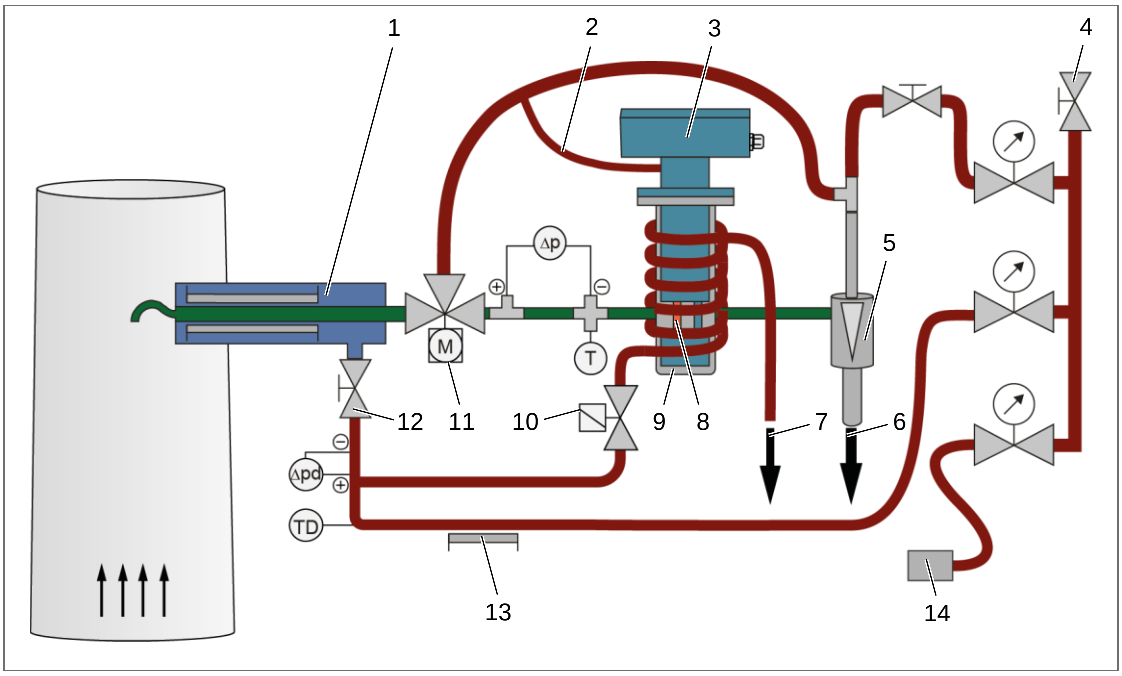

Fig. 2: Functional scheme with compressed-air supply

1 Probe tube with heating
2 Purge air of probe
3 Probe electronics
4 Pressured-air inlet with shut-off valve
5 Injector
6 Exhaust air
7 Heating air of measuring chamber
8 Laser beam
9 Measuring chamber
10 Magnetic valve
11 3/2-way ball valve
12 Valve of dilution air supply
13 Heating for dilution air
14 Probe ventilation (e.g. Vortex Cooler)

3.1 Gas path

The PFM 06 ED extracts continuously a partial stream from the exhaust. Thereby the suction is realised by an injector. Because the measuring gas possesses mostly a strong concentration of the particles to be measured, a defined (heated) dilution air stream is added to the sucked partial gas stream at the probe inlet of the PFM 06 ED (see section 3.2, page 18), whereby the measuring gas is diluted in a defined way. At the same time the gas mixture is heated up to an adjustable temperature where the humidity evaporates and the particles to be measured have a solid state. This sucked, diluted and heated measuring gas afterwards passes the measuring chamber and leaves the measuring device. The following figure shows the path of the measuring gas.

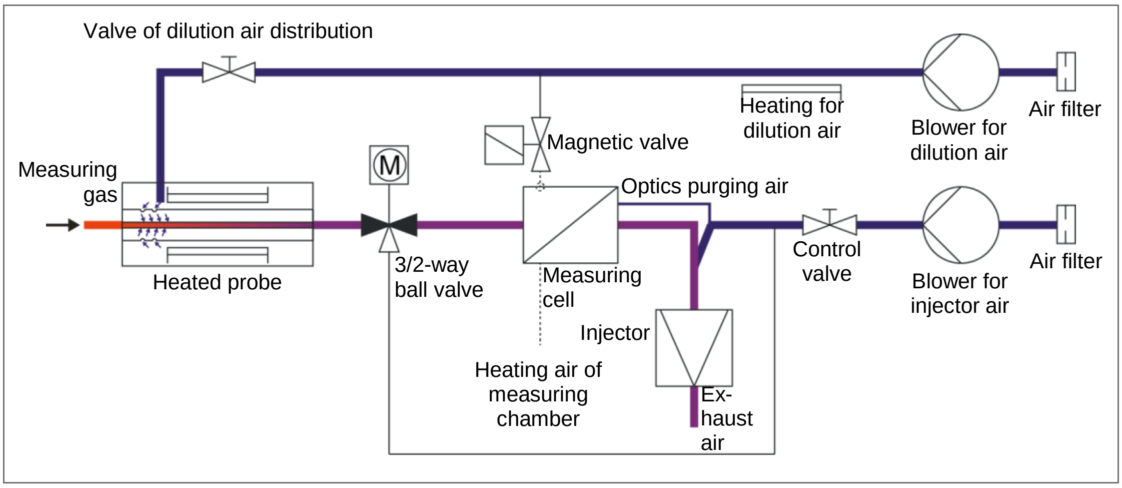

Fig. 3: Gas flow diagram at measuring mode

For diagnostics and cleaning reasons the PFM 06 ED carries out a purging operation autonomously. Thereby a zero and reference point control (see section 3.3, page 18) and a cleaning of the gas paths leading measuring gas is made via the 3/2-way ball valve (see section 3.4, page 19). Through the opened magnetic valve the measuring chamber is heated indirectly by the gas stream. A presentation of the purging operation is shown in Fig. 4.

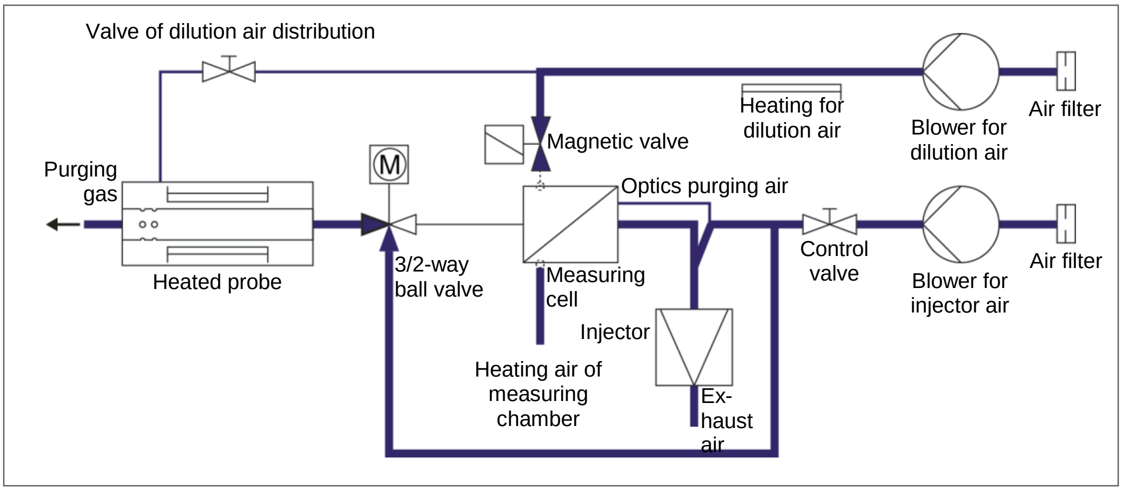

Fig. 4: Gas flow diagram at purging mode

3.2 Dilution

The dilution is a degree for the concentration of the sucked measuring gas. The dilution is specified in %.

$$D = \left(\frac{\dot{M}_m}{\dot{M}}\right) \cdot 100 \approx \left(\frac{F_M}{F}\right) \cdot 100$$

D	dilution [%]
$\dot{M}_m$	sucked mass flow [kg/s]
$\dot{M}$	mass flow in measuring chamber [kg/s]
F_M	sucked measuring gas volume flow [m³/h]
F	volume flow in measuring chamber [m³/h]

Normal values for the dilution D are in the range between 30% and 70%. Respective values are defined as follows:

- D = 100% ➝ no dilution air
- D = 0% ➝ only dilution air (theoretic value → F_M = 0 m³/h)

3.3 Zero and reference point and zero offset correction

For assuring an exact measurement the PFM 06 ED carries out an automatic zero and reference point control in the interval of 4 h (beginning at switch-on). At first the current zero point of the sensor is tested. Afterwards the relating reference value is subtracted from the reference point value of the sensor (see chapter 8 “Calibration”, page 41). Thereby the sensor is tested for correct function.

At the current output the relating reference value for the zero point (0 ≙ 4 mA) and for the reference point (70% ≙ 15.2 mA) is successively output for respectively 3 min.

NOTICE
In the case of too large differences a maintenance/error message is generated (see chapter 11, page 83).

The following figure shows an example of the signal sequence at output of the dust concentration (0...15 mg/m³).

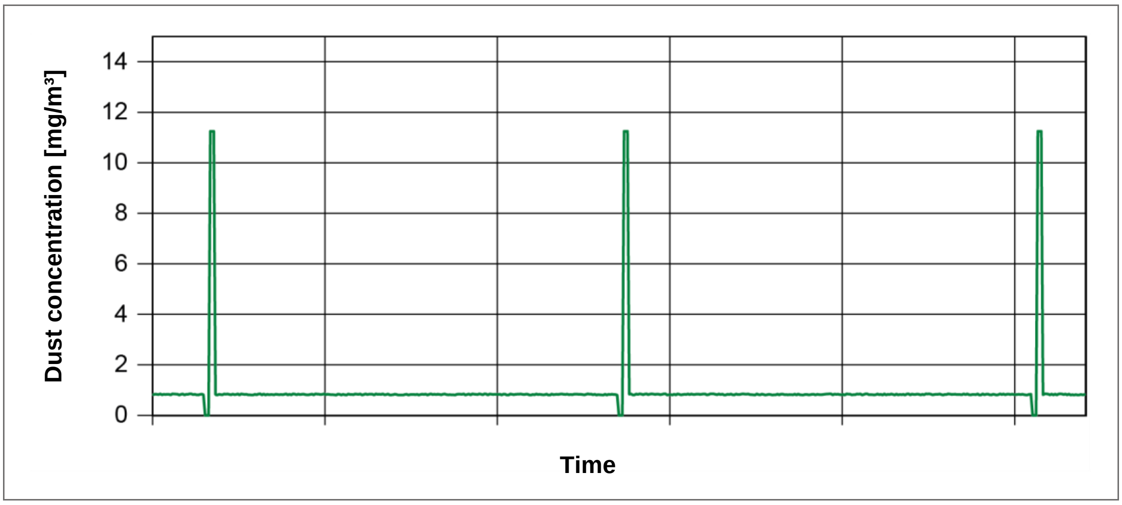

Fig. 5: Signal sequence of zero and reference point control, example

After starting the zero and reference point control, the PFM 06 ED moves to purging operation. Thereby the dust-free scattered light value of the optical sensor is determined. As average value with the last five measurements it is evaluated in purging operation as zero offset correction value. By this zero offset correction value the scattered light values (raw signal cal) of the optical sensor are corrected.

After switch-on at the beginning and at the end of preheating the zero offset correction value is determined.

NOTICE
If the device is already in purging operation, the zero offset correction value can be determined new without averaging by manual starting of the zero and reference point control (see section 9.5.6 “Check zero and reference point”, page 64).

CAUTION
Wrong zero offset correction value of the scattered light!
After demounting and mounting of the optical sensor (e.g. at linearity control) or through premature abort of preheating measuring faults can be caused.
Determine the zero offset correction value of the scattered light manually new or carry out a reset with re-preheating.
In the case of a correct zero offset correction value a raw signal cal in the range of -0.3 V to 0.3 V is displayed during purging operation.

3.4 Self-diagnostics and cleaning

For self-diagnostics and cleaning of the elements of the probe which are leading measuring gas, the 3/2-way ball valve and the magnetic valve for heating of the measuring pot are set to purging operation in the interval of 4 h (standard value) and the probe is cleaned. Thereby the 3/2-way ball valve and the magnetic valve are exerted automatically.

This cleaning procedure is also carried out in the case of a detected pollution (volume flow too low). Thereby the volume flows of the measuring chamber and of the dilution air are controlled. If the respective preset minimal value is undershot, a maintenance request message is displayed via the display of the operating device. If the respective volume flow is lower than 80% of this value, an error message is output and by a purging process the increasing of the volume flow is attempted.

NOTICE
For purposes of diagnostics or maintenance the 3/2-way ball valve can also be operated manually (see section 4.1.1, page 21).

4 Design

The dust concentration measuring device PFM 06 ED consists of a probe with GRP weather protection casing and a separate operating device where the electrical distribution is located. The operating device as well as two blowers for dilution and injector air are attached in a profile rack.

4.1 Probe

The probe of the PFM 06 ED is integrated in a two-part GRP weather protection casing which is fixed directly at the flange of the probe. The probe tube is a heated double jacket tube with integrated dilution. This is realised by a mixing jet. The probe ventilation, depending on application, is carried out by a casing fan or thermostatted by a Vortex Cooler (option) supplied with compressed-air respectively nitrogen.

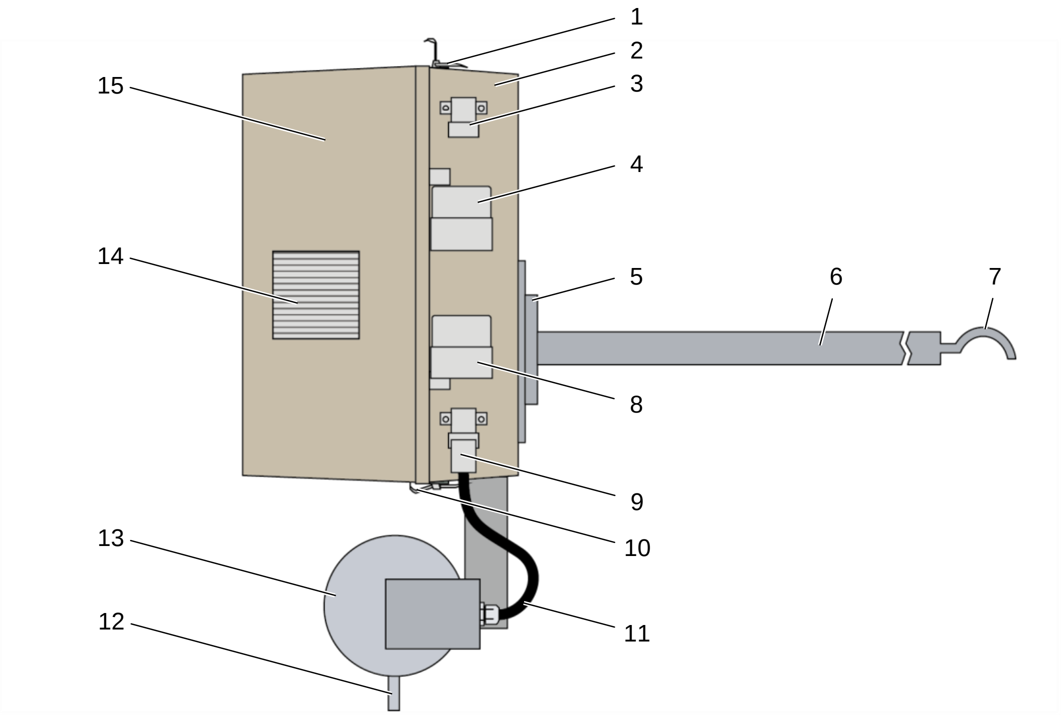

Fig. 6: Design of probe (with closed weather protection casing), view from right side

1 Casing lock, opened
2 Casing
3 Plug-in connector for probe ventilation (optional)
4 Plug-in connector for signal transfer
5 Flange
6 Heated probe tube
7 Sampling bow
8 Plug-in connector for power supply
9 Plug-in connector for dilution air heating
10 Casing lock, closed
11 Connecting cable for dilution air heating
12 Connection for dilution air
13 Dilution air heating
14 Inlet filter of probe ventilation
15 Casing cap

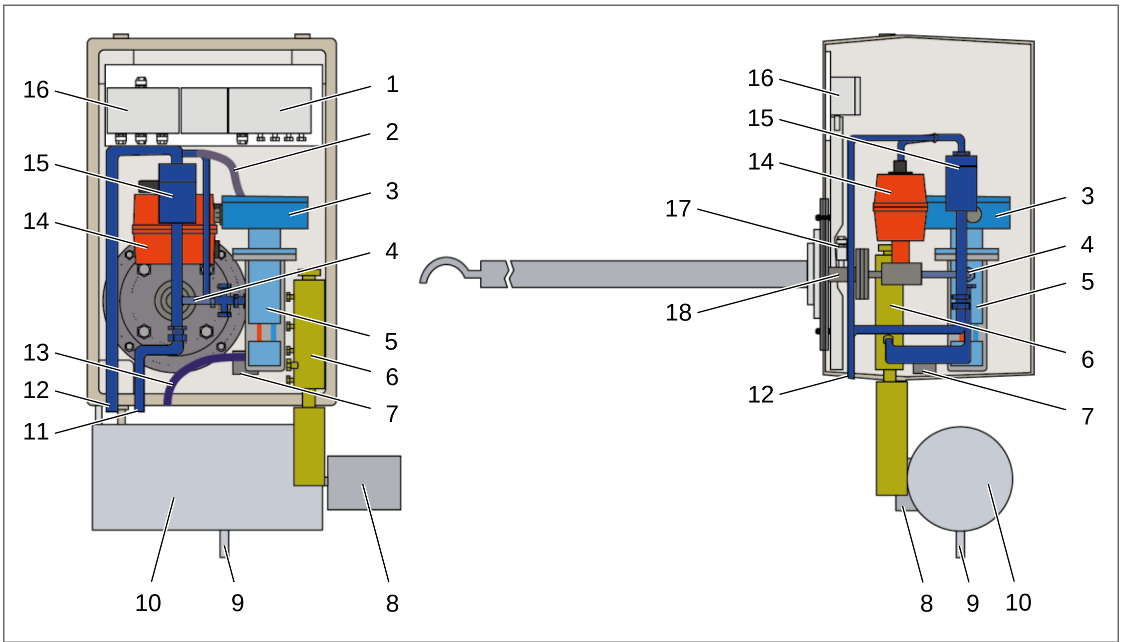

Fig. 7: Internal design of the probe

1 ∆p measurement
2 Purging air of optical sensor
3 Probe electronics
4 Connecting tube between ball valve and measuring chamber
5 Measuring chamber
6 Dilution air distribution with valve
7 Magnetic valve
8 Temperature limiter of dilution air heating
9 Connection for dilution air
10 Dilution air heating
11 Exhaust connection
12 Connection for injector air
13 Outlet for heating air of measuring chamber
14 3/2-way ball valve
15 Injector
16 Temperature measurement
17 Connection for probe heating
18 Dilution air feed (oppositely hidden)

4.1.1 3/2-way ball valve

The 3/2-way ball valve is located inside the probe. There it regulates the self-diagnostics and the cleaning of the elements which are leading measuring gas (see section 3.4, page 19).

Via the operating device the 3/2-way ball valve can be set to measuring operation by pressing the key – and it can be set to purging operation by pressing the key + (see Fig. 22, page 46).

NOTICE
The functions of the keys + and – for moving the ball valve are only active at chosen measuring value display in standard mode, service mode and at alternative displays.

Alternatively, the rotary switch for measuring/purging operation (see Fig. 8, page 22) can be operated manually. For this, the handwheel (see Fig. 9) at the side of the 3/2-way ball valve housing must additionally be set to manual operation.

CAUTION

Ball valve position!

Depending on the chosen operation mode, wrong position of the 3/2-way ball valve causes pollution of the measuring chamber or measuring faults.

- For works of mounting, commissioning and shutdown the 3/2-way ball valve must be set to manual and purging operation.
- For continuous dust concentration measurement the 3/2-way ball valve must be set to measuring and automatic operation.

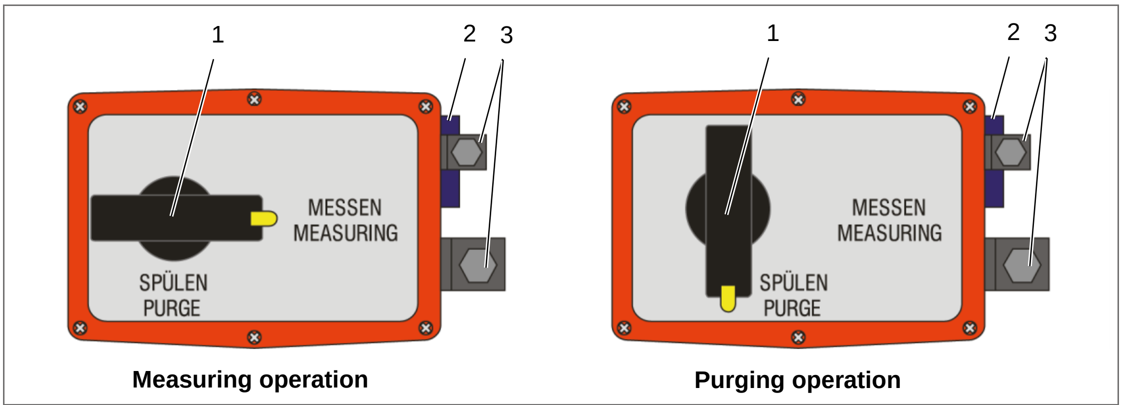

Fig. 8: 3/2-way ball valve, view from top

1 Rotary switch “PURGE/MEASURING”
2 Handwheel “AUTO/MAN”
3 Plug-in connector

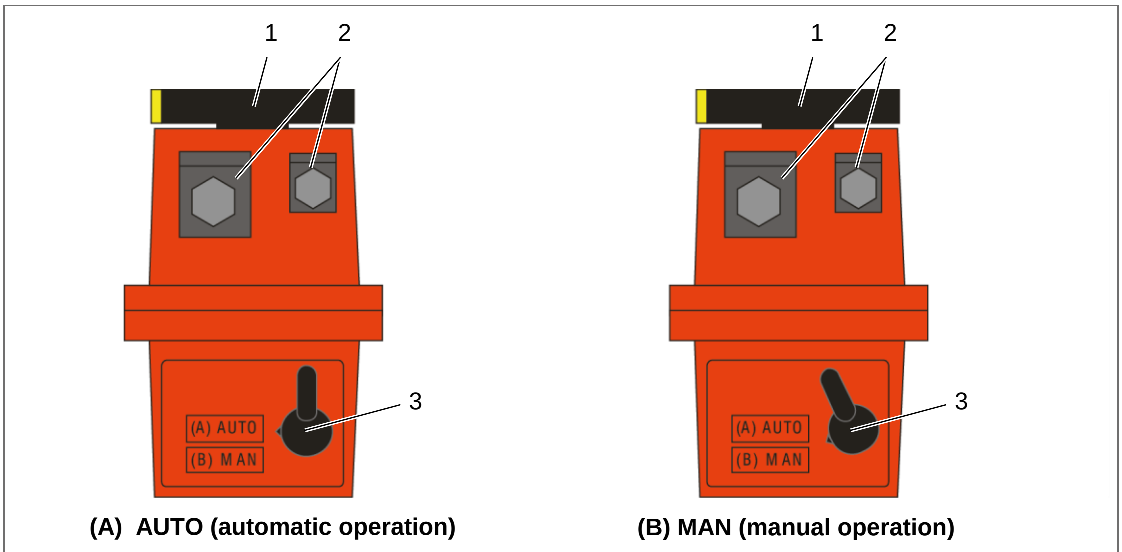

Fig. 9: 3/2-way ball valve, view from right side

1 Rotary switch “PURGE/MEASURING”
2 Plug-in connector
3 Handwheel “AUTO/MAN”

NOTICE

After changeover from purging to measuring operation the device still stays in maintenance mode for 1 min. Thereby the LED “Maintenance” at the operating device lights and the message “maintenance state” is displayed.

4.1.2 Probe ventilation

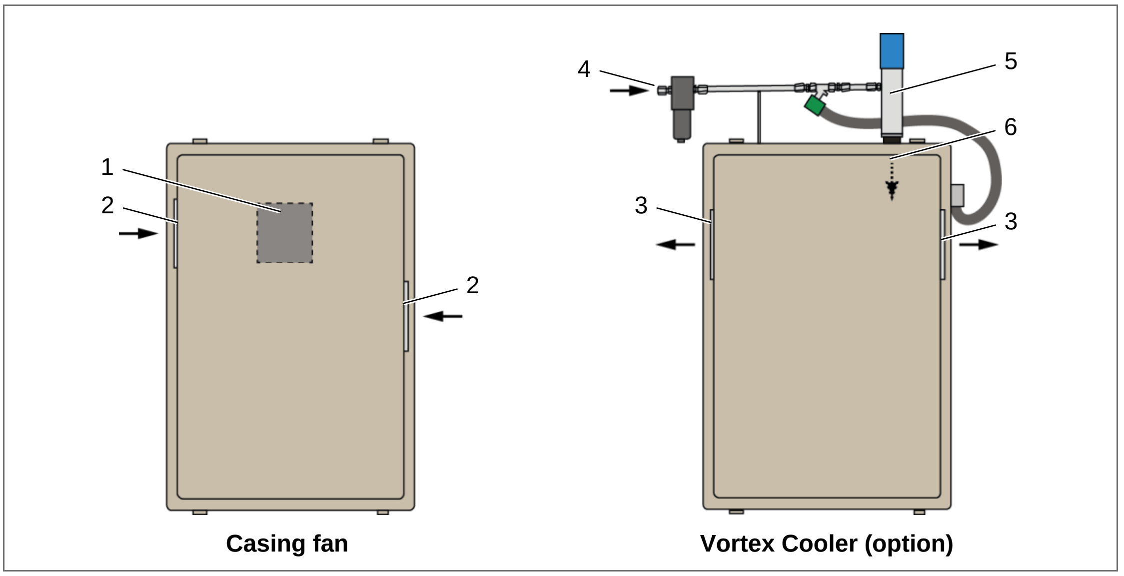

Fig. 10: Variants of probe ventilation

1 Casing fan (at rear side of casing)
2 Inlet filter (at casing cap)
3 Outlet filter (at casing cap)
4 Compressed-air/nitrogen supply
5 Vortex Cooler
6 Inlet of cooling medium (inside the casing)

4.1.3 Dimensions of probe

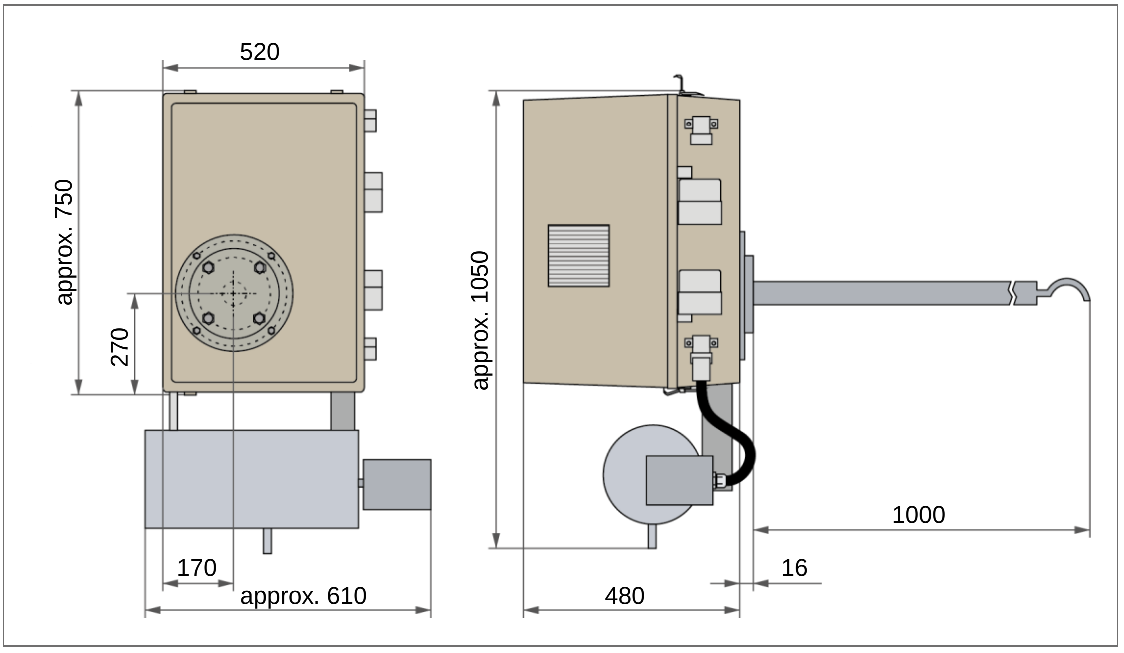

Fig. 11: Dimensions of probe (specifications in mm)

4.2 Profile rack with operating device and blowers

The operating device and the two blowers are tightly mounted and wired on a conjoint profile rack. The rack serves the fixing at a wall or on the floor.

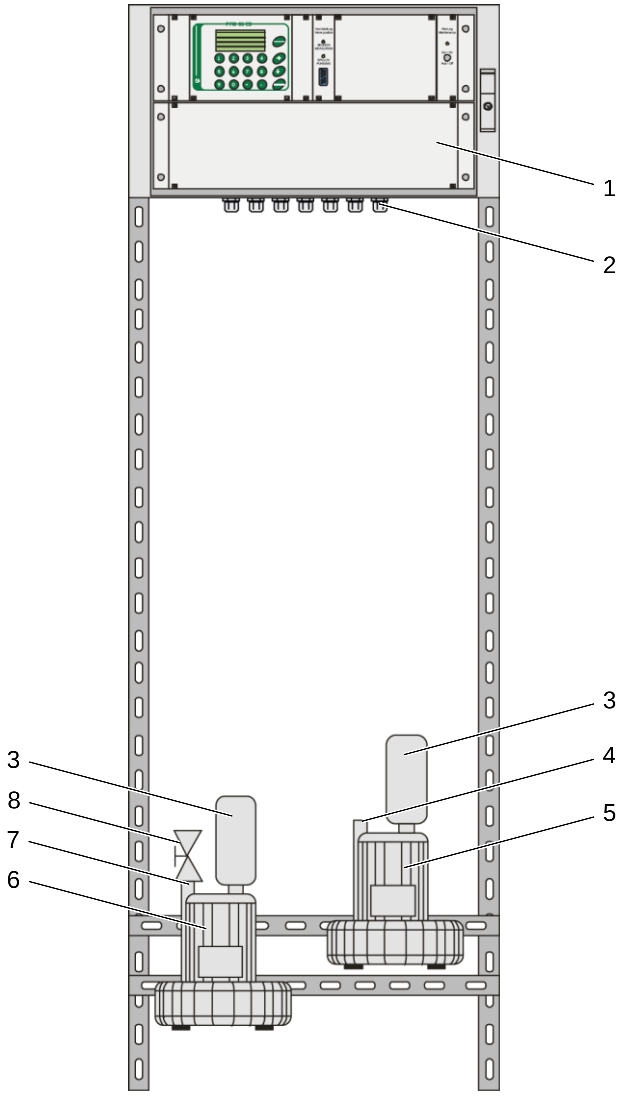

Fig. 12: Design of the profile rack with operating device and blowers

1 Operating device
2 Cable entry (7x)
3 Air filter
4 Connection for dilution air
5 Blower for dilution air
6 Blower for injector air
7 Connection for injector air
8 Control valve of injector air blower

4.2.1 Dimensions of the profile rack

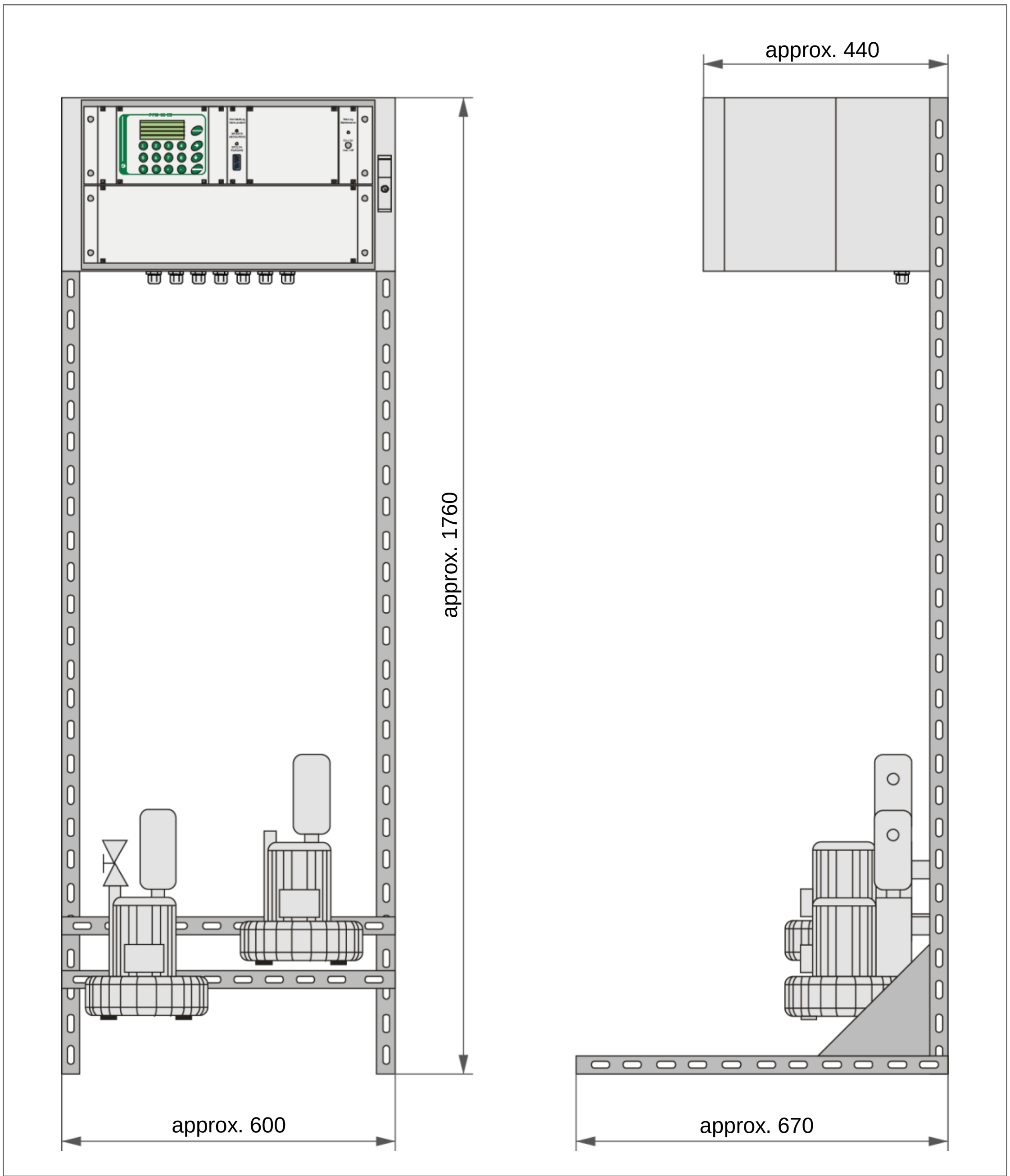

Fig. 13: Dimensions of the profile rack, incl. operating device and blowers (specifications in mm)

4.2.2 Operating device

The operating device is constructed as three-part 19" protection casing. In the operating device the complete electrical connection, the signal evaluation as well as displaying and output of the measured dust values is made. Display and operating unit are accessible by opening the fore door (with inspection window). Via the second door the access to the mounting plate (see section 4.3.2, page 27) is enabled.

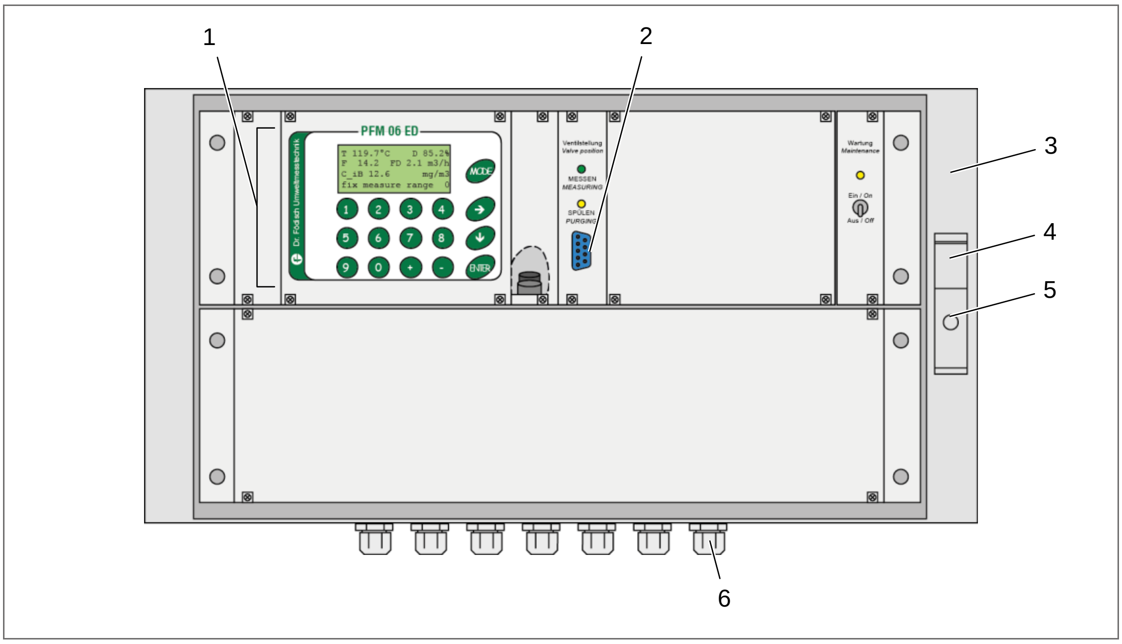

Fig. 14: Design of the operating device

1 Range of display and operating elements
2 Connection of RS232 interface
3 Door (2x, one after another)
4 Opening lever
5 Pivot for control cabinet key
6 Cable entries

NOTICE
Display and operating elements, see section 9.1, page 46.

4.3 Connection assignment

4.3.1 Probe

At the right side of the probe casing (see Fig. 6, page 20) there are three sockets for electrical connection of the probe:

- plug-in connector for signals (4)
- plug-in connector for power supply (8)
- plug-in connector for dilution air heating (9)

NOTICE
The pin assignments of the plug-in connectors have to be gathered from the circuit diagrams (see appendix).

4.3.2 Operating device (mounting plate)

Inside the operating device the mounting plate is installed. It can be reached by opening the two doors at the operating device with the aid of a control cabinet key.

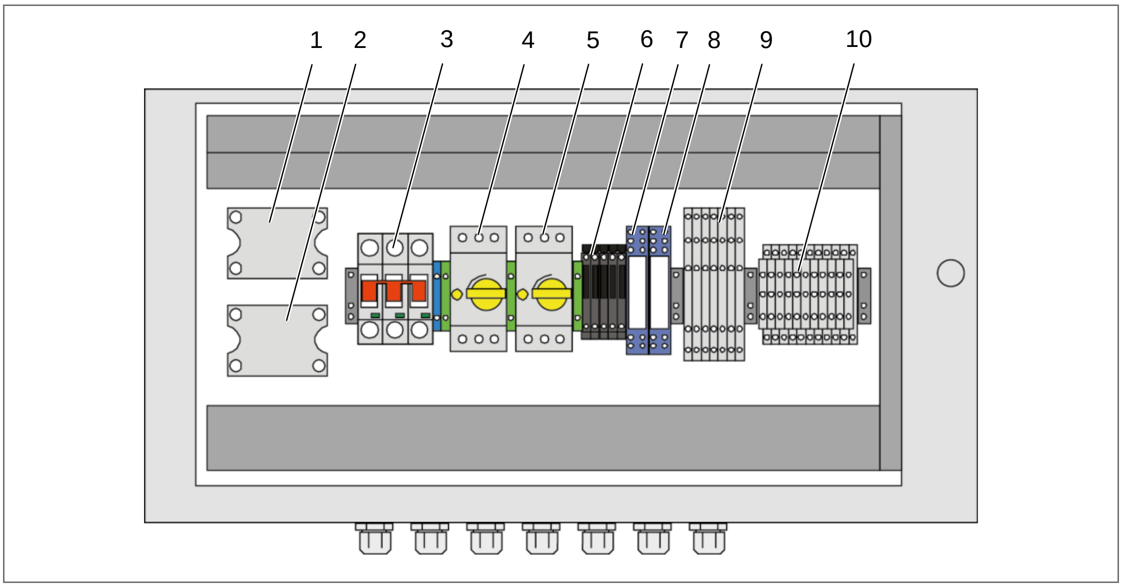

Fig. 15: Mounting plate inside the operating device

1 Relay –K01
2 Relay –K03
3 Main fuse –F01
4 Motor protection switch –S01
5 Motor protection switch –S02
6 Micro-fuses –F02 to –F06
7 Relay –K02
8 Relay –K04
9 Terminal strip –X01
10 Terminal strip –X03

Designation	Type
-F01	Main fuse of power feed 10 A
-F02	Micro-fuse for power supply unit 5 AT
-F03	Micro-fuse for probe heating 4 AT
-F04	Micro-fuse for 3/2-way ball valve / magnetic valve 2 AT
-F05	Micro-fuse of dilution air heating 4 AT
-F06	Micro-fuse for casing fan of probe 0.25 AT
-K01	Solid state relay for probe heating
-K02	2-way relay for 3/2-way ball valve /magnetic valve
-K03	Solid state relay of dilution air heating
-K04	2-way relay for casing fan of probe
-S01	Motor protection switch of blower for injector air
-S02	Motor protection switch of blower for dilution air
-X01	Terminal strip for voltage distribution
-X03	Terminal strip for signal transfer

Tab. 1: Connections and switches at mounting plate

NOTICE
The assignment of the terminal strips and plug-in connectors have to be gathered from the circuit diagrams (see appendix).

5 Transport and scope of supply

5.1 Transport

The probe of the PFM 06 ED is stored in a handling case. The probe tube is additionally packed in a protective covering. Blowers and operating device are mounted at the profile rack and are transported separately from the probe.

CAUTION
Through heavy percussion at transport (e.g. by falling) susceptible elements can be damaged.
Chose appropriate means of transport.

1. Check the device as well as the packaging material for transport damage.
2. Document possibly existing damage.

5.2 Scope of supply

According to the legal sales contract the respective scope of supply is specified in the shipping documents attached to the supply.

The scope of supply includes:

- 1 probe with GRP weather protection casing
- 1 welding flange with adapter gasket DN 80 PN 6, nominal width 100 mm (incl. separate or integrated return) + each with 4 fixing screws and nuts
- 1 optical sensor
- 1 operating device
- 1 control cabinet key
- 2 blowers respectively side duct impellers (not in the case of compressed-air/nitrogen supply)
- 1 profile rack for fixture of the operating device and both blowers
- 1 operation manual
- 3 hoses with 2 hose clamps in each case
- options:
  - 1 stationary heating installation for blowers
  - 1 Vortex Cooler
  - 1 compressed-air/nitrogen distribution unit
  - 1 weather protection casing for operating device
  - 1 brush set for cleaning of gas-leading parts

NOTICE
Depending on purchase configuration deviations in technical design are possible.

1. Check the scope of supply for completeness and intactness.
2. Remove the packaging material.
3. Keep the packaging material for possible re-use.

6 Mounting

DANGER
Open ducts with harmful gases, high temperatures or high pressure can cause serious health damage, injury or death.
Mounting of the welding flange must solely be executed at standstill of the system.

CAUTION
Ball valve position!
Wrong position of the 3/2-way ball valve causes pollution of the measuring chamber.
From the beginning of mounting until commissioning the 3/2-way ball valve must be set as follows:

- handwheel in manual operation (position “(B) MAN”, see Fig. 9, page 22)
- rotary switch in purging operation (position “PURGE”, see Fig. 8, page 22)

6.1 Ambient conditions

CAUTION
The device must not be operated in potentially explosive atmosphere.

For correct operation of the device the following ambient conditions apply:

- Ambient temperature: -20...+50 °C
- Relative humidity: max. 90% (non-condensing)
- Protection against wetness
- Location free of percussion

6.2 Requirements for installation place

The installation point of the probe must meet the requirements of the local valid guidelines (e.g. EN 13284-1; see Fig. 16):

- run-in zone: min. 5-fold length of the duct diameter
- run-out zone: min. 2-fold length of the duct diameter
- measuring line preferably in vertical duct

In case of doubt the installation place can be defined by a responsible measuring institute (measuring point according to §§ 26/28 BImSchG).

NOTICE
Dr. Födisch Umweltmesstechnik AG recommends run-in and run-out zones which correspond to the 5-fold length of the duct diameter (see Fig. 16).

NOTICE
To get a representative determination of the dust load above the cross section of the duct, a homogenous dust and stack gas distribution must be existent at the measuring point.

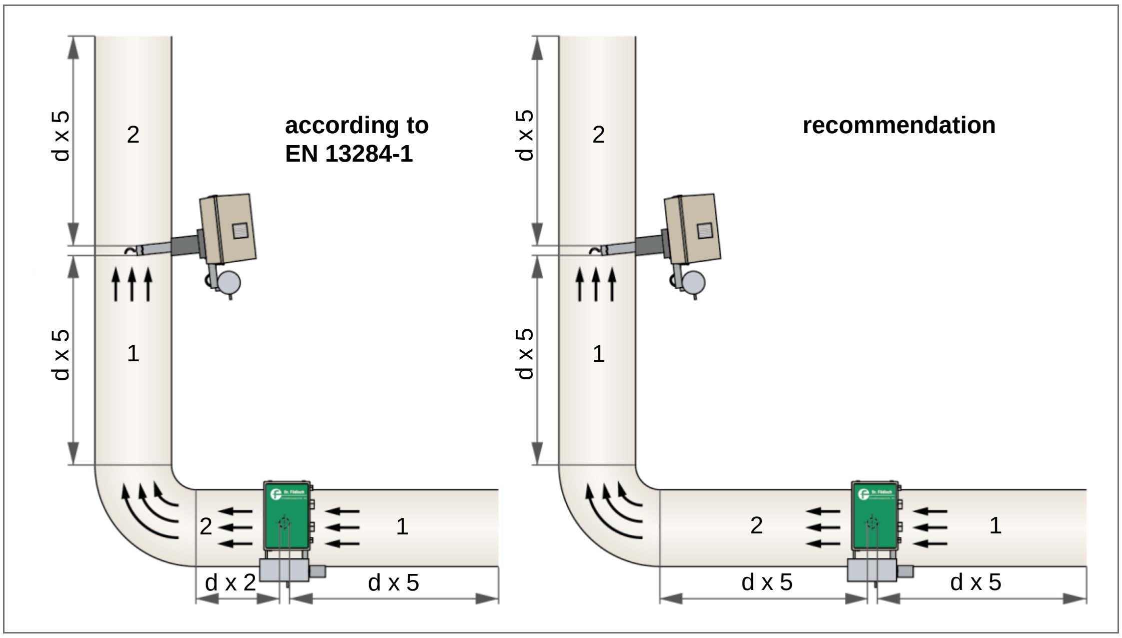

Fig. 16: Run-in and run-out zone at the measuring point

1 Run-in zone
2 Run-out zone

6.3 Mounting of welding flange

CAUTION
Wrong installation of the welding flange can cause measuring faults.
The welding flange must be grounded. For this, it must be involved into the local potential equalisation.

CAUTION
In case of wrong installation, condensate accruing during operation can deposit at the duct wall.
To avoid deposits, the installation angle must be between 2° and 5°, so that at the later installation of the probe the probe tube tends downwards into the duct.

6.3.1 Welding flange with separate return

1. Furnish the measuring point allowing for the necessary run-in and run-out zone (see section 6.2, page 32).
2. Mount the welding flange (3, Fig. 17) at the provided aperture in the duct wall (1).
3. Mount the return flange (5) at the provided aperture in the duct wall (1).

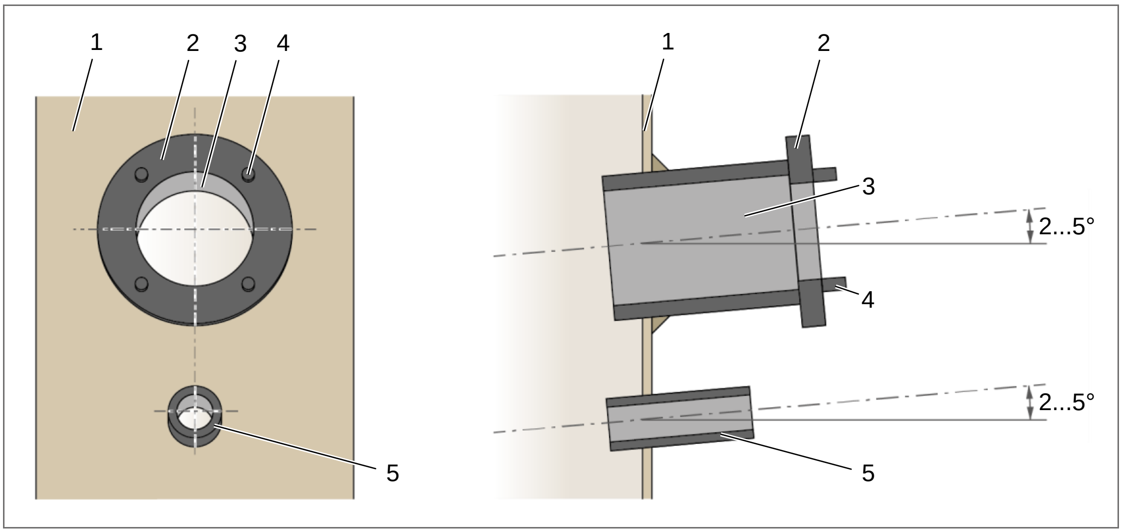

Fig. 17: Mounting of welding flange – variant with separate return

1 Duct wall
2 Adapter
3 Welding flange
4 Thread bolt (4x)
5 Return flange

6.3.2 Welding flange with integrated return

1. Furnish the measuring point allowing for the necessary run-in and run-out zone (see section 6.2, page 32).
2. Mount the welding flange (3, Fig. 18) with integrated return flange (5) at the provided aperture in the duct wall (1).

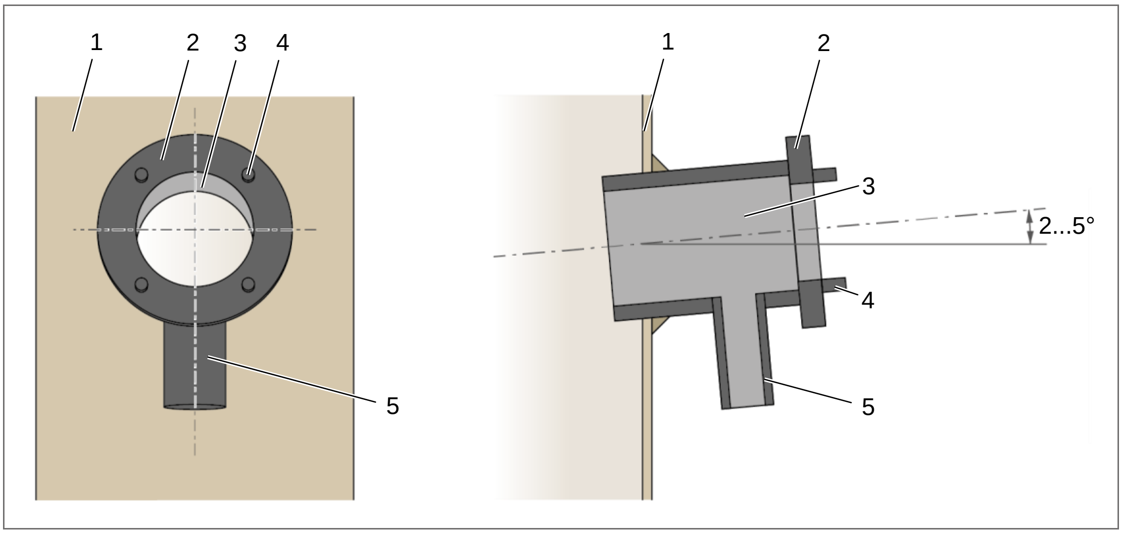

Fig. 18: Mounting of welding flange – variant with integrated return

1 Duct wall
2 Adapter
3 Welding flange
4 Thread bolt (4x)
5 Return flange

6.4 Installation of the probe at the duct

CAUTION

Uncontrolled gas inlet causes pollution of the device.

Before the beginning of mounting, check that the valve of dilution air distribution (6, Fig. 7, page 21) is closed.

NOTICE

The optical sensor is delivered separately. For eased mounting of the probe at the duct the optical sensor should be left demounted at the beginning.

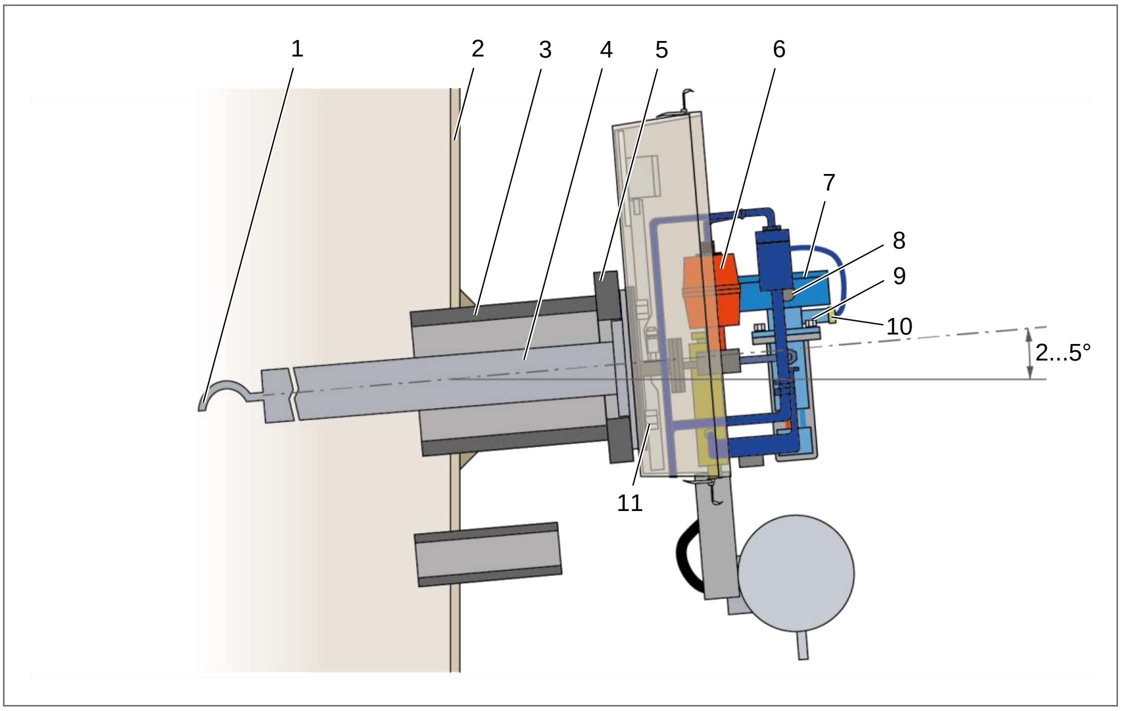

Fig. 19: Installation of probe at the duct (side view, example with separate return)

1 Sampling bow
2 Duct wall
3 Welding flange
4 Probe tube
5 Adapter
6 3/2-way ball valve
7 Optical sensor
8 Electrical connection of optical sensor
9 Fixing nuts of optical sensor (4x)
10 Connection of optical purging air
11 Fixing nut for thread bolt (4x)

1. Open the casing of the probe. For this, release the four casing locks and take the casing cap off (see Fig. 19).

NOTICE

Depending on the incident flow direction of the exhaust gas in the duct, the alignment of the sampling bow must be adjusted if necessary. The probe itself must not be turned.

2. Check the alignment of the sampling bow (1). If necessary, turn the sampling bow until the opening is aligned towards the flow direction.

3. Furnish the adapter (5, Fig. 19, page 35) with a gasket.
4. Hoist the probe to the measuring point and lead the probe tube (4) through the welding flange (3) into the duct, until the probe abuts at the adapter (5).
5. Fix the device by furnishing fixing nuts (11) at the four thread bolts of the welding flange inside the probe. Tighten the fixing nuts with the aid of a screw driver.
6. Mount the optical sensor (7):
   a) Insert the optical sensor vertically into the support in the probe.
   b) Connect the electrical connection (8) of the optical sensor as well as the connection of the optical purging air (10).
   c) Screw the fixing nuts (9) at the support of the optical sensor and tighten them.
7. Set the 3/2-way ball valve (6) to manual and purging operation.

6.5 Placement of operating device and blowers

The operating device and the two blowers are tightly mounted and wired on a conjoint profile rack. For placement the profile rack can be fixed easily at a wall or on the floor.

NOTICE
The distance between the probe and the two blowers should be as short as possible. For a larger distance between probe and operating device the blowers can be mounted on a separate rack and placed near the probe.

1. Place the profile rack at the installation place respectively attach the profile rack at the wall.

WARNING
Imbalance of the operating device!
At opened operating device the profile rack can topple.
Personal damage in terms of simple up to serious injury as well as material damage at the operating device and the blowers can be caused.
Care for a safe fixing of the profile rack on the floor respectively at the wall.

2. Screw the profile rack tight at the installation place.

DANGER
Poisonous substances!
In the case of insufficient ventilation or outage of a blower poisonous gases can escape through the respective blower and cause most serious injury or death.
In the case of operation of the device in closed rooms the blowers must be placed in a separate sector with sufficient ventilation.

3. At operation of the device in closed rooms:
   a) Demount the two blowers from the profile rack.
   b) Place the two blowers in a separate sector with sufficient ventilation.

6.6 Connection of gas and electrical connections

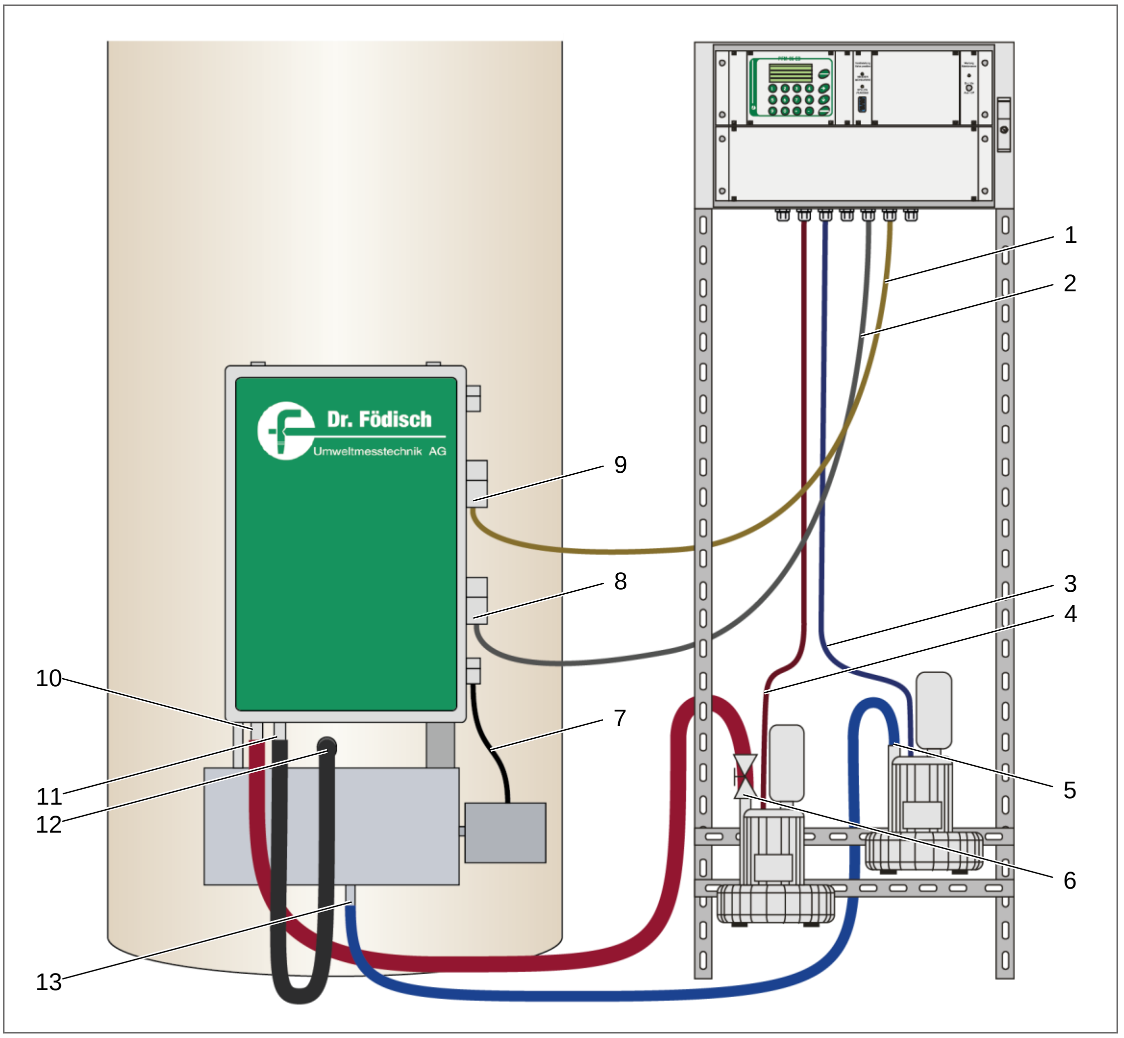

Fig. 20: Gas and electrical connections

1 Low-voltage cable for signal transfer
2 Power cord for probe
3 Power cord for dilution air blower
4 Power cord for injector air blower
5 Dilution air connection at blower outlet
6 Injector air connection at blower outlet
7 Power cord for dilution air heating
8 Plug-in connector for power supply
9 Plug-in connector for signal transfer
10 Connection for injector air
11 Exhaust connection
12 Exhaust connection at return flange (example: separate return in duct wall)
13 Connection for dilution air

6.6.1 Gas connection

1. Connect the hose for injector air with the respective connection (6, Fig. 20) at the blower outlet and secure it with a hose clamp.
2. Connect the hose for dilution air with the respective connection (5) at the blower outlet and secure it with a hose clamp.
3. Connect the exhaust hose with the exhaust connection (11, Fig. 20, page 37) of the probe and secure it with a hose clamp.

DANGER
Poisonous substances!
Poisonous gases can cause serious health damage or death.
Irritating of eyes, skin or respiratory system organs can be caused.
Make all necessary safety arrangements (e.g. personal protective equipment) to assure riskless handling in the environment of the exhaust duct.

4. Connect the exhaust hose with the return flange (12) respectively lay the exhaust hose outdoor.
5. Connect the hose for injector air with the respective connection (10) at the probe inlet and secure it with a hose clamp.
6. Connect the hose for dilution air with the respective connection (13) at the probe inlet and secure it with a hose clamp.

6.6.2 Electrical connection

DANGER
Hazardous voltage!
Parts of the device can be energised with hazardous voltage.
Danger of electric shock.
Any work at the device must solely be executed by qualified personnel.

1. Ensure that the external pre-fuse is disconnected.
2. Connect the plug of the low-voltage cable for signal transfer (1, Fig. 20, page 37; white coating) with the respective plug-in connector (9) at the probe and secure it with the aid of the fastening clip at the socket.
3. Connect the plug of the power cord for the probe (2; grey coating) with the respective plug-in connector (8) at the probe and secure it with the aid of the fastening clip at the socket.
4. If applicable, connect the optional probe ventilation (Vortex Cooler) with the respective plug-in connector (3, Fig. 6, page 20) at the probe.
5. Open the two doors at the operating device one after another with the aid of the delivered control cabinet key. To open the first door, turn the opening lever 90° clockwise.
6. Connect the external power supply with the main fuse:
   a) Connect the PE conductor with the connection “PE”.
   b) Connect the N conductor with the connection “N”.
   c) Connect the L1 conductor with the connection “2”.
   d) Connect the L2 conductor with the connection “4”.
   e) Connect the L3 conductor with the connection “6”.
7. Connect the cables of signal transfer (circuit diagrams: see appendix).
8. Lay all cables inside the lower cable trench and fit the cover.

7 Commissioning

DANGER
Hazardous voltage!
Parts of the device can be energised with hazardous voltage.
Danger of electric shock.
Any work at the device must solely be executed by qualified personnel.

CAUTION
Ball valve position!
Wrong position of the 3/2-way ball valve causes pollution of the measuring chamber.
From the beginning of mounting until commissioning the 3/2-way ball valve must be set as follows:

- handwheel in manual operation (position "(B) MAN", see Fig. 9, page 22)
- rotary switch in purging operation (position "PURGE", see Fig. 8, page 22)

1. Connect the external pre-fuse.
2. Check the value and the phase relation of the voltage (400 V) at the input of the operating device.
3. Switch on the main fuse -F01 (3, Fig. 15, page 27) at the mounting plate by setting the red lever upwards.

CAUTION
Temperatures under 0 °C!
Freezing of the blowers can result in damages.
When using a stationary heating installation, a waiting time of 2 h must be observed before switching on the motor protection switches of the two blowers.

4. Set the 3/2-way ball valve via the handwheel to automatic operation (position "(A) AUTO").
5. Close the micro-fuses -F02, -F03 and -F04 (6).
6. Switch on the motor protection switch of the blower for injector air by turning the rotary switch -S01 (4) clockwise to the position "I".
7. Check the start-up and the rotational direction of the blower for injector air.
8. Switch on the motor protection switch of the blower for dilution air by turning the rotary switch -S02 (5) clockwise to the position "I".
9. Check the start-up and the rotational direction of the blower for dilution air.
10. Open the valve of dilution air distribution (6, Fig. 7, page 21) until the volume flow of the dilution air is min. 1 m³/h. For this, check the value FD at the display of the operating device.
11. Close the micro-fuses -F05 and -F06 (6).
    › The PFM 06 ED starts preheating in purging operation. The duration is approx. 30 min (settings of preheating mode: see section 9.5.11, page 70).
    › At the operating device the LED "PURGING" (3, Fig. 22, page 46) lights.
12. Close the two doors at the operating device one after another and lock them with the aid of the delivered control cabinet key. To close the first door, prior to this, turn the opening lever 90° anti-clockwise.

CAUTION

Danger through abort of preheating!

Premature abort of preheating can cause condensate formation, pollution of the measuring chamber or measuring faults.

Determine the zero offset correction value of the scattered light manually new (see section 3.3 “Zero and reference point and zero offset correction”, page 18) or carry out a reset with re-preheating.

In the case of a correct zero offset correction value a raw signal cal in the range of -0.3 V to 0.3 V is displayed during purging operation.

NOTICE

After changeover from purging to measuring operation the device still stays in maintenance mode for 1 min. Thereby the LED “Maintenance” at the operating device lights and the message “maintenance state” is displayed.

13. Wait until the preheating has finished and the LED “MEASURING” (4, Fig. 22, page 46) at the operating device lights.

WARNING

Hot surface!

Several device parts can develop high temperatures.

Burn of skin can be caused!

For protection against possible injury protective gloves must be worn.

14. Adjust the volume flow of the measuring chamber. For this, open the control valve (8, Fig. 12, page 24) at the injector air blower and check the display at the operating device.
15. Set the volume flow of the dilution so that the dilution D of the measuring gas is between 30% and 70%. For this, regulate the valve of dilution air distribution (6, Fig. 7, page 21) inside the probe and check the display at the operating device.
16. Secure the valve of dilution air distribution against inadvertent adjusting (e.g. with hose clamp).
17. Check the displayed measuring values for plausibility:
    - temperature T (nominal value): e.g. 120 °C → volume flow F: 8...12 m³/h
    - temperature TD (nominal value): e.g. 120 °C → volume flow FD: 3...6 m³/h
18. Hoist the casing cap of the probe to the casing and fix it by the four casing locks.
19. Carry out a device calibration (see section 9.6.1, page 73).
20. Check and save all software parameters and settings by output to a terminal programme via the serial interface (see section 9.5.12, page 72).

8 Calibration

NOTICE

- The device calibration regards solely the electronics of the optical sensor. At delivery the device is already pre-calibrated.
- The dust calibration is at the discretion of the customer.

The operating for calibration has to be carried out pursuant to section 9.6 from page 73.

8.1 Device calibration

The PFM 06 ED as highly sensitive measuring device is subject to the fine fluctuations of the used elements and materials. To detect possible errors at the optical sensor early enough, the reference point is compared and determined at current time. A calibration of the zero point of the optical sensor is not required.

CAUTION
Ball valve position!
Wrong position of the 3/2-way ball valve causes pollution of the measuring chamber.
For device calibration the 3/2-way ball valve must be set as follows:

- handwheel in automatic operation (position “(A) AUTO”, see Fig. 9, page 22)
  → The 3/2-way ball valve is automatically set to purging operation.

8.2 Function test

The function test comprises the following terms:

- visual inspection of the measuring system
- check of linearity with reference material
- manual zero and reference point control with control of long-time drift
- check of data transfer to the evaluation system (analogue and status signals)

8.3 Dust calibration

NOTICE
Before dust calibration a function test must be carried out by a responsible measuring institute (§§ 26/28 BImSchG).

At the PFM 06 ED there is a different mathematical correlation of the physical variables in comparison to the optical in-situ measurement. The optical raw signal cal (scattered light) is dependent on the dust concentration and the adjusted dilution of the sucked measuring gas. Thereby a pollution-caused falling volume flow is corrected.

Due to the different technological circumstances at the respective measuring points, the acquisition of the measuring values is subject to several interferences. Variable influencing factors are amongst others the type of dust, the gas velocity and the temperature. The dusts which have to be measured vary for example in particle size, density, particle shape, charge and other dust and gas characteristics. Therewith a different characteristic curve of the output signal related to the dust concentration results for each case of application of a PFM 06 ED. Therefore a calibration of the device signals by gravimetric comparison measurements according to DIN EN 13284-1 is required. The results of the calibration, i.e. the parameters, can be input directly via the operating of the device. The output is a signal which is proportional to the dust concentration.

Execution of calibration

NOTICE
Before beginning the calibration the proper input of the device and operating parameters has to be made sure.

- adjust device parameters: see section 9.5.2, page 58
- adjust operating parameters: see section 9.5.3, page 60

1. Make sure that the device calibration has been carried out correct.
2. Carry out a gravimetric comparison measurement.

CAUTION
Wrong calibration can cause measuring faults in operation.
If during gravimetric comparison measurement the maintenance request message “sensor range too low” or an error message of any kind is output, the dust calibration must be retried.
If the automatic zero and reference point control (see section 3.3, page 18) is carried out during calibration, there is no data for comparison measurement. The dust calibration must be retried.

NOTICE
In the case of a regulatory dust measurement the calibration must be carried out by a responsible measuring institute (§§ 26/28 BImSchG).

3. Constitute average values from the determined analogue signal $C_{i.B.}$ about the whole period of the gravimetric measurement.

NOTICE
For calibration the following constants are used:
- A = 1 (preset)
- D = 0 (preset)
- C = 1 (factory-configured, cannot be changed)

The mathematical correlation is derived from the following equation:

$$C_{i.B.} = A \cdot \left[\frac{cal \cdot F}{F - C \cdot F_D} \right] + D$$

$C_{i.B.}$	dust concentration [mg/m³]
cal	raw signal [V] (correlate with scattered light)
F	volume flow in measuring chamber [m³/h]
F_D	volume flow of dilution air [m³/h]
A, D	constants from dust calibration
C	constant from device calibration

The correlation of the measuring gas is presented in the following way:

$$F_M \approx F - F_D$$

F_M	sucked measuring gas volume flow [m³/h]
F	volume flow in measuring chamber [m³/h]
F_D	volume flow of dilution air [m³/h]

Provided that A = 1 and D = 0, a raw signal $C_{i.B.}$' is displayed which is used for calibration.

$$C_{i.B.} = A \cdot C_{i.B.}' + D$$

$C_{i.B.}$	dust concentration [mg/m³]
$C_{i.B.}$'	raw signal $C_{i.B.}$
A, D	constants from dust calibration

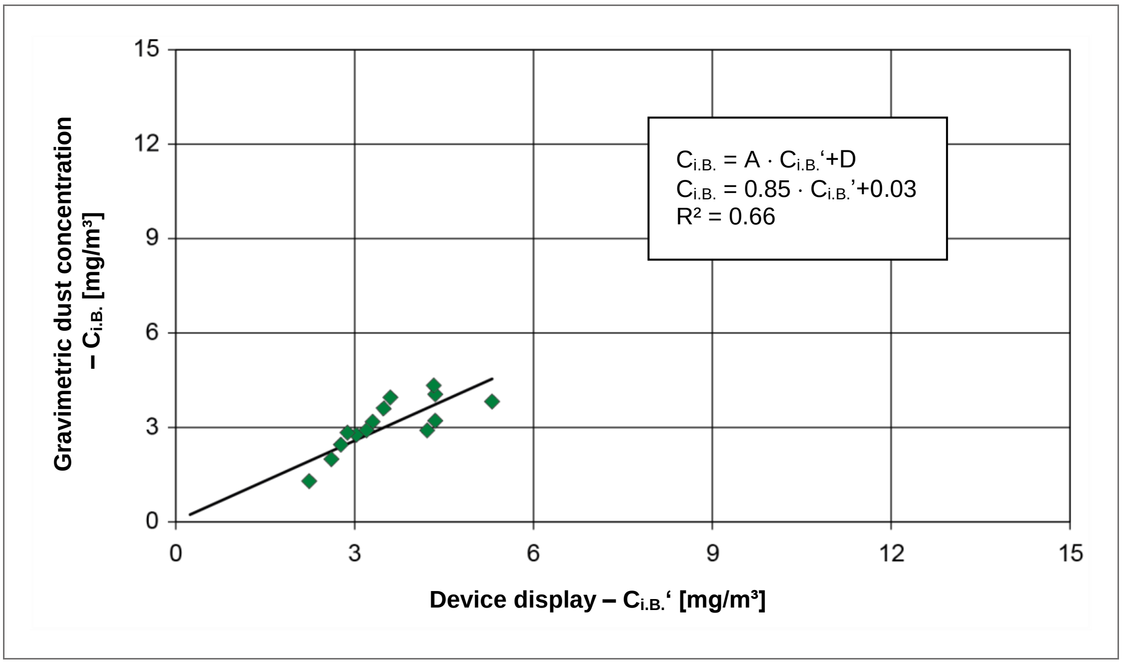

Fig. 21: Gravimetric calibration, example

4. Determine the calibrating constants A and D from the correlation (see also Tab. 2 “Value table for calibration”, page 45).
5. Input the calibrating constants A and D via the respective menu (see section 9.6.2, page 76).
   › The values which are output via the measuring value display as well as the signals of the analogue outputs are assimilated to the actual dust content in the system.

Meas. signal PFM 06 ED at A = 1 and D = 0 [mA]	Device display PFM 06 ED at A = 1 and D = 0 [mg/m³]	Gravimetric dust concentration $C_{i.B.}$ [mg/m³]
4.00 *	0.00 *	0.00 *
5.81	2.24	1.29
7.52	4.35	4.05
7.41	4.22	2.90
5.81	2.24	1.29
6.24	2.77	2.45
6.45	3.03	2.76
6.11	2.61	1.99
6.82	3.49	3.60
6.33	2.88	2.83
8.29	5.31	3.82
6.91	3.60	3.95
7.52	4.35	3.21
6.67	3.30	3.18
7.50	4.33	4.33
7.41	3.20	2.91
* zero point hypothesis		

Tab. 2: Value table for calibration

9 Operating

CAUTION

Incorrect operating can cause false measuring results, interferences in measuring process or material damage.

Basic requirement for safe operating is exact knowledge of the functioning of the PFM 06 ED as well as knowledge of measuring method and system-dependent details.

For operation with the software the instructions and calibration directions must be obeyed exactly. The respective setting possibilities as well as their effects and parameter limits have to be observed.

Operating must solely be executed by qualified personnel (see section 2.2 “Requirements for personnel”, page 11).

Operating at the optical sensor is not allowed.

CAUTION

Ball valve position!

Wrong position of the 3/2-way ball valve can cause measuring faults.

For continuous dust concentration measurement the 3/2-way ball valve must be set as follows:

- handwheel in automatic operation (position “(A) AUTO”, see Fig. 9, page 22)

  → The 3/2-way ball valve is automatically set to measuring operation.

9.1 Display and operating elements

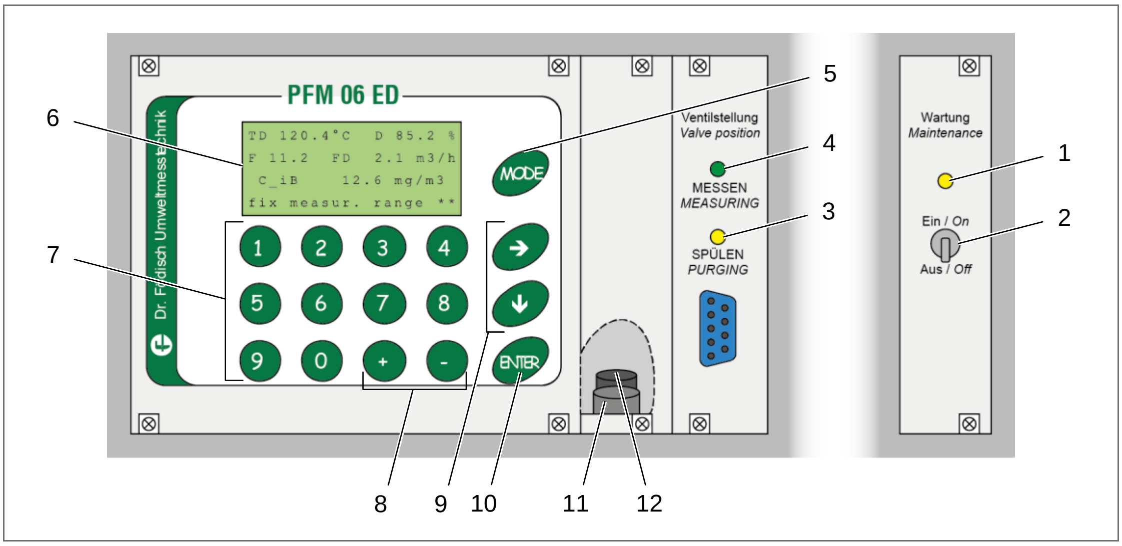

Fig. 22: Display and operating elements

1 LED “Maintenance”
2 Maintenance switch On/Off
3 LED “PURGING”
4 LED “MEASURING”
5 Key “MODE”
6 Text display
7 Numerical keys
8 Keys + and –
9 Keys → and ↓
10 Key “ENTER”
11 Set push-button (behind cover)
12 Reset push-button (behind cover)

The operating device at the profile rack of the PFM 06 ED contains a 4-line text display as well as all keys necessary for operating. Beside the navigation and acknowledging keys a numerical keyboard serves the reviewing and adjusting of the parameters. The assignment of the keys is supported in the menu via displaying the key name which is designated in angle brackets.

Text display

Via the text display (6, Fig. 22, page 46) the measuring values as well as the menu navigation are presented. The measuring value display shows the current measuring values – selectively on standard or service mode. The last line is defined by the status line. There additional messages as well as states of probe and dilution air heating are displayed.

Indicating lights

LED “MEASURING”

The LED “MEASURING” (4, Fig. 22) lights when the device is set to measuring operation via the 3/2-way ball valve (see section 4.1.1, page 21).

LED “PURGING”

The LED “PURGING” (3) lights when the device is set to purging operation via the 3/2-way ball valve (see section 4.1.1, page 21).

LED “Maintenance”

The LED “Maintenance” (1) lights when the device is set to maintenance mode via the maintenance switch or at active parameter input respectively calibration. In the status line of the text display (6) “maintenance state” is displayed.

Maintenance switch

By exerting the maintenance switch (“On”/“Off”; 2, Fig. 22) the device is set to maintenance mode. At activated maintenance mode the belonging LED (1) lights up. In the status line of the text display “maintenance state” is displayed.

Key “MODE”

By pressing the key “MODE” (5, Fig. 22) there is a switchover between different display possibilities.

Keys → and ↓

By pressing the keys → and ↓ (9, Fig. 22) the navigation within the menu levels is made. The key → enables moving forward in the menu as well as between the numeral positions at parameter input. Via the key ↓ it is possible to move back from a subordinate menu to the superordinate menu level and to move forward there. Furthermore, the key ↓ serves the decreasing of numerical values at parameter inputs.

Key “ENTER”

By pressing the key “ENTER” (10, Fig. 22) the selection to subordinate menu levels or to parameter input is done. Settings at parameter input or function selection are acknowledged via the key “ENTER”. Thereby there is an automatic change back to the superordinate menu item.

Keys + and –

The keys + and – (8, Fig. 22, page 46) have different functions. At activated parameter input they serve the change of algebraic sign of the numerical values.

At selected measuring value display in standard mode, service mode and at alternative displays the 3/2-way ball valve is set by pressing the keys:

- Key +: The 3/2-way ball valve is set to purging operation.
- Key –: The 3/2-way ball valve is set to measuring operation.

NOTICE
3/2-way ball valve: see section 4.1.1, page 21.

Numerical keys

The numerical keys (7, Fig. 22) have different functions. Basically, they serve the input of numerical values at parameter setting. Within the menu navigation there are respective selection possibilities which are assigned to the single numerical keys, so that the access to the desired subordinate menu level is enabled. In exceptional cases (e.g. selection of mode output, switch-on/-off of functions) the respective selection is activated directly.

Set push-button

The set push-button (11, Fig. 22) is placed covered at the operating device. It is located behind the cover rightwards next to the operating keys (fore push-button). For pressing the push-button the cover must be untightened and removed via the two screws.

By pressing the set push-button the access for changing parameters and for calibrating is enabled. Thereby the device is set to “maintenance state”.

NOTICE
- Change parameters: see section 9.5, page 57
- Calibrating: see section 9.6, page 73

Reset push-button

The reset push-button (12, Fig. 22) is placed covered at the operating device. It is located behind the cover rightwards next to the operating keys (behind the set push-button). For pressing the push-button the cover must be untightened and removed via the two screws.

By pressing the reset push-button the device is switched off and switched on again. Thereby failure/error messages are reset and all evaluations are restarted.

NOTICE
Alternative to pressing the reset push-button an interruption of the power supply can be carried out by disconnecting and re-connecting the external pre-fuse.

9.2 Menu

9.2.1 Menu navigation

The operating of menu navigation can vary depending on the current menu item. For a correct navigation hence the designations of key explanation have to be observed. The respective key is named in angle brackets after the respective function (see Fig. 23 and Fig. 24).

In the main menu paths “basic settings”, “change parameters” and “calibrating” the forward-switching is carried out via the key → and the selection of the current menu item (change into the subordinate menu level) is done by the key “ENTER”.

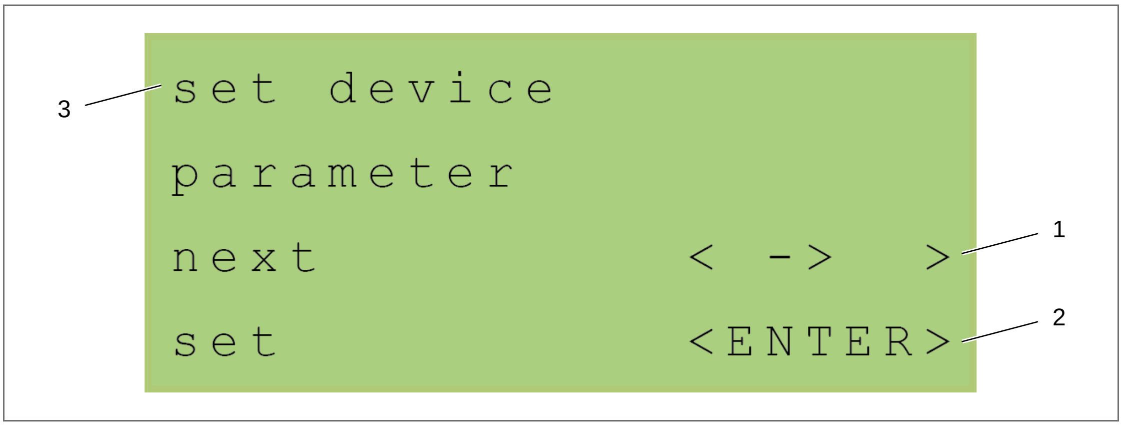

Fig. 23: Navigation, main menu path

1 Forward-switching to next menu item via key →
2 Selection of the current menu item via key “ENTER”
3 Name of the current menu resp. of the adjustable parameter

Within the subordinate menus the selection at several available sub-menus resp. at parameter selection is carried out via assigned numerical keys.

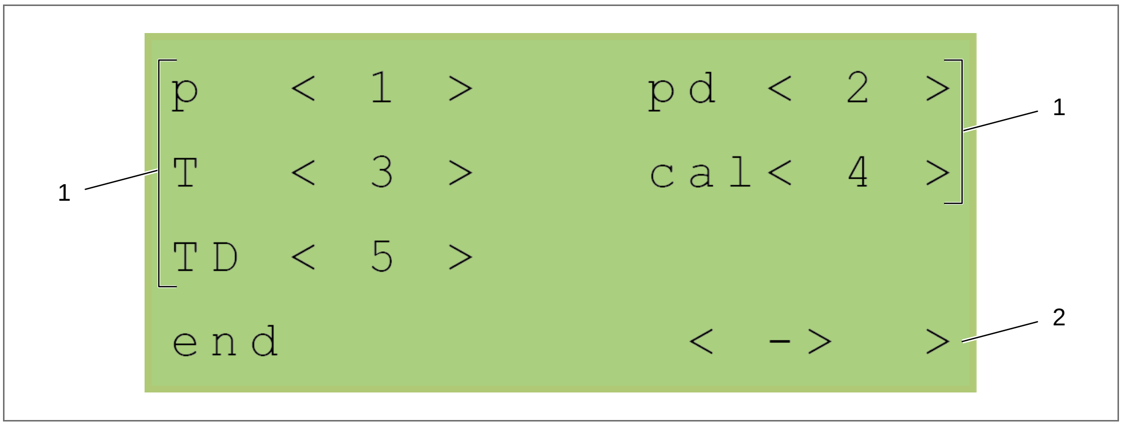

Fig. 24: Navigation, selection menu

1 Selection of the subordinate menu via numerical keys
2 Forward-switching to superordinate menu item via key →

9.2.2 Menu structure

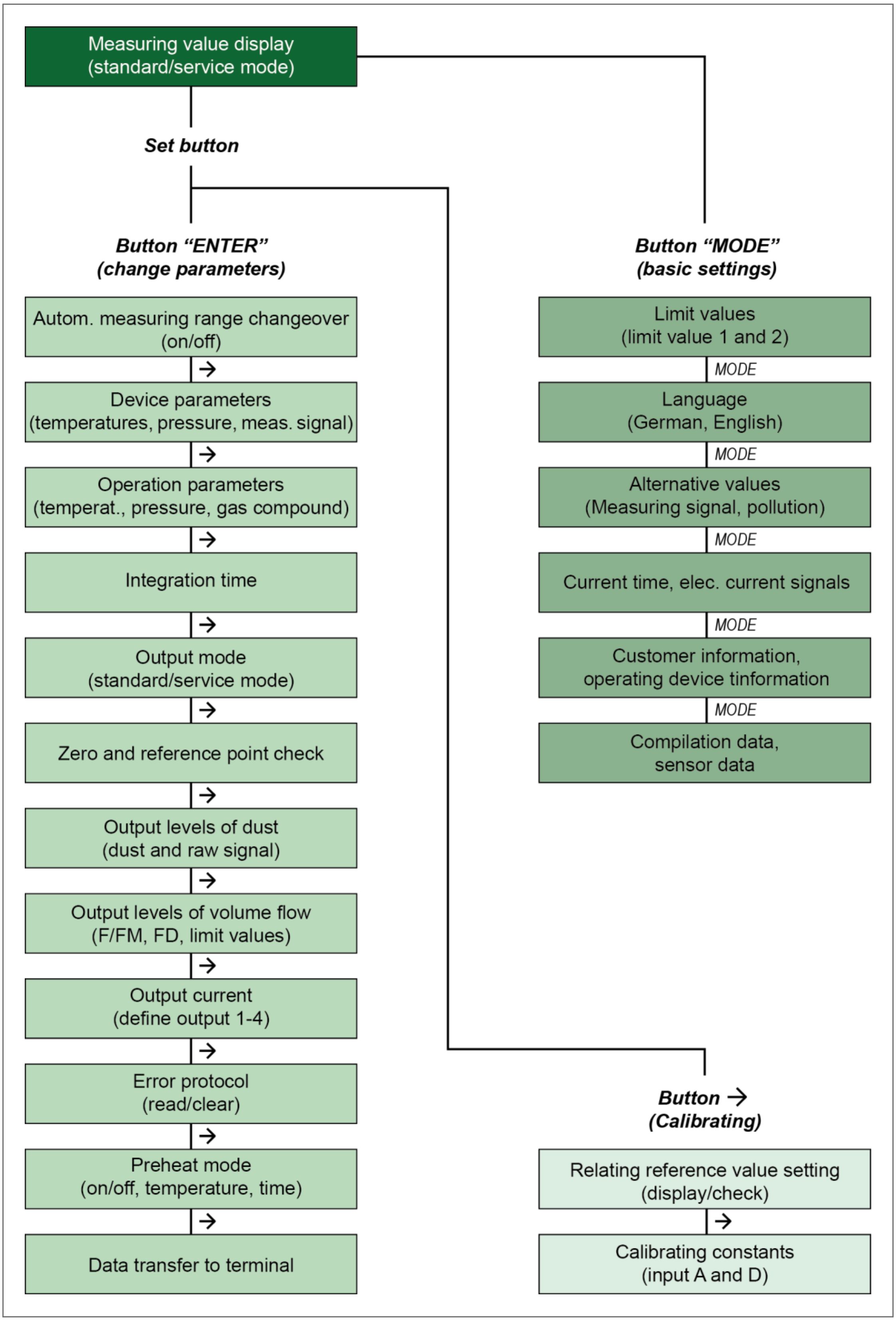

Fig. 25: Menu structure (summarised presentation)

9.2.3 Value input

At the input of parameters the respective numerical value is displayed in the menu in mathematical notation. The current cursor position flashes in short intervals.

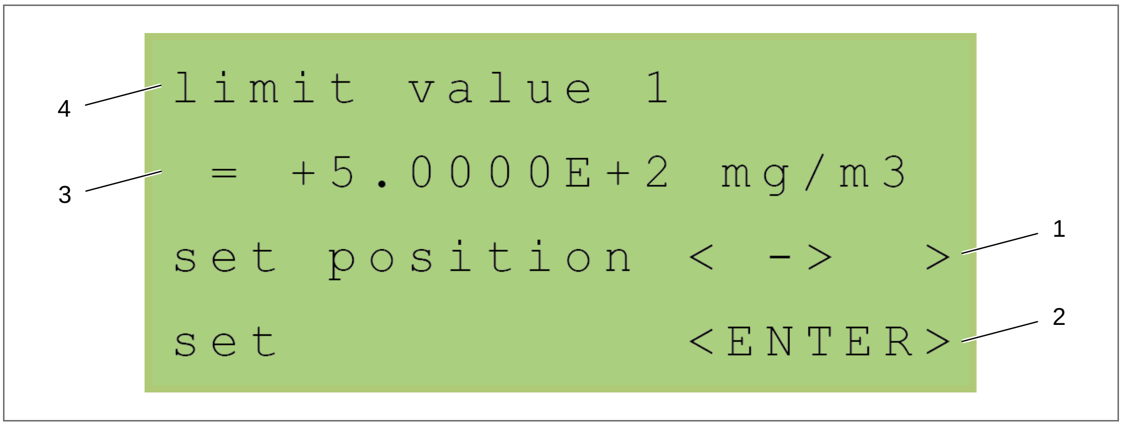

Fig. 26: Value input (example: limit value 1)

1 Forward-switching to next position of numerical value via key →
2 Acknowledgement of the adjusted value via key “ENTER”
3 Input parameter
4 Name of the adjustable parameter

NOTICE
Observe the kind of input of the parameter. It is ruled by the exponent (E+/- ...). Examples for parameter input are shown in the following table.

Input	Translation	Actual value
+3.0000E+2	$\triangleq 3{,}0000 \cdot 10^{2}$	300
–4.0000E+0	$\triangleq -4{,}0000 \cdot 10^{0}$	–4
+5.0000E–1	$\triangleq 5{,}0000 \cdot 10^{-1}$	0,5

Tab. 3: Examples for parameter input

Procedure

1. Press the key → to select the respective cursor position of the numerical value.
2. Input he desired parameter:
   a) Press the key + or – to determine the algebraic sign and the exponent.
   b) Press the respective numerical keys to determine the numerical value.
3. Set the input value by pressing the key “ENTER”.
   › The input value is adopted.
   › The display changes back to the superordinate menu level.

9.3 Mode output of the measuring value display

Concerning the measuring value display in the main view there are two possibilities which are selectable for mode output – standard mode and service mode. Thereby in each case the respective measuring values are output. In both cases the last line is defined by the status line. In it additional information or upcoming failure/error messages are displayed.

NOTICE
The selection of mode output is carried out via the menu of parameter changes (see section 9.5.5, page 63).

9.3.1 Standard mode

In standard mode the corrected value of dust concentration at the measuring point, the current set measuring range, the temperature and the volume flow are displayed.

Fig. 27: Standard mode

1 Dilution of measuring gas
2 Volume flow of dilution air
3 Dust concentration of the exhaust
4 Probe heating switched on
5 Dilution air heating switched on
6 Status line with message
7 Volume flow in the measuring chamber
8 Temperature in the measuring chamber resp. of the dilution air (changeover of display)

NOTICE
The measuring values for temperature and volume flow refer to operating data in the measuring device.

9.3.2 Service mode

In service mode the uncorrected measuring value of the dust sensor (raw signal cal) as well as the temperatures and differential pressures of the analogue outputs as 4...20 mA are output.

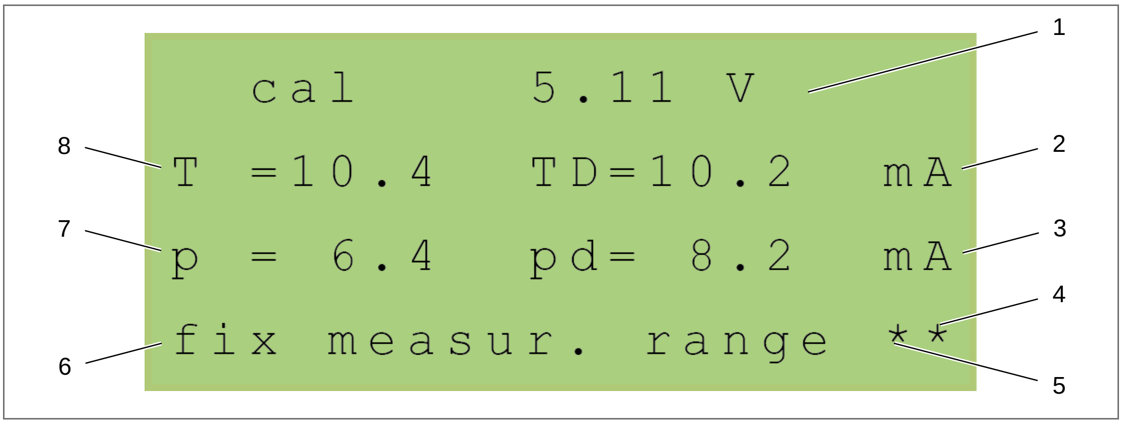

Fig. 28: Service mode

1 Measuring value of the dust sensor
2 Temperature of dilution air
3 Differential pressure of dilution air (through measuring orifice)
4 Probe heating switched on
5 Dilution air heating switched on
6 Status line with message
7 Differential pressure through measuring chamber
8 Temperature in the measuring chamber

NOTICE
The automatic measuring range switchover for the dust concentration $C_{i.B.}$ is not active.

9.4 Basic settings

9.4.1 Input limit values

```
input limit values
limit value 1   < 1 >
limit value 2   < 2 >
end           < -> >
```

Fig. 29: Limit value selection

The PFM 06 ED is able to monitor two independent limit values. Thereby in standard mode a maintenance request message is output for signalling of the reached limit value and in the case of an exceeded limit value an error message is output via the text display.

1. Press the key “MODE”.
   › The menu “input limit values” is opened.
2. Select the desired limit value. For this, press the respective numerical key (limit value 1 = key 1; limit value 2 = key 2).
   › The menu for parameter input is opened.
3. Input the desired parameter and acknowledge the input with “ENTER” (value input: see section 9.2.3, page 51).
4. Press the key → to return back to the measuring value display after setting of the input value.

9.4.2 Select language

```
display language is
english
change        < -> >
set          <ENTER>
```

Fig. 30: Language selection

For presentation of the menu contents the displayed language can be changed.

Dependent on the device configuration the languages German and English can be displayed.

1. Press the key “MODE” 2 times.
   › The menu “display language is english” is opened.
2. Select the desired language by pressing the key →.
   › All menus are immediately changed to the selected language.
   › After a few seconds the display returns back to the measuring value display.

9.4.3 Special displays

Display alternative values

```
TD 120.4°C  D 85.2 %
F 11.2   FD  2.1 m3/h
cal    1.20V pl    4 %
fix measur. range **
```

Alternatively to the general measuring value display, further measuring values can be presented (e.g. sucked measuring gas volume, measuring signal, pollution).

Fig. 31: Alternative values

1. Press the key “MODE” 3 times.
   › An alternative measuring value display is opened. This basically corresponds to the measuring value display in standard mode (see Fig. 27, page 52). Instead of the dust concentration of the exhaust (C_iB) the raw signal (cal) and the pollution (pl) are displayed in the alternative display.
2. To return back to the general measuring value display, press the key → or ↓.

Display time and current signals

```
time:  10:20  02.12.15
I1=  5.20  I3=  4.00 mA
I2=  5.72  I4=14.31 mA
fix measur. range **
```

Alternatively to the general measuring value display, the date and time as well as momentary current signals of the outputs can be presented.

Fig. 32: Time and current signals

1. Press the key “MODE” 4 times.
   › A menu with information of momentary time and the current signals of the outputs is opened. The last line of the menu is defined by the status line.

NOTICE
The analogue outputs are defined as follows:
I1 = raw signal cal
I2 = dust concentration c_iB (in service mode: pollution (pl))
I3 = free-allocable
I4 = free-allocable

2. To return back to the general measuring value display, press the key → or ↓.

Display customer and operating device information

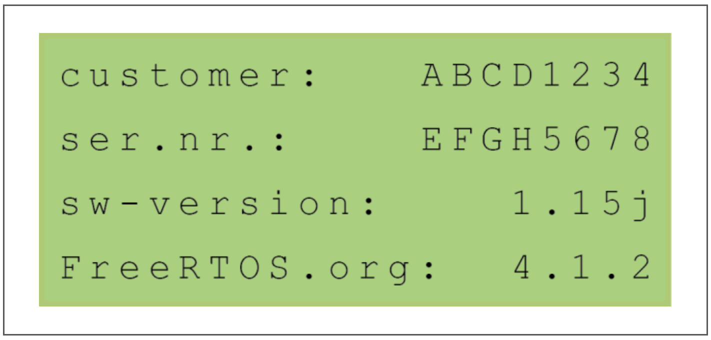

Fig. 33: Customer and operating device information

Alternatively to the general measuring value display, details of the customer as well as device and software information of the operating device can be presented.

1. Press the key “MODE” 5 times.
   › A menu with customer and operating device information is opened.
2. To return back to the general measuring value display, press the key → or ↓.

Display compilation and sensor data

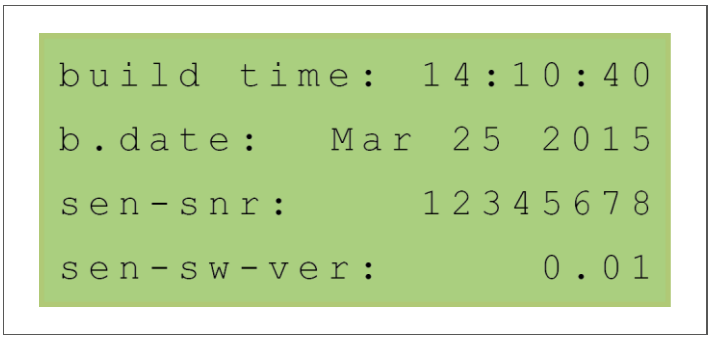

Fig. 34: Compilation and sensor data

Alternatively to the general measuring value display, details of compilation date of the operating device as well as the software version and serial number of the optical sensor can be presented.

1. Press the key “MODE” 6 times.
   › A menu with compilation and sensor data is opened.
2. To return back to the general measuring value display, press the key → or ↓.

9.5 Change parameters

The menu path for changing of parameters is activated via the set push-button (11, Fig. 22, page 46). For this, the cover above at the operating device must be screwed off. During the parameter input the device is automatically set to the maintenance mode. Thereby the normal operation is not interrupted.

9.5.1 Switch on/off the automatic measuring range switchover

```
set auto range


next            < -> >
set            <ENTER>
```

Fig. 35: Automatic measuring range switchover

Via the menu the automatic measuring range switchover for $C_{i.B.}$ can be switched on or off.

> **NOTICE**
> The setting of the measuring ranges for the dust and raw signal is described in section 9.5.7 at page 65.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key “ENTER” to open the menu “set auto range”.
   › The menu for function selection of the automatic measuring range switchover is opened.

```
off *  =          < 0 >
on     =          < 1 >


next            < -> >
```

Fig. 36: Automatic measuring range switchover
– function selection

4. Select the desired function. For this, press the respective numerical key (off = key 0; on = key 1).
   › The automatic measuring range switchover is switched on/off.
   › The active function selection is designated by the symbol *.

NOTICE
- When the measuring range switchover is switched on, the following applies: measuring range 1 = range of value $C_{i.B.}$; measuring range 2 = 3x range of value $C_{i.B.}$
- When the measuring range switchover is switched off, the following applies: measuring range = range of value $C_{i.B.}$

5. To return back to the general measuring value display, press key → repeatedly.

9.5.2 Set device parameters

```
set device
parameter
next          < ->  >
set          <ENTER>
```

Fig. 37: Device parameters

Via the menu device-internal parameters, e.g. for pressure and temperature, can be adjusted.

CAUTION
Changing of the device parameters can cause measuring faults.
The device parameters are factory-configured and are used for evaluation.
Changes must solely be executed by qualified specialised personnel.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key →, until the menu item “set device parameter” is displayed.
4. Press the key “ENTER” to open the menu “set device parameter”.
   › The menu for parameter selection is opened.

```
p  < 1 >      pd < 2 >
T  < 3 >      cal< 4 >
TD < 5 >
end           < ->  >
```

Fig. 38: Device parameters – parameter selection

5. Select the desired parameter. For this, press the respective numerical keys corresponding to the relative menu structure:

Superordinate parameter selection	Key	Subordinate parameter selection	Key
p (absolute pressure in the measuring chamber)	1	**range p** (measuring range of Δp transmitter in the measuring chamber)	1
		factor K (construction factor of Δp measure-ment in the measuring chamber)	2
pd (absolute pressure of dilution air)	2	**range pd** (measuring range of Δp transmitter of the dilution air)	1
		factor KD (construction factor of Δp measure-ment of the dilution air relating to the probe)	2
T (temperature in the measuring chamber)	3	**range T** (measuring range of temperature measurement in the measuring chamber)	1
cal (raw signal)	4	**K_C** (calibrating factor)	1
TD (temperature of dilution air)	5	**range TD** (measuring range of temperature measurement of the dilution air)	1

6. Input the desired parameter and acknowledge the input with “ENTER” (value input: see section 9.2.3, page 51).
7. Press the key → to return back to the superordinate menu.
8. To return back to the general measuring value display, press key → repeatedly.

9.5.3 Set operation parameters

```
set operation
parameter
next          < ->  >
set           <ENTER>
```

Fig. 39: Operation parameters

Via the menu measuring location-specific parameters for temperature, pressure and gas compound in the exhaust duct can be adjusted.

CAUTION

Changing of the operation parameters can cause measuring faults.

The operation parameters are measuring location-specific parameters and in the line with commissioning they must be assimilated to the system preconditions at location.

After changing of operation parameters the dust signal $C_{i.B.}$ must be re-calibrated (see section 8.3 “Dust calibration”, page 42).

Changes must solely be executed by qualified specialised personnel.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “set operation parameter” is displayed.
4. Press the key “ENTER” to open the menu “set operation parameter”.
   › The menu for parameter selection is opened.

```
T   < 1 >     p   < 2 >
Vol< 3 >     R   < 4 >

end           < ->  >
```

Fig. 40: Operation parameters – parameter selection

5. Select the desired parameter. For this, press the respective numerical keys corresponding to the relative menu structure:

Superordinate parameter selection	Key	Subordinate parameter selection	Key
T (temperature)	1	**TM** *[device-internal operand]* (average measuring gas temperature at the measuring point in the exhaust duct)	1
p (pressure)	2	**p** (absolute pressure in the measuring chamber)	1
		pd (absolute pressure of dilution air)	2
		pm (static pressure at the measuring point in the exhaust duct)	3
Vol (gas compound of the measuring gas) *[The nitrogen content in the exhaust is evaluated.]*	3	**O2** (volumetric content of oxygen)	1
		CO2 (volumetric content of carbon dioxide)	2
		H2O (volumetric content of water vapour)	3
R (constant)	4	**RD** (gas constant of dilution air)	1

6. Input the desired parameter and acknowledge the input with “ENTER”
(value input: see section 9.2.3, page 51).

Operation parameter “Vol” – examples for gas compounds		
Component	**Dry air**	**Combustion exhaust gas**
O_2	21 vol. %	11 vol. %
CO_2	0 vol. %	20 vol. %
H_2O	0 vol. %	14 vol. %
N_2	79 vol. %	55 vol. %

7. Press the key → to return back to the superordinate menu.
8. To return back to the general measuring value display, press key → repeatedly.

9.5.4 Set integration time

```
set integration
time
next            < ->  >
set             <ENTER>
```

Fig. 41: Integration time

Via the menu a smoothing of the measuring values is adjustable.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key →, until the menu item “set integration time” is displayed.
4. Press the key “ENTER” to open the menu “set integration time”.
   › The current value of integration time is displayed.

```
time =  1.0 s

increment       < ->  >
set             <ENTER>
```

Fig. 42: Integration time – time adjustment

5. Press the key → resp. key ↓ to increase respectively to decrease the value of integration time.

NOTICE
The integration time can only be defined as integer. The displayed decimal has no relevance.

6. Press the key “ENTER” to adopt the input value.
7. To return back to the general measuring value display, press key → repeatedly.

9.5.5 Select output mode

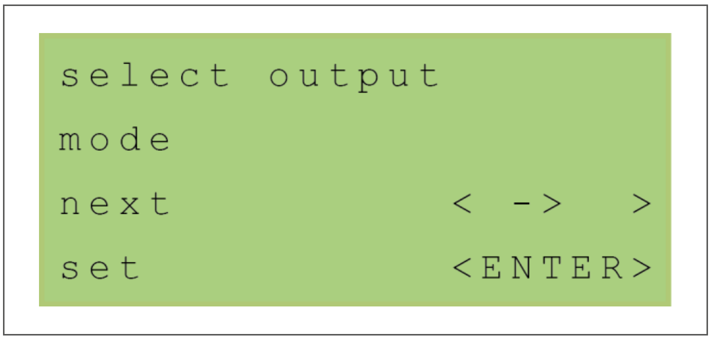

Fig. 43: Output mode

Via the menu a switchover between the output possibilities standard mode and service mode can be carried out.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “select output mode” is displayed.
4. Press the key “ENTER” to open the menu “select output mode”.
   › The menu for selection of the output mode is opened.

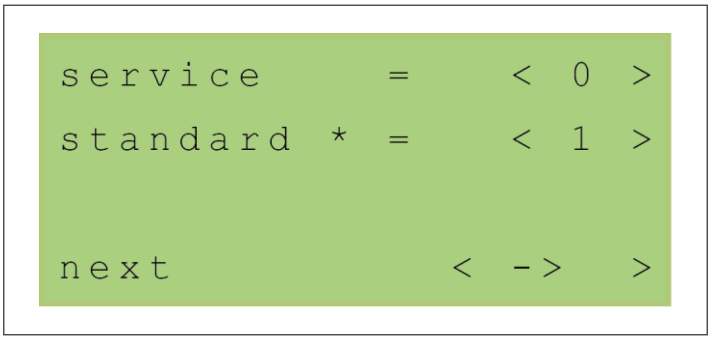

Fig. 44: Output mode – mode selection

5. Select the desired mode. For this, press the respective numerical key (service = key 0; standard = key 1).
   › The respective mode is adopted.
   › The active mode is designated by the symbol *.

NOTICE
If the service mode is selected, the device is automatically set to the maintenance mode.

6. To return back to the general measuring value display, press key → repeatedly.

9.5.6 Check zero and reference point

```
check zero and
reference point
next          < ->  >
check        <ENTER>
```

Via the menu a test of the zero and reference point of the optical sensor can be carried out (no storage of the check values). Thereby the difference between the current and the calibrated values in mV is output.

Fig. 45: Zero and reference point test

NOTICE
If the device is already in purging operation, the zero offset correction value can be determined new without averaging by manual starting of the zero and reference point control (see section 3.3, page 18).

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “check zero and reference point” is displayed.
4. Press the key “ENTER” to start the testing process.
   › Zero and reference point are checked. The start process is signalised by the display “start ...” and takes approx. 1 min.
   › For respectively 3 min the relating reference value for zero point and reference point for the dust-free scattered light value of the optical sensor is displayed. In the meantime no operating is possible.

```
check reference              check reference

zero point    000.0 %        ref. value    000.0 %
```

Fig. 46: Zero and reference point test – relating reference value for zero point and reference point

NOTICE
The displayed value signalises the device state as follows:
- value less than 4% ≙ normal operation
- value between 4% and 6% ≙ maintenance request
- value greater than 6% ≙ error

5. To return back to the general measuring value display, press key → repeatedly.

9.5.7 Select output ranges for dust and raw signal

```
select output
level of dust
next          < ->  >
set          <ENTER>
```

Via the menu the measuring ranges for dust and raw signal can be defined.

Fig. 47: Output ranges of dust

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “select output level of dust” is displayed.
4. Press the key “ENTER” to open the menu “select output level of dust”.
   › The menu for parameter selection is opened.

```
select range
c_iB            < 1 >
voltage cal     < 2 >
next          < ->  >
```

Fig. 48: Output ranges of dust – parameter selection

5. Select the desired parameter. For this, press the respective numerical key:
   - c_iB (measuring range for dust) = key 1
   - voltage cal (measuring range for raw signal) = key 2

   › The menu for parameter input is opened.
6. Input the desired parameter and acknowledge the input with “ENTER” (value input: see section 9.2.3, page 51).
7. To return back to the general measuring value display, press key → repeatedly.

9.5.8 Select output ranges for volume flow

```
select output
level of flow
next              <  ->  >
set              <ENTER>
```

Fig. 49: Output ranges for volume flow

Via the menu the measuring ranges for the volume flows of measuring gas and dilution air can be defined.

1. Press the set push-button.
2. Press the key "ENTER" to open the main menu path "change parameters".
3. Press the key → repeatedly, until the menu item "select output level of flow" is displayed.
4. Press the key "ENTER" to open the menu "select output level of flow".
   › The menu for parameter selection is opened.

```
flow F/FM            <  1  >
flow FD              <  2  >

next              <  ->  >
```

Fig. 50: Output ranges for volume flow – parameter selection

5. Select the desired parameter. For this, press the respective numerical keys corresponding to the relative menu structure:

Superordinate parameter selection	Key	Subordinate parameter selection	Key
flow F/FM (volume flow in the measuring chamber)	1	**flow F/FM** (measuring range of volume flow in the measuring chamber)	1
		flow min. Fmin (limit value for volume flow monitoring of the measuring chamber)	2
		Fmin over ... (F / FM) *[Function selection]* (limit value test of volume flow monitoring for volume flow F or FM)	*[3]*
flow FD (volume flow of the dilution air)	2	**flow FD** (measuring range of volume flow of dilution air)	1
		flow min. FDmin (limit value for volume flow monitoring of dilution air)	2

6. Input the desired parameter and acknowledge the input with “ENTER”
(value input: see section 9.2.3, page 51).

NOTICE
At undershooting of the set limit values for Fmin and FDmin a maintenance request message is output. If the value is less than 80% of the set limit value, an error message is generated.
The volume flow F can be increased by back-purging of the probe.

7. Press the key → to return back to the superordinate menu.
8. To return back to the general measuring value display, press key → repeatedly.

9.5.9 Check current outputs

```
check output
current
next          < ->  >
set           <ENTER>
```

Fig. 51: Current outputs

Via the menu the current signals for the analogue outputs can be defined by predetermined values (e.g. for commissioning, error search and function test).

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “check output current” is displayed.
4. Press the key “ENTER” to open the menu “check output current”.
   › The menu for input of the parameter for output 1 is opened.

```
output 1
 = +0.0000E+0 mA
set position  < ->  >
set           <ENTER>
```

Fig. 52: Current outputs – output 1

5. Input the desired parameter and acknowledge the input with “ENTER” (value input: see section 9.2.3, page 51).
   › The menu for input of the parameter for the next output is opened.
6. Repeat step 5 for every further output.
7. Acknowledge the end of the check of current outputs. For this, press the key “ENTER”.

NOTICE
The analogue outputs are set with the input values. The values can be measured at the terminal strip with the aid of a multimeter.

8. To return back to the general measuring value display, press key → repeatedly.

9.5.10 Read/clear error protocol

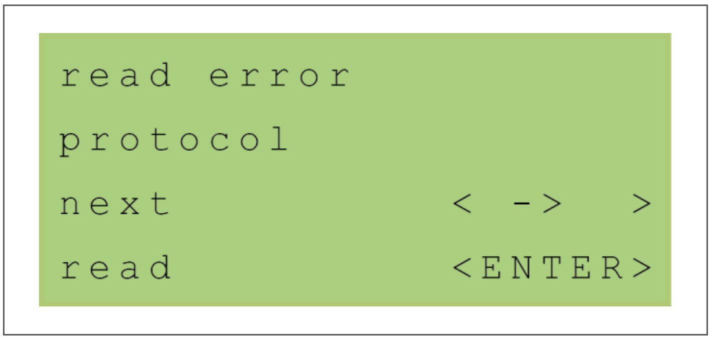

Fig. 53: Error protocol

Via the menu the last 100 error and event messages can be displayed in order of their occurring. Every event possesses a consecutive number, an event code as well as a designation of date and time.

> **NOTICE**
> Optionally, in the menu the clearing of the complete error protocol is possible.

1. Press the set push-button.
2. Press the key "ENTER" to open the main menu path "change parameters".
3. Press the key → repeatedly, until the menu item "read error protocol" is displayed.
4. Press the key "ENTER" to open the error protocol.
   › The freshest entry of the error protocol is displayed.

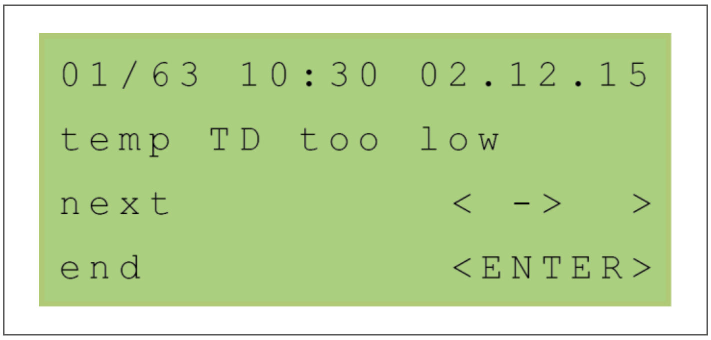

Fig. 54: Error protocol – entry

5. Press the key → resp. key ↓ to change forwards/backwards between all entries.
6. Press the key "ENTER" to leave the error protocol.
   › A query for clearing the error protocol is displayed.

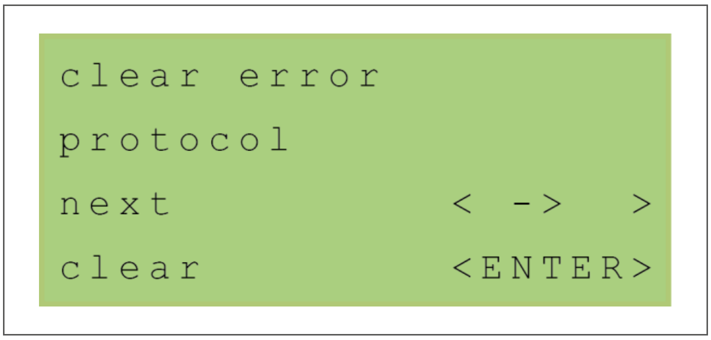

Fig. 55: Error protocol – clearing

7. If necessary, clear the error protocol. For this, press the key “ENTER”.
   › All entries of the error protocol are deleted. The process is graphically displayed by a progress bar.
8. To return back to the general measuring value display, press key → repeatedly.

9.5.11 Preheating mode

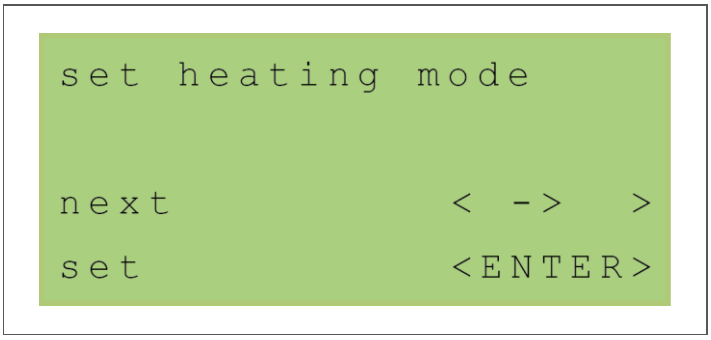

Fig. 56: Preheating mode

Via the menu all setting for preheating of the probe can be made. If preheating mode is switched on, the preheating starts at switch-on of the device. In the status line of the text display the preheating time is displayed.

NOTICE
During preheating no measuring gas is sucked. After switch-on at the beginning and at the end of preheating the zero offset correction value is determined.
The preheating can be aborted prematurely by pressing the key –.

CAUTION
Danger through abort of preheating!
Premature abort of preheating can cause condensate formation, pollution of the measuring chamber or measuring faults.
Determine the zero offset correction value of the scattered light manually new (see section 3.3 “Zero and reference point and zero offset correction”, page 18) or carry out a reset with re-preheating.
In the case of a correct zero offset correction value a raw signal cal in the range of -0.3 V to 0.3 V is displayed during purging operation.

9.5.11.1 Switch on/off preheating mode

```
off              < 0 >
on  *            < 1 >
set values       < 2 >
next           < ->  >
```

Fig. 57: Preheating mode – function selection

Via the menu the preheating mode can be switched on and off.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “set heating mode” is displayed.
4. Press the key “ENTER” to open the menu “set heating mode”.
5. Select the desired function. For this, press the respective numerical key (off = key 0; on = key 1).
   › The preheating mode is switched on/off.
   › The active function selection is designated by the symbol *.
6. To return back to the general measuring value display, press key → repeatedly.

9.5.11.2 Set temperatures and time of preheating mode

```
T  temperature   < 1 >
TD temperature   < 2 >
preheat time     < 3 >
next           < ->  >
```

Fig. 58: Preheating mode – temperatures and time

Via the menu the temperatures and the duration of preheating can be defined.

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “set heating mode” is displayed.
4. Press the key “ENTER” to open the menu “set heating mode”.
5. Press the numerical key “2” to change the parameters of preheating mode.

6. Select the desired parameter. For this, press the respective numerical key:
   - T (nominal value for temperature in the measuring chamber) = key 1
   - TD (nominal value for temperature of dilution air) = key 2
   - preheat time (duration of preheating after switch-on) = key 3

   › The menu for input of the parameter is opened.
7. Input the desired parameter and acknowledge the input with “ENTER” (value input: see section 9.2.3, page 51).
8. To return back to the general measuring value display, press key → repeatedly.

9.5.12 Transfer data to terminal

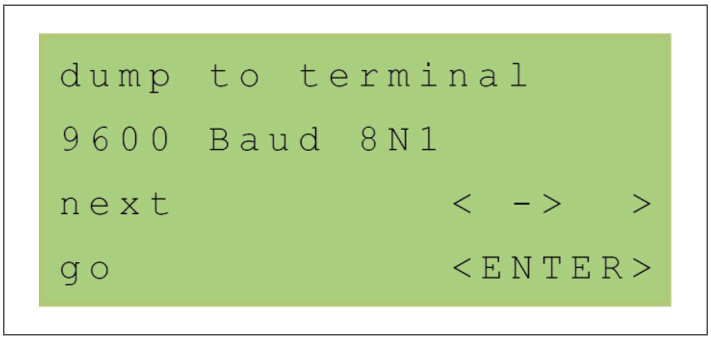

Fig. 59: Data transfer

Via the menu the parameters of the PFM 06 ED can be output via the RS232 interface to a terminal programme.

NOTICE
For data transfer the operating device must be connected with a PC via the connection of the RS232 interface (2, Fig. 14, page 26).

The following settings must be observed:
9600 baud, 8 bit, no parity, 1 stop bit

1. Press the set push-button.
2. Press the key “ENTER” to open the main menu path “change parameters”.
3. Press the key → repeatedly, until the menu item “dump to terminal” is displayed.
4. Press the key “ENTER” to start the sending process.
   › The data is transferred to the terminal. The process is graphically displayed by a progress bar.
   › The respective terminal programme can be set in a way that the received parameters are saved as a text file and used for documentation of the setting values.
5. To return back to the general measuring value display, press key → repeatedly.

9.6 Calibrating

The menu path for calibrating is activated via the set push-button (11, Fig. 22, page 46). For this, the cover above at the operating device must be screwed off. During the calibration the device is automatically set to the maintenance mode.

NOTICE
Observe the notices, explanations and instruction in chapter 8 “Calibration” from page 41.

9.6.1 Relating reference value of the sensor

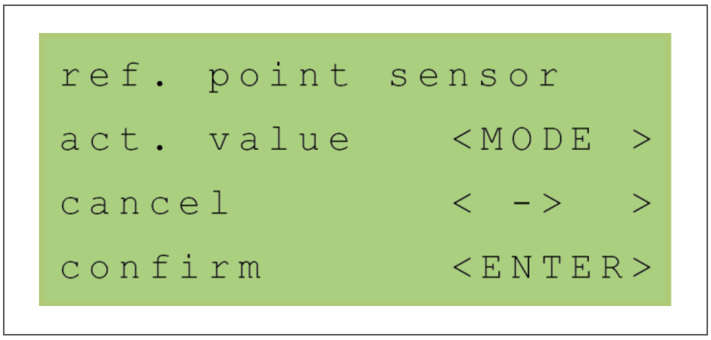

Fig. 60: Relating reference value of the sensor

Via the menu the current relating reference value of the optical sensor can be displayed respectively set.

9.6.1.1 Display current relating reference value

Via the menu (see Fig. 60, page 73) the current relating reference value of the optical sensor can be displayed.

NOTICE
The current relating reference value should be between 60% and 80%.

1. Press the set push-button.
2. Press the key → to open the main menu path “calibrating”.
3. Press the key → repeatedly, until the menu item “ref. point sensor” is displayed.
4. Press the key “MODE”.
   › The current relating reference value is displayed.

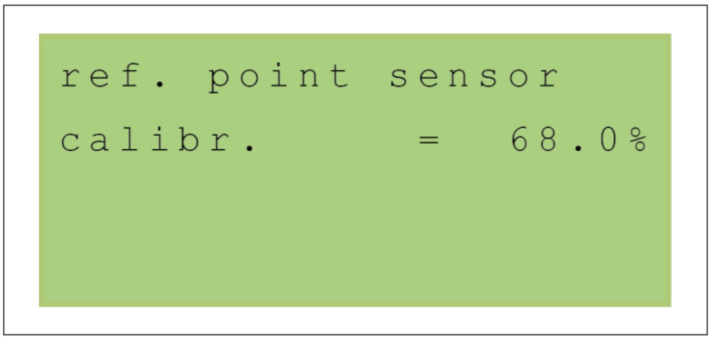

Fig. 61: Relating reference value of the sensor – current value

   › After a few seconds the display returns back to the superordinate menu level.
5. To return back to the general measuring value display, press the key →.

9.6.1.2 Set relating reference value

Via the menu (see Fig. 60, page 73) a new relating reference value can be set. Thereby the optical sensor carries out a zero and reference point check.

CAUTION
A new relating reference value may only be set at commissioning or after a change of the optical sensor.

1. Press the set push-button.
2. Press the key → to open the main menu path “calibrating”.
3. Press the key → to open the menu “ref. point sensor”.
4. Press the key “ENTER” to start the calibration.
   › The start process is signalised by the display “start ...” and takes 2 min. Thereby different values are displayed.

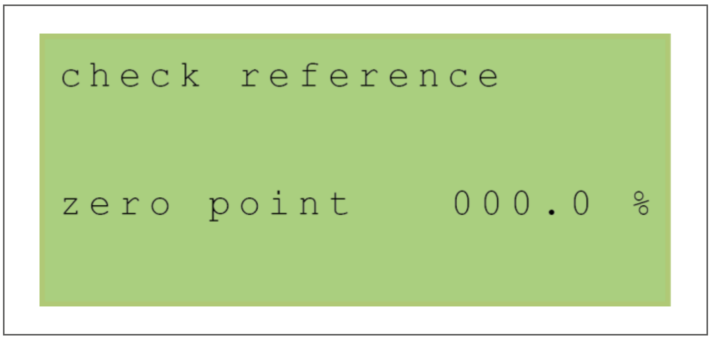

Fig. 62: Relating reference value of the sensor – display values

5. Wait until the test has finished. For this, check the menu display.
6. To display the new relating reference value, press the key “MODE”.
   › The current relating reference value is displayed. After a few seconds the display returns back to the superordinate menu level.

NOTICE
The new relating reference value should be between 60% and 80%.

7. To return back to the general measuring value display, press the key →.

9.6.2 Input calibrating constants A and D

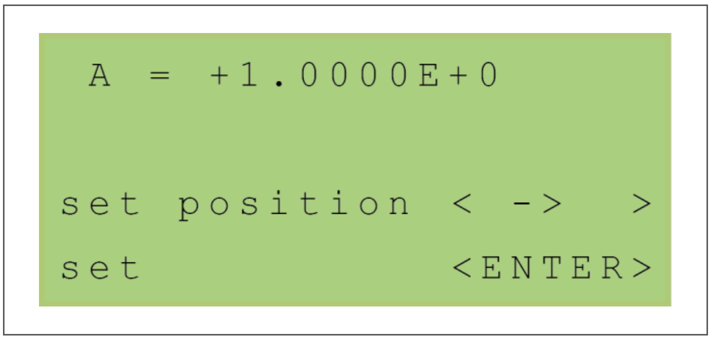

Fig. 63: Calibrating constants

Via the menu the determined calibrating constants for dust calibration (see section 8.3, page 42) can be input.

1. Press the set push-button.
2. Press the key → to open the main menu path “calibrating”.
3. For setting of the calibrating constants, press the key “ENTER”.
4. Select the desired calibrating constant. For this, press the respective numerical key (constant A = key 1; constant D = key 2).
   › The menu for parameter input is opened.
5. Input the desired parameter and acknowledge the input with “ENTER” (value input: see section 9.2.3, page 51).
   › The values which are output via the measuring value display as well as the signals of the analogue outputs are assimilated to the actual dust content in the system.
6. To return back to the general measuring value display, press key →.

10 Maintenance/Upkeep

NOTICE
The provision of guarantee items requires an execution of maintenance in due form.
Any work at the device must solely be executed by qualified personnel.
Executed maintenance/upkeep work must be documented according to the legal regulations.

The maintenance work has the following target:

- Preservation of measurement precision
- Warranty of operational safety
- Increment of service life

The following maintenance/upkeep work must be executed:

Component	Action	Maintenance interval
Measuring system, general	Visual inspection	1 month
Differential pressure measurement	Check and cleaning	1 month
Optical sensor	Manual zero and reference point control	1 month
	Plausibility test of volume flow in the measuring chamber	1 month
	Plausibility test of volume flow of dilution air	1 month
	Check and cleaning	1 month
	Plausibility test of measuring chamber temperature	3 months
	Plausibility test of dilution air temperature	3 months
Injector	Cleaning	3 months
Probe, incl. casing and all components	Cleaning	3 to 6 months
Probe ventilation	Check / cleaning resp. exchange of filters	3 to 6 months / if required
Blower for injector air and dilution air	Check of suck-in openings, Exchange of suction filters	3 to 6 months / if required

Tab. 4: Maintenance/upkeep work

NOTICE
The frequency of cleaning work which has to be carried out depends on the chosen measuring point resp. the measuring medium, particularly the dust concentration, and the environmental and climatic conditions.

NOTICE
Once a year a function test must be carried out by a responsible measuring institute (§§ 26/28 BImSchG). This comprises the following terms:

- visual inspection of the measuring system
- check of linearity with reference material
- manual zero and reference point control with control of long-time drift
- check of data transfer to the evaluation system (analogue and status signals)

10.1 Visual inspection of the measuring system

1. Check the measuring point (e.g. duct wall, welding flange, adapter) for visible damages.
2. Check the probe, operating device and blowers for visible damages.
3. Check all gas-leading hoses for kinks and cracks resp. break.
4. Check all cables for kinks and cracks resp. break.
5. Exchange possibly damaged components.

10.2 Manual zero and reference point control

1. Switch the device to maintenance mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “On”.
2. Set the device to purging operation. For this, press the key + at the operating device.
3. Check the zero and reference point via the menu (see section 9.5.6, page 64).
4. Document the values which are displayed in the menu.

CAUTION
If the result of the zero and reference point test exceeds the relating reference value of 4%, there is an error at the optical sensor.

5. To clear a possible error at the optical sensor, proceed as follows:
   a) Check the optical sensor for pollution. If necessary, clean it.
   b) Check the optical sensor for correct mounting.
   c) Increase the temperature of dilution air resp. of the measuring chamber.
   d) Check the relating reference value.

NOTICE
Once a year a control of long-time drift must be carried out by a responsible measuring institute (§§ 26/28 BImSchG).

10.3 Plausibility tests

CAUTION
Incorrect execution of plausibility tests can cause false measuring results.
The device stays in measuring mode (The LED “MEASURING” lights.). There must not be any error messages.
Also at the optical sensor in the probe there must not be any errors. The LED at the sensor head must not light.

1. Check the volume flow in the measuring chamber (F) via the measuring value display in standard mode (see Fig. 27, page 52) as well as the adjusted parameters of the output ranges (see section 9.5.8, page 66).
2. Check the volume flow of the dilution air (FD) via the measuring value display in standard mode (see Fig. 27, page 52) as well as the adjusted parameters of the output ranges (see section 9.5.8, page 66).
3. Check the temperature in the measuring chamber (T) via the measuring value display in standard mode (see Fig. 27, page 52) as well as the adjusted device parameters (see section 9.5.2, page 58).
4. Check the temperature of the dilution air (TD) via the measuring value display in standard mode (see Fig. 27, page 52) as well as the adjusted device parameters (see section 9.5.2, page 58).

10.4 Cleaning of the probe and all integrated components

1. Switch the device to maintenance mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “On”.
2. Open the casing of the probe. For this, release the four casing locks and take the casing cap off (see Fig. 19, page 35).

WARNING
Hot surface!
Several device parts can develop high temperatures.
Burn of skin can be caused!
For protection against possible injury protective gloves must be worn.

3. Open the probe insulation. For this, release the five locking clamps at the sides and take the insulation off.
4. Detach the connecting tube between ball valve and measuring chamber (4, Fig. 7, page 21). For this, release the screwing with the aid of a screw-wrench.
5. Check resp. clean the probe resp. the appropriate component according to the actions which are described as follows and after this continue with step 6 on page 81.

Check and cleaning of the differential pressure measurement

I. Clean the line of differential pressure measurement (connecting tubes between ball valve and measuring chamber – 4, Fig. 7, page 21).

II. Release the hose clamps at the hoses for dilution air, injector air and exhaust (see Fig. 20, page 37) and pull the hoses off.

III. Clean all hoses with compressed-air.

IV. Mount all hoses again at the respective connections (see section 6.6.1, page 37).

Check and cleaning of the optical sensor

WARNING
Laser beam!
The optical sensor has a hazardous radiation power.
Serious eye injury up to blindness can be caused.
To remove the optical sensor, prior to this, it must be electrically disconnected.

I. Demount the optical sensor (7, Fig. 19, page 35):
   a) Release the four fixing nuts (9).
   b) Release the connection of the optical purging air (10) as well as the electric connection (8) of the optical sensor.
   c) Pull the optical sensor vertically out of the support and remove it from the probe.

II. Clean the optical sensor with a clean, lint-free cloth.

III. Mount the optical sensor (7):
   a) Insert the optical sensor vertically into the support in the probe.
   b) Connect the electrical connection (8) of the optical sensor as well as the connection of the optical purging air (10).
   c) Screw the fixing nuts (9) at the support of the optical sensor and tighten them.

Cleaning of the injector

I. Release the screwing at the tubing of the injector (15, Fig. 7, page 21) and remove it.

II. Unscrew the injector with the aid of special tool. (The injector consists of three parts.)

III. Clean the injector with the aid of a cleaning brush.

NOTICE
Alternatively, a cleaning can be carried out via an ultrasonic bath.

IV. Screw all parts of the injector together with the aid of special tool.

V. Insert the injector at the installation place and screw it tight at the tubing.

Cleaning of the probe tube

I. Clean the probe tube with the aid of a cleaning brush. Rotate it clockwise at moving in and out.

Cleaning of other components

I. Clean all tubes and parts which are leading measuring gas.

II. Clean the interior of the probe casing with a clean, lint-free cloth.

6. Insert the connecting tube between ball valve and measuring chamber (4, Fig. 7, page 21) and screw it tight with the aid of a screw-wrench.
7. Place the insulation at the probe and fix it by the five locking clamps.
8. Hoist the casing cap of the probe to the casing and fix it by the four casing locks.
9. Switch the device to operation mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “Off”.

10.5 Check and cleaning of the probe ventilation

Casing fan (standard)

1. Switch the device to maintenance mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “On”.
2. Open the micro-fuse -F06 (6, Fig. 15, page 27).
3. Release the ventilation grids at the inlet filter resp. at the outlet filter of the probe casing (see Fig. 10, page 23) by unfolding them downwards.
4. Check the condition of the filter mats.
5. Clean possibly polluted filter mats respectively exchange them.
6. Close the ventilation grids at the inlet filter resp. at the outlet filter and let them latch in place.
7. Close the micro-fuse -F06.
8. Switch the device to operation mode. For this, set the maintenance switch at the operating device to “Off”.

Vortex Cooler (option)

1. Switch the device to maintenance mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “On”.
2. Open the casing of the probe. For this, release the four casing locks and take the casing cap off (see Fig. 19, page 35).

WARNING
Hot surface!
Several device parts can develop high temperatures.
Burn of skin can be caused!
For protection against possible injury protective gloves must be worn.

3. Check if at the air inlet of the Vortex Cooler (5, Fig. 10, page 23) cooling medium gets in.

NOTICE
In case of outage of the cooling or at pollution the Vortex Cooler must be exchanged.

4. Hoist the casing cap of the probe to the casing and fix it by the four casing locks.
5. Switch the device to operation mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “Off”.

10.6 Check of suck-in openings and exchange of the suction filters

1. Switch the device to maintenance mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “On”.
2. Switch off the two motor protection switches of the blowers by turning the rotary switches -S01 (4, Fig. 15, page 27) and -S02 (5) anti-clockwise to the position “0”.
3. Check the suck-in openings of the two blowers for injector air and dilution air.
4. Clean possibly polluted filters of the suck-in openings respectively exchange them.

CAUTION
Impurity or foreign material in the blower can cause damages.
Observe that at filter exchange neither impurities nor foreign material get into the suck-in openings.
Possibly open suck-in openings must be covered by a protective grid according to DIN EN 294 and free of deposit.

5. Switch on the two motor protection switches of the blowers by turning the rotary switches -S01 (4, Fig. 15) and -S02 (5) clockwise to the position “I”.
6. Switch the device to operation mode. For this, set the maintenance switch (2, Fig. 22, page 46) at the operating device to “Off”.

11 Error search and failure clearance

CAUTION
Failure clearance must be executed by qualified personnel.

For monitoring and signalling of error states the PFM 06 ED outputs status signals. These are provided for the display of the operating device in clear text and for the status contacts (circuit diagrams: see appendix) as potential-free contact. Differed are the error messages “state error” and “maintenance request” as well as the messages “maintenance state” and simple info messages.

Current messages are displayed via the status line in the display of the operating device (see section 9.1, page 46). In the case of several upcoming messages, these are displayed mutually one after another.

1. Check all messages which are mutually displayed in the status line in the display of the operating device.
   › If the message “state error” is displayed, an error resp. failure is existent. The respective cause has to be eliminated immediately.
   › If the message “maintenance request” is displayed, there is no failure. For prevention of failures at early stage, it is recommended to check the causes of the message and to initiate the respective actions if necessary.
2. Compare the displayed messages with the following itemised tables:
   › Message display “state error” (see section 11.1, page 84)
   › Message display “maintenance request” (see section 11.2, page 87)

NOTICE
Messages in the status line without the characterisations “state error” or “maintenance request” serve the simple information without compulsory actions to be carried out.

3. Carry out the respective actions for failure clearance. If necessary, use the attached circuit diagrams (see appendix).
4. Press the set pushbutton or set the 3/2-way ball valve to measuring operation by pressing the key –.
   › All current messages in the status line disappear. Possibly further existing messages are displayed in the status line again.
   › Entries of elapsed error/failure messages can be still reviewed in the error protocol (see section 9.5.10, page 69).
5. Check the error/failure message for disappearance.

NOTICE
At upcoming messages which are not itemised in the following tables or in case of upcoming errors or failure which cannot be eliminated – also in spite of observance of the operation manual, please refer to Dr. Födisch Umweltmesstechnik AG (contact details: see cover inside).

11.1 Message display "state error"

state error		
Message	**Cause**	**Action**
current input p	current input for pressure of measuring chamber (p) not in valid range (3.5...20.5 mA)	• Check the wiring to the pressure sensor p. • Check the p transmitter and carry out a test for the analogue signal. • Check the hoses of the pressure sensor p for deposit resp. cross-section constrictions as well as for correct fit.
current input pd	current input for pressure of dilution (pd) not in valid range (3.5...20.5 mA)	• Check the wiring to the pressure sensor pd. • Check the pd transmitter and carry out a test for the analogue signal. • Check the hoses of the pressure sensor pd for deposit resp. cross-section constrictions as well as for correct fit.
current input T	current input for temperature of measuring chamber (T) not in valid range (3.5...20.5 mA)	• Check the wiring to the transmitter and to the PT100. • Check the PT100 for electric function. • Check the transmitter and carry out a test for the analogue signal.
current input TD	current input for temperature of dilution (TD) not in valid range (3.5...20.5 mA)	• Check the wiring to the transmitter and to the PT100. • Check the PT100 for electric function. • Check the transmitter and carry out a test for the analogue signal.
data packet lost	transmission error between optical sensor and evaluation electronics	• Check the wiring to the optical sensor. • Check the function of the optical sensor.
err. measur. data *(superordinate message)*	missing or invalid measuring values	• Check all other state messages and compare them with the messages itemised in this table. • Carry out the respective actions for failure clearance.
error current inp. *(superordinate message)*	one or more current inputs not in valid range (3.5...20.5 mA)	• Check the state messages "current input ..." and compare them with the messages itemised in this table. • Carry out the respective actions for failure clearance.
error sensor	no connection to the optical sensor	• Carry out a reset (by pressing the reset push-button or by temporary interruption of the power supply). • Check the wiring to the optical sensor. • Check the error output at the display of the optical sensor.

state error		
Message	**Cause**	**Action**
failure ball valve	failure at moving the 3/2-way ball valve	• Check the fuse and the wiring of the 3/2-way ball valve • Check the position of the handwheel. Set it to automatic operation. • Check the rotatability of the 3/2-way ball valve manually if necessary. For this, set the handwheel to manual operation. • Check if the LED “MEASURING” resp. “PURGING” lights. › The LEDs light as soon as the 3/2-way ball valve has reached the correct end positions.
flow F or FM too low	• flow of measuring chamber undershoots the preset minimal flow intensely (F < 80% of Fmin) • flow of sucked measuring gas undershoots the preset minimal flow intensely (FM < 80% of Fmin)	• Check the function of the blower for injector air as well as the respective fuse and the motor protection switch -S01. • Check the hose connections for tight fit. • Check the gas path for pollution. • Check the hose connections of the pressure sensor p.
flow Fd too low	flow of purging air undershoots the preset minimal purging air flow intensely (FD < 80% of FDmin)	• Check the function of the blower for dilution air as well as the respective fuse and the motor protection switch -S02. • Check the hose connections for tight fit. • Check the gas path for pollution. • Check the hose connections of the pressure sensor pd.
pollution	degree of pollution exceeds 60%	• Check the optical sensor for pollution. If necessary, clean it. • Check the optical sensor for correct mounting. • Check the temperature limiter in the probe casing rightwards next to the purging air heating.
reference point	result of reference point test exceeds 6%	• Check the optical sensor for pollution. If necessary, clean it. • Increase the temperature of dilution air resp. of the measuring chamber. • Check the relating reference value.
sensor temperature	sensor temperature exceeds 65 °C	• Check the probe ventilation. • Check the fuse -F06 of the probe ventilation. • Check the air inlet/outlet filters of the probe.

state error		
Message	**Cause**	**Action**
temp T too low	temperature of measuring chamber undershoots the preset minimal temperature intensely ($T < 70\%$ of $T_{nominal}$)	• Check the fuse -F03. • Check the switching of the heating via the LED at relay -K01 at the mounting plate in the operating device. • Check the PT100 resp. the temperature. • Check the flow F. › If F < 1 m³/h and not in purging operation, the heating for T is switched off.
temp Td too low	temperature of purging air undershoots the preset minimal purging air temperature intensely ($TD < 70\%$ of $TD_{nominal}$)	• Check the fuse -F05. • Check the switching of the heating via the LED at relay -K03 at the mounting plate in the operating device. • Check the PT100 resp. the temperature. • Check the flow FD. › If FD < 0.5 m³/h, the heating for TD is switched off.
zero point	result of zero point test exceeds 6%	• Check the optical sensor for pollution. If necessary, clean it. • Check the optical sensor for correct mounting.

11.2 Message display “maintenance request”

maintenance request		
Message	**Cause**	**Action**
flow F or FM too low	• flow of measuring chamber undershoots the preset minimal flow (F < Fmin) • flow of sucked measuring gas undershoots the preset minimal flow (FM < Fmin)	• Check the function of the blower for injector air as well as the respective fuse and the motor protection switch -S01. • Check the hose connections for tight fit. • Check the gas path for pollution. • Check the hose connections of the pressure sensor p.
flow Fd too low	flow of purging air undershoots the preset minimal purging air flow (FD < FDmin)	• Check the function of the blower for dilution air as well as the respective fuse and the motor protection switch -S02. • Check the hose connections for tight fit. • Check the gas path for pollution. • Check the hose connections of the pressure sensor pd.
pollution	degree of pollution exceeds 30%	• Check the optical sensor for pollution. If necessary, clean it. • Check the optical sensor for correct mounting. • Check the temperature limiter in the probe casing rightwards next to the purging air heating.
range Ix too low (x = 1...4)	output range at current output Ix too low	• Check the respective current output. • Correct the measuring range.
reference point	result of reference point test exceeds 4%	• Check the optical sensor for pollution. If necessary, clean it. • Increase the temperature of dilution air resp. of the measuring chamber. • Check the relating reference value.
sensor range too low	sensor input value too high, i.e.: • dust loading too high • sensors polluted or defective • measuring chamber polluted or defective	• Clean the sensors resp. exchange them if necessary. • Clean the measuring chamber. • Increase the dilution resp. the temperature.
sensor temperature	sensor temperature exceeds 55 °C	• Check the probe ventilation. • Check the fuse -F06 of the probe ventilation. • Check the air inlet/outlet filters of the probe.

maintenance request		
Message	**Cause**	**Action**
temp T too low	• temperature of measuring chamber undershoots the preset minimal temperature ($T < 80\%$ of $T_{nominal}$) • heating for T switched off	• Check the fuse -F03. • Check the switching of the heating via the LED at relay -K01 at the mounting plate in the operating device. • Check the PT100 resp. the temperature. • Check the flow F. › If $F < 1$ m³/h and not in purging operation, the heating for T is switched off.
temp Td too low	temperature of purging air undershoots the preset minimal purging air temperature ($TD < 80\%$ of $TD_{nominal}$)	• Check the fuse -F05. • Check the switching of the heating via the LED at relay -K03 at the mounting plate in the operating device. • Check the PT100 resp. the temperature. • Check the flow FD. › If $FD < 0.5$ m³/h, the heating for TD is switched off.
zero point	result of zero point test exceeds 4%	• Check the optical sensor for pollution. If necessary, clean it. • Check the optical sensor for correct mounting.

11.3 Message display “maintenance state”

maintenance state		
Message	**Cause**	**Action**
–	maintenance switch is switched on Notice: At switched-on maintenance switch the 3/2-way ball valve is not automatically set to purging operation if F_X < 80% of F_{Xmin} and a sensor error	• Set the maintenance switch at the operating device to “Off”.
–	zero and reference control is active (see section 3.3, page 18)	• Wait until the zero and reference point control has finished.
–	optical sensor in maintenance state	• Wait until the zero and reference point control has finished.
purging operation	device in purging operation / self-diagnostics and cleaning via 3/2-way ball valve (see section 3.4, page 19)	• Check the position of the 3/2-way ball valve. › After changeover from purging to measuring operation the device still stays in maintenance mode for 1 min.
remote control	device in maintenance via remote control (serial communication) Notice: The keys at the operating device are without function.	• Wait until the remote control has finished. • If required, disconnect the cable of the RS232 connection. › After a waiting time of 2 min the operating at the device is re-enabled.
service display	• service mode is active • parameter input is active	• Set the device to standard mode (see section 9.3 “Mode output of the measuring value display”, page 52) via the menu. • Finish the parameter input by pressing the key “ENTER”.

11.4 Info messages

Message	Cause	Action
auto reset ERR	error message has been reset automatically.	–
auto reset MR	maintenance request message had been reset automatically.	–
error A/D conversion	error at analogue-digital conversion	• Carry out a reset (by pressing the reset push-button or by temporary interruption of the power supply).
error current inp.	one or more current inputs not in valid range (3.5...20.5 mA)	• Check all messages of the message display “state error” (see section 11.1, page 84).
error EEPROM	error at writing or reading in the internal parameter storage	• Carry out a reset (by pressing the reset push-button or by temporary interruption of the power supply).
error read RTC	error at reading of the internal time (irrelevant for the function of the device; local time used)	–
fix measur. range	fix measuring range at current output I2 ($C_{i.B.}$)	–
Initialisation	initialisation of the sensor is active	• Wait until the initialisation has finished.
limit value 1 ON	limit value 1 exceeded	• Wait until the limit value 1 is below the limit again. › The message disappears after 10 s.
limit value 2 ON	limit value 2 exceeded	• Wait until the limit value 2 is below the limit again. › The message disappears after 10 s.
measuring range = ...	automatic measuring range switchover is on, the current measuring range is displayed	–
new parameters	parameters have been changed (automatic revaluation)	–
newly calibrated	device calibration has been carried out	–
power on	• device has been switched on • a reset via the operating device has been released	–

Message	Cause	Action
preheating ... min	device heats up	• Wait until the preheating has finished. Observe the displayed remaining time.
program error	error in the programme	• Carry out a reset (by pressing the reset push-button or by temporary interruption of the power supply).
purging operation	device in purging operation / self-diagnostics and cleaning via 3/2-way ball valve (see section 3.4, page 19)	• Check the position of the 3/2-way ball valve. › After changeover from purging to measuring operation the device still stays in maintenance mode for 1 min.
remote control	device in maintenance via remote control (serial communication) Notice: The keys at the operating device are without function.	• Wait until the remote control has finished. • If required, disconnect the cable of the RS232 connection. › After a waiting time of 2 min the operating at the device is re-enabled.
reset MR/error	current messages have been reset after leaving the service menu	–
secured access	access to the secured area enabled by service personnel of Dr. Födisch Umweltmesstechnik AG	• Working at the device must solely be executed by service personnel of Dr. Födisch Umweltmesstechnik AG.
service display	• service mode is active • parameter input is active	• Set the device to standard mode (see section 9.3 “Mode output of the measuring value display”, page 52) via the menu. • Finish the parameter input by pressing the key “ENTER”.
wait for sensor	Optical sensor builds up the communication to the system	• Wait until the message disappears.

12 Shutdown and Disposal

CAUTION
Ball valve position!
Wrong position of the 3/2-way ball valve causes pollution of the measuring chamber.
From the beginning of shutdown until finishing of disassembling the 3/2-way ball valve must be set as follows:

- handwheel in manual operation (position “(B) MAN”, see Fig. 9, page 22)
- rotary switch in purging operation (position “PURGE”, see Fig. 8, page 22)

12.1 Shutdown

DANGER
Hazardous voltage!
Parts of the device can be energised with hazardous voltage.
Danger of electric shock.
Any work at the device must solely be executed by qualified personnel.

Case 1 – Permanent shutdown (with subsequent disassembling)

1. Open the casing of the probe. For this, release the four casing locks and take the casing cap off (see Fig. 19, page 35).
2. Set the 3/2-way ball valve (6) to manual and purging operation.
3. Close the valve of dilution air distribution (6, Fig. 7, page 21) inside the probe.
4. Open the two doors at the operating device one after another with the aid of the delivered control cabinet key. To open the first door, turn the opening lever 90° clockwise.
5. Open the micro-fuses -F02 to -F06 (6, Fig. 15, page 27).
6. Switch off the main fuse -F01 (3) by setting the red lever downwards.
7. Disconnect the external pre-fuse.
8. Go on with the disassembling of probe, operating device and blowers (see section 12.2, page 93).

Case 2 – Temporary shutdown (remaining of probe in the duct)

1. Open the casing of the probe. For this, release the four casing locks and take the casing cap off (see Fig. 19, page 35).
2. Set the 3/2-way ball valve (6) to manual and purging operation.
3. Set the volume flow of the dilution so that the dilution between 3 m^3/h and 5 m^3/h. For this, regulate the valve of dilution air distribution (6, Fig. 7, page 21) inside the probe and check the display at the operating device.
4. Open the two doors at the operating device one after another with the aid of the delivered control cabinet key. To open the first door, turn the opening lever 90° clockwise.

5. Open the micro-fuses -F03 and -F04 (6, Fig. 15, page 27).

NOTICE
The micro-fuses -F02, -F05 and -F06 for control, dilution air heating and casing fan stay closed. Furthermore, the motor protection switches -S01 and -S02 of the two blowers stay switched on.

6. Close the two doors at the operating device one after another and lock them with the aid of the delivered control cabinet key. To close the first door, prior to this, turn the opening lever 90° anti-clockwise.

12.2 Disassembling

DANGER
Open ducts with harmful gases, high temperatures or high pressure can cause serious health damage, injury or death.
Disassembling of welding flange and probe must solely be executed at standstill of the system.

WARNING
Hot surface!
Several device parts can develop high temperatures.
Burn hazard!
Before working at the device it must be cooled down.

NOTICE
A disassembling of probe, operating device and blowers has only to be carried out in the case of permanent shutdown (see section 12.1, page 92).

1. Release the cover at the lower cable trench inside the operating device.
2. Disconnect the cables of signal transfer.
3. Disconnect all cables of the external power supply from the main fuse.
4. Close the two doors at the operating device one after another and lock them with the aid of the delivered control cabinet key. To close the first door, prior to this, turn the opening lever 90° anti-clockwise.
5. Release the fastening clip at the socket of the power supply at the probe and disconnect the plug of the power cord from the plug-in connector.
6. Release the fastening clip at the socket of the signal transfer at the probe and disconnect the plug of the low-voltage cable from the plug-in connector.
7. Release the hose clamps.
8. Pull off the hose of dilution air from the connections at the probe inlet and at the blower outlet.
9. Pull off the hose of injector air from the connections at the probe inlet and at the blower outlet.
10. Pull off the exhaust hose from the probe and, if necessary, from the return flange.
11. Demount the profile rack from the installation place.

NOTICE
For eased disassembling of the probe from the duct the optical sensor should be demounted.

WARNING
Laser beam!
The optical sensor has a hazardous radiation power.
Serious eye injury up to blindness can be caused.
To remove the optical sensor, prior to this, it must be electrically disconnected.

12. Demount the optical sensor (7, Fig. 19, page 35):
    a) Release the four fixing nuts (9).
    b) Release the connection of the optical purging air (10) as well as the electric connection (8) of the optical sensor.
    c) Pull the optical sensor vertically out of the support and remove it from the probe.
13. Demount the probe from the duct:
    a) Release the four fixing nuts (11, Fig. 19, page 35) inside the probe with the aid of a screw driver.
    b) Hoist the probe out of the welding flange (3).
    c) Remove the gasket from the adapter (5).
    d) Hoist the casing cap of the probe to the casing and fix it by the four casing locks.
14. Demount the welding flange and, if necessary, the return flange (in case of separate return) from the duct.
15. Store all cables and hoses safe and care for correct storage of probe, optical sensor, operating device and blowers.

12.3 Storage

For correct storage of the device the following ambient conditions apply:

- Ambient temperature: -20...+50 °C
- Relative humidity: max. 90% (non-condensing)
- Protection against wetness
- Location free of percussion

12.4 Disposal

NOTICE
The disposal must be executed according to the country-specific legal environmental protection regulations. The device must be treated as hazardous waste.

13 Technical data

13.1 General technical data

General	
Power supply	3L, N, PE, 400 V AC 50 Hz, 4 kVA (max. 5x 4 mm²)
Protection class	1
Operational availability	after 5 to 15 min (without preheating)
Measuring method	dust: optical dust measurement with laser beam (scattered light), extractive
Measuring range	dust in operation: 0...15 mg/m³ (max. 500 mg/m³)
Calibration	via gravimetric comparison measurement
Media temperature	max. 180 °C
Exhaust humidity	rel. humidity: 100%
Pressure against ambience	-30...+2 hPa
Ambient temperature	-20...+50 °C
Flow of measuring gas	6...12 m³/h (sucked measuring gas and dilution air)
Operating device	steel sheet housing on profile rack (incl. blowers) approx. 600 mm x 1760 mm x 670 mm (w x h x d), approx. 90 kg, IP 65; display: 4-line LC display
Probe	extractive sampling with GRP weather protection casing approx. 610 mm x 1050 mm x 1500 mm (w x h x d), approx. 65 kg, IP 55 immersion depth: max. 1000 mm; probe cable length max. 25 m
Flange	DN 80 PN 6, special type: tube ∅ 100 mm
Analogue outputs	4 x 4...20 mA, galvanically separated with common ground, burden max. 1 kΩ
Digital outputs	6 x potential-free contact, max. 35 V UC, 0.4 A
Digital input	optional, external switch contact for switchover of measuring/purging
Clip contacts	max. 2.5 mm²
Suitability test	DIN EN 15267, QAL1, ID: 0000035014

Tab. 5: General technical data

13.2 Pneumatic connection

NOTICE
Modification by customer must come to an agreement with Dr. Födisch Umweltmesstechnik AG (contact details: see cover inside).

Pneumatic connections	
Hose of injector air	max. 15 m (standard: 5 m), d_i = 25 mm
Hose of dilution air	max. 15 m (standard: 5 m), d_i = 13 mm
Optional compressed-air/nitrogen connections (supply with instrument air):	
Dilution air supply	admission pressure max. 1 ... 2 bar, consumption 3 ... 5 m³/h (output pressure of pressure regulating valve: set value 0.2 bar)
Injector air supply	admission pressure max. 1 ... 2 bar, consumption 40 ... 50 m³/h (output pressure of pressure regulating valve: set value 0.2 bar)
Casing purging (Vortex Cooler)	admission pressure max. 2 ... 3 bar, consumption max. 60 m³/h (output pressure of pressure regulating valve: set value min. 2 bar)

Tab. 6: Pneumatic connections

13.3 Factory settings

Basic settings		
Parameter	**Description**	**Factory setting**
Limit values (see section 9.4.1, page 54)	setting of limit values 1 and 2	
• Limit value 1	1. limit value for dust in standard condition dry	500 mg/m³
• Limit value 2	2. limit value for dust in standard condition dry	500 mg/m³

Tab. 7: Factory settings – Basic settings

Change parameters		
Parameter	**Description**	**Factory setting**
Automatic measuring range switchover (see section 9.5.1, page 57)	switch-on/-off of automatic measuring range switchover for dust in standard condition dry	off
Device parameters (see section 9.5.2, page 58)	setting of device parameters for volume flow and temperature	
• range p	measuring range of Δp transmitter in the measuring chamber	10 mbar
• factor K	construction factor of Δp measurement in the measuring chamber	device-specific factory calibration
• range pd	measuring range of Δp transmitter of the dilution air	70 mbar
• factor KD	construction factor of Δp measurement of the dilution air relating to the probe	0.3
• range T	measuring range of temperature measurement in the measuring chamber	300 °C
• K_C	calibrating factor (factory calibration)	1.0
• range TD	measuring range of temperature measurement of the dilution air	300 °C

Change parameters		
Parameter	**Description**	**Factory setting**
Operation parameters (see section 9.5.3, page 60)	setting of operation parameters	
• T (TM)	average measuring gas temperature at the measuring point in the exhaust duct (device-internal operand – no nominal value!)	80 °C
• p	absolute pressure in the measuring chamber	980 mbar
• pd	absolute pressure of dilution air	1070 mbar
• pm	static pressure at the measuring point in the exhaust duct	1000 mbar
• O2	volumetric content of oxygen in the measuring gas	21 vol. %
• CO2	volumetric content of carbon dioxide in the measuring gas	0 vol. %
• H2O	volumetric content of water vapour in the measuring gas	0 vol. %
• RD	gas constant of dilution air	287 J/(kg K)
Integration time (see section 9.5.4, page 62)	setting of integration time for smoothing of the measuring values	10 s
Output mode (see section 9.5.5, page 63)	selection between standard and service mode	standard mode
Output ranges for dust (see section 9.5.7, page 65)	setting of measuring ranges for dust and raw signal	
• c_iB	measuring range for dust	15 mg/m³
• cal	measuring range for raw signal	20 V

Change parameters		
Parameter	**Description**	**Factory setting**
Output ranges for volume flow (see section 9.5.8, page 66)	setting of measuring ranges for the volume flow of measuring gas and dilution air	
• flow F/FM	measuring range of volume flow in the measuring chamber	20 m³/h
• flow min. Fmin	limit value for volume flow monitoring of the measuring chamber	5 m³/h
• flow FD	measuring range of volume flow of dilution air	20 m³/h
• flow min. FDmin	limit value for volume flow monitoring of dilution air	2 m³/h
Preheating mode (see section 9.5.11, page 70)	Setting of probe heating	
• preheating	preheating at switch-on	on
• T	nominal value for temperature in the measuring chamber	140 °C
• TD	nominal value for temperature of dilution air	140 °C
• preheating time	duration of preheating after switch-on	30 min

Tab. 8: Factory settings – Change parameters

Calibrating		
Parameter	**Description**	**Factory setting**
Relating reference value of the sensor (see section 9.6.1, page 73)	determination of relating reference value	
Calibrating constants (see section 9.6.2, page 76)	setting of constants for evaluation of the dust concentration C_iB	
• A	parameter A	1
• D	parameter D	0

Tab. 9: Factory settings – Calibrating

14 Spare and wear parts

For placing a purchase order of spare and wear parts please refer to Dr. Födisch Umweltmesstechnik AG (contact details: see cover inside).

Element/Component	Item number
3/2-way ball valve	ETLF 13019
Adapter gasket DN 80 PN 6	ETLF 13202
Cleaning brush set	ETLF 13342
Filter element	ETLF 13375
Fixing screw set for sampling bow	ETLF 13249
Gasket for screwing-in of ball valve	ETLF 99129
Gasket set for injector	ETLF 13225
Gasket, PTFE	ETLF 13169
Hose for dilution air (5 m)	ETLF 13367
Hose for injector air (5 m)	ETLF 13366
Optical sensor	ETLF 13351
Pressure transmitter 10 mbar	ETLF 13216
Pressure transmitter 70 mbar	ETLF 13217
Probe tube	ETLF 13084
Sampling bow (Ø 9 mm)	ETLF 13259
Temperature sensor PT100	ETLF 13125

Tab. 10: Spare and wear parts

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16 Appendix

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Auftraggeber:

Projekt: PFM 06 ED

Auftragsnummer: Serie

Messgrössen:

Ausführender Betrieb: Dr. Födisch Umweltmesstechnik AG
Zwenkauer Str. 159
04420 Markranstädt

Schranknummer:

				Datum	10.09.2014			Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
				Bearb.	Bude				Deckblatt	Serie	+
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf					Zeichnungsnummer:	Blatt: 1
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33

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AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 2
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0030	terminal diagram: =probe-X16
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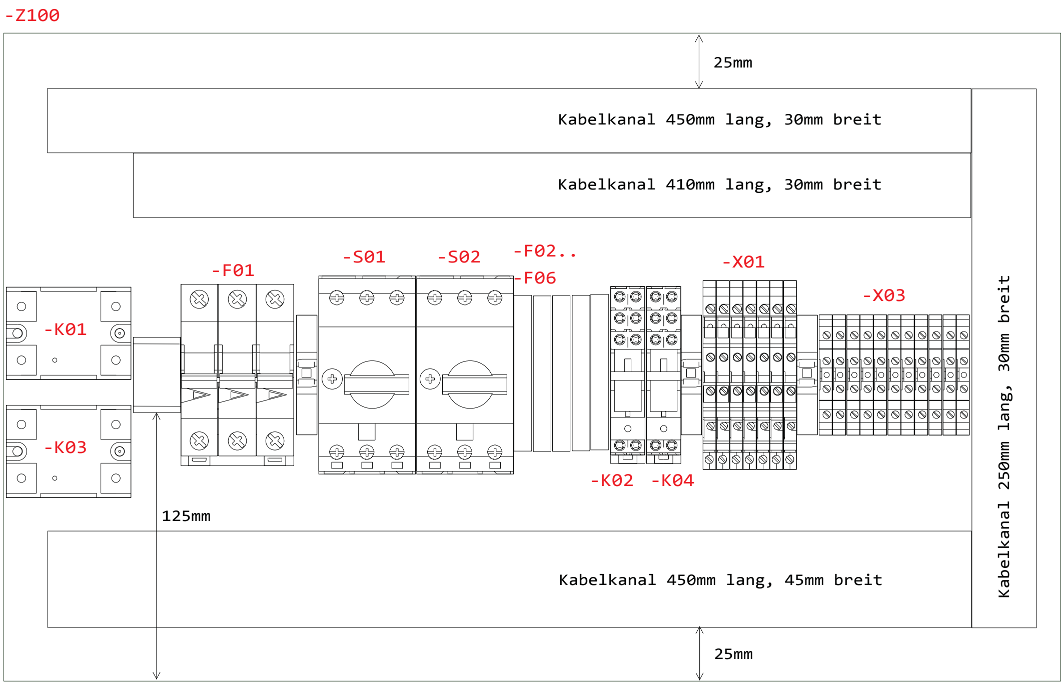
0032	Materialliste
0033	Materialliste



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AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			Zeichnungsnummer:
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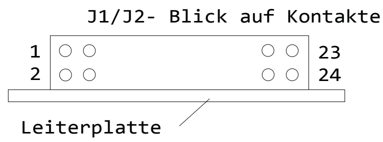

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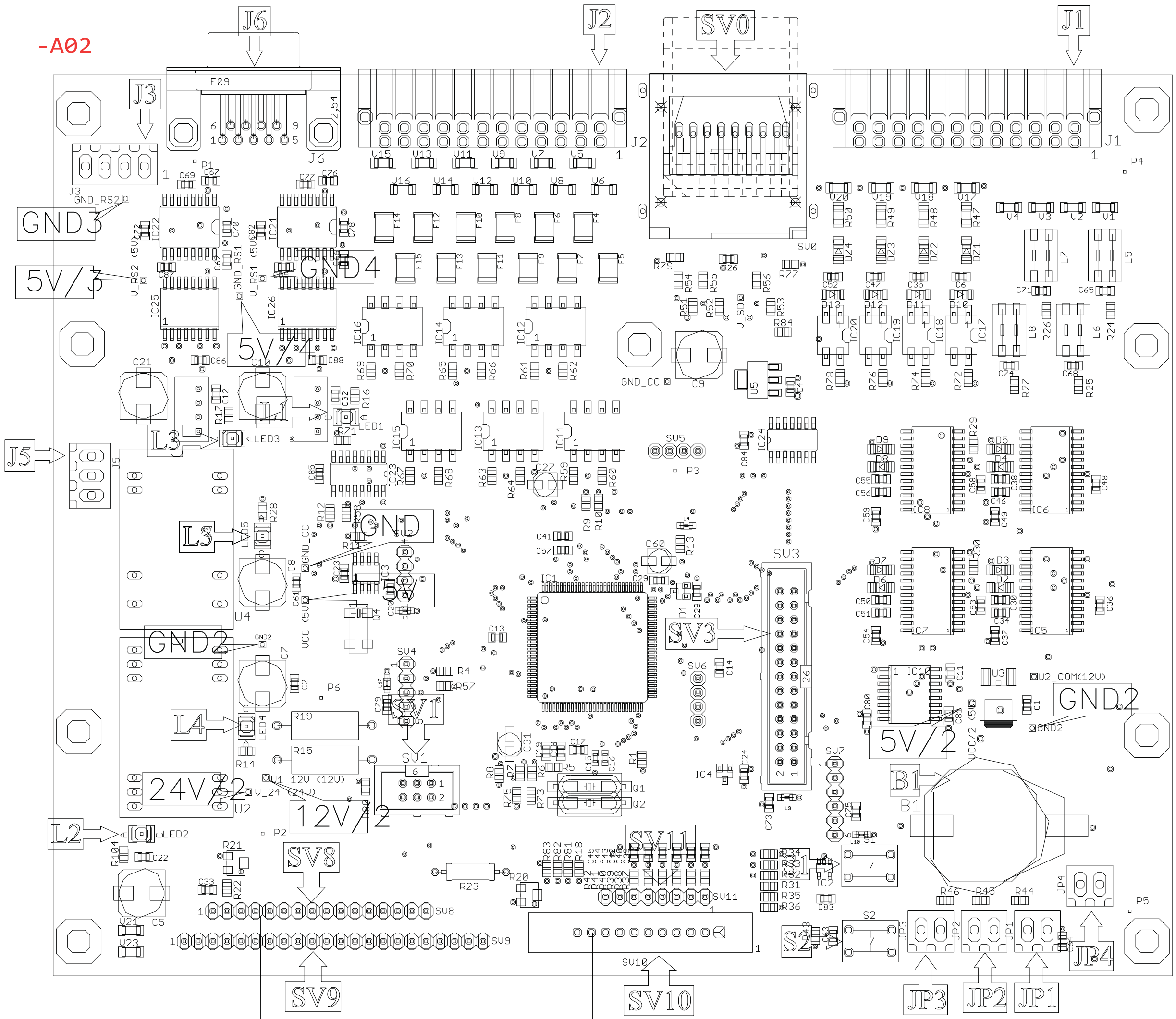

-P01

Display

-S01

Tastatur

Anschlüsse:

JP1: LED Messen
JP2: LED Spülen
JP3: LED Wartung
JP4: Schalter Wartung
JP1-JP3/1: LED Anode

AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014	Dr. Födisch Umweltmesstechnik AG		PFM 06 ED Aufbau Signalplatine	Projektnummer: Serie
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude				Zeichnungsnummer:
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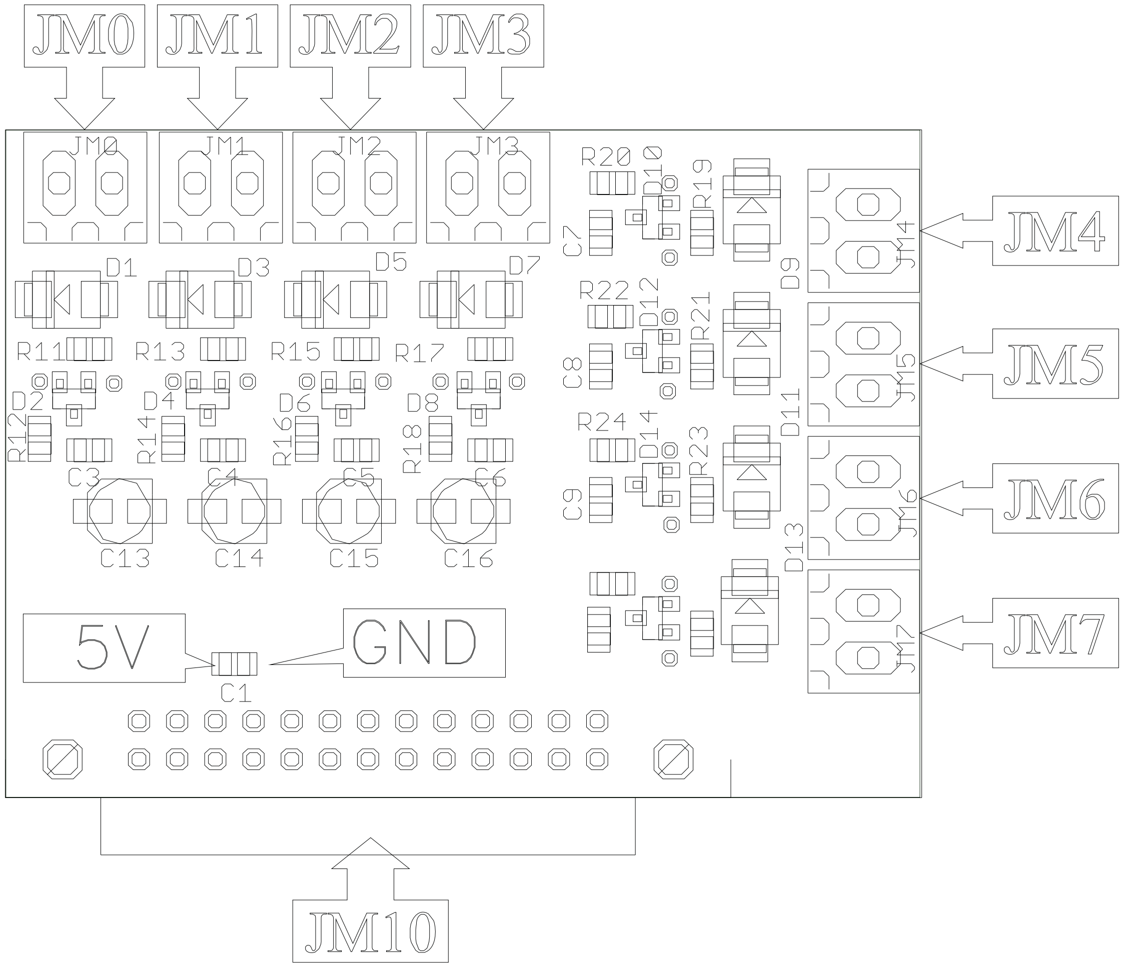
-A03



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AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			Zeichnungsnummer:
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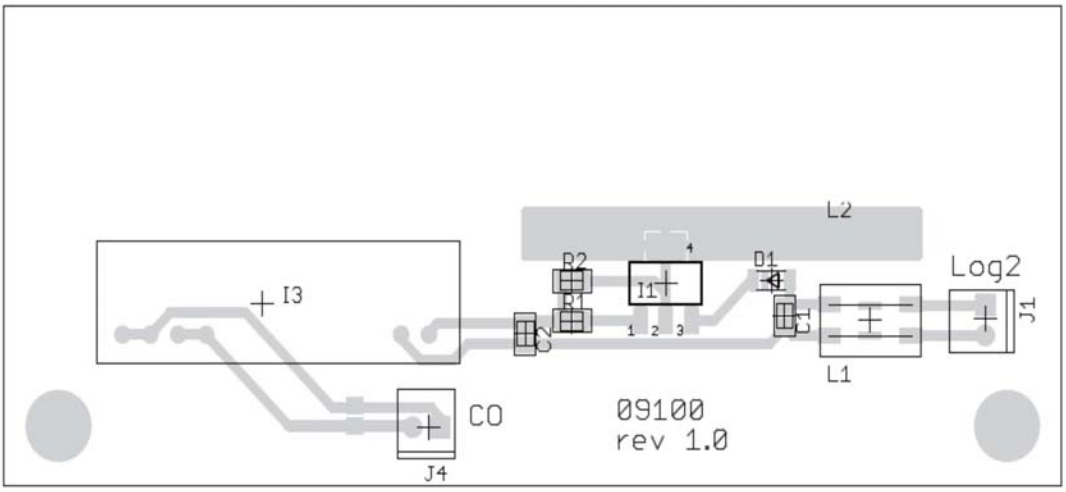
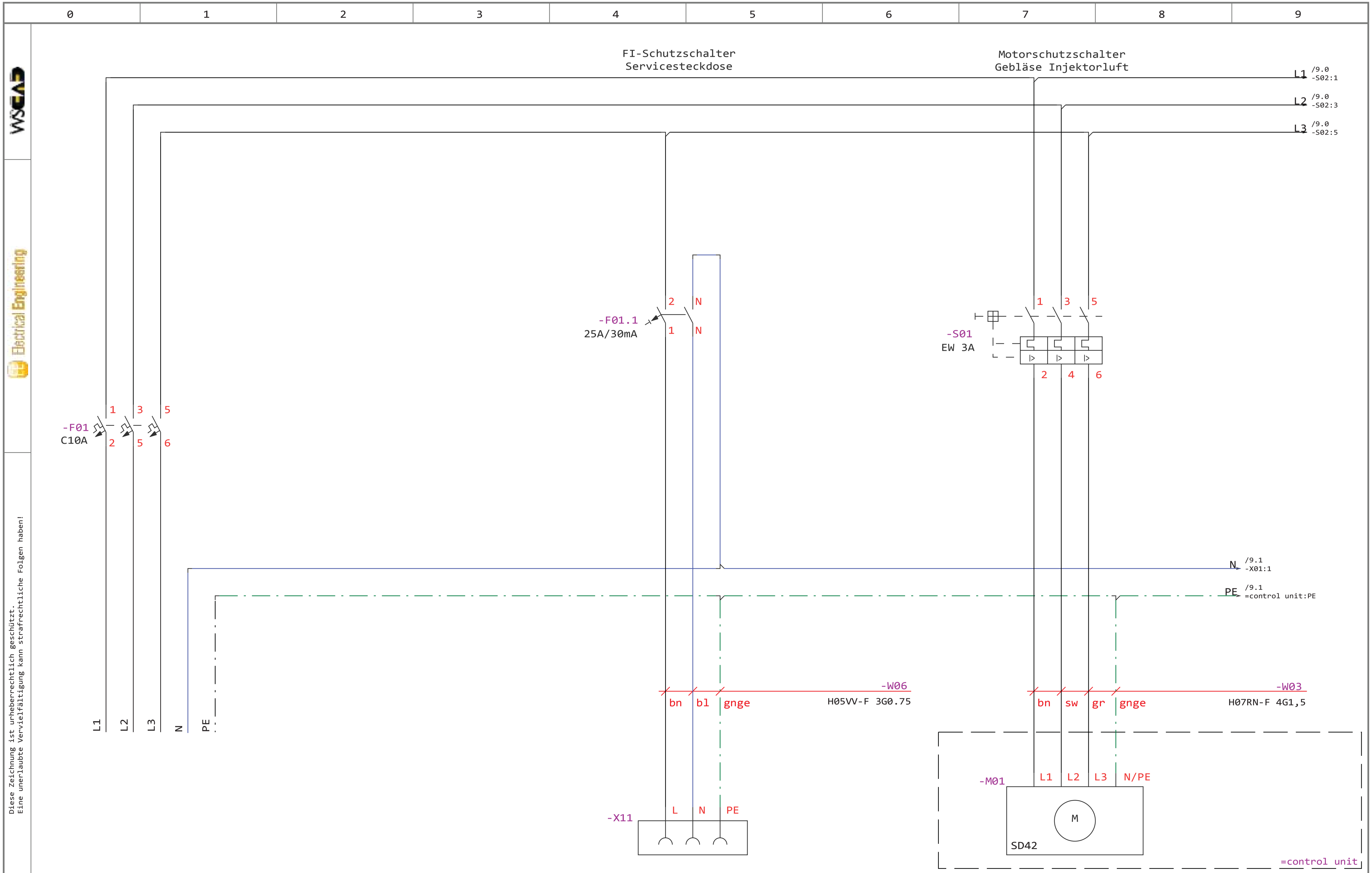
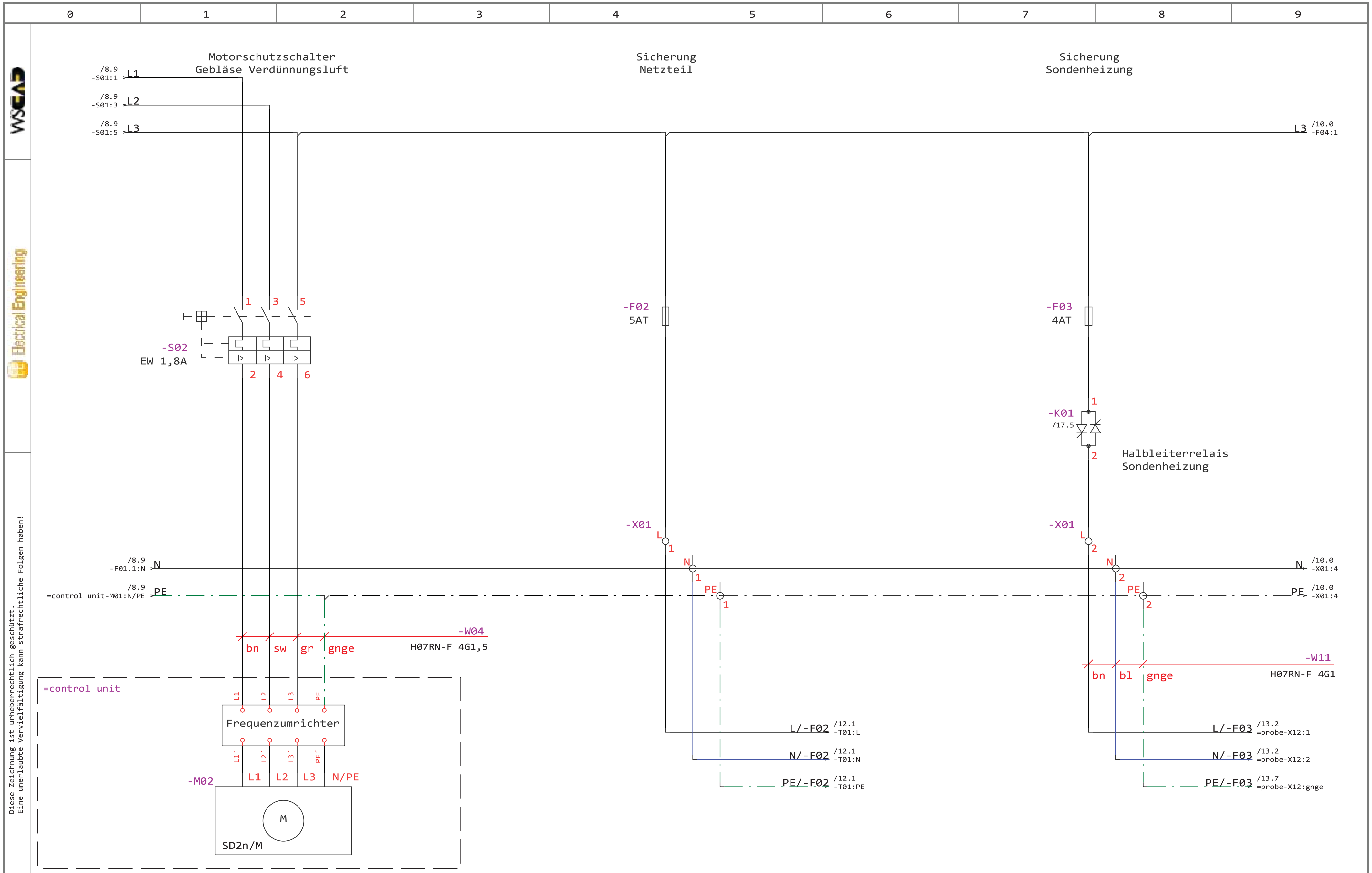
-A05



				Datum	10.09.2014	Dr. Födisch Umweltmesstechnik AG		PFM 06 ED	Projektnummer: Serie
AC	-F4	14.03.2017	Ro.Kronberg	Bearb.	Bude			Platine lext	Zeichnungsnummer:
AB	Rev.02	29.02.2016	Ro.Kronberg	Gepr.	Krummsdorf				=PFM06ED +
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Ers.f	Ers.d		Blatt: 7 von: 33

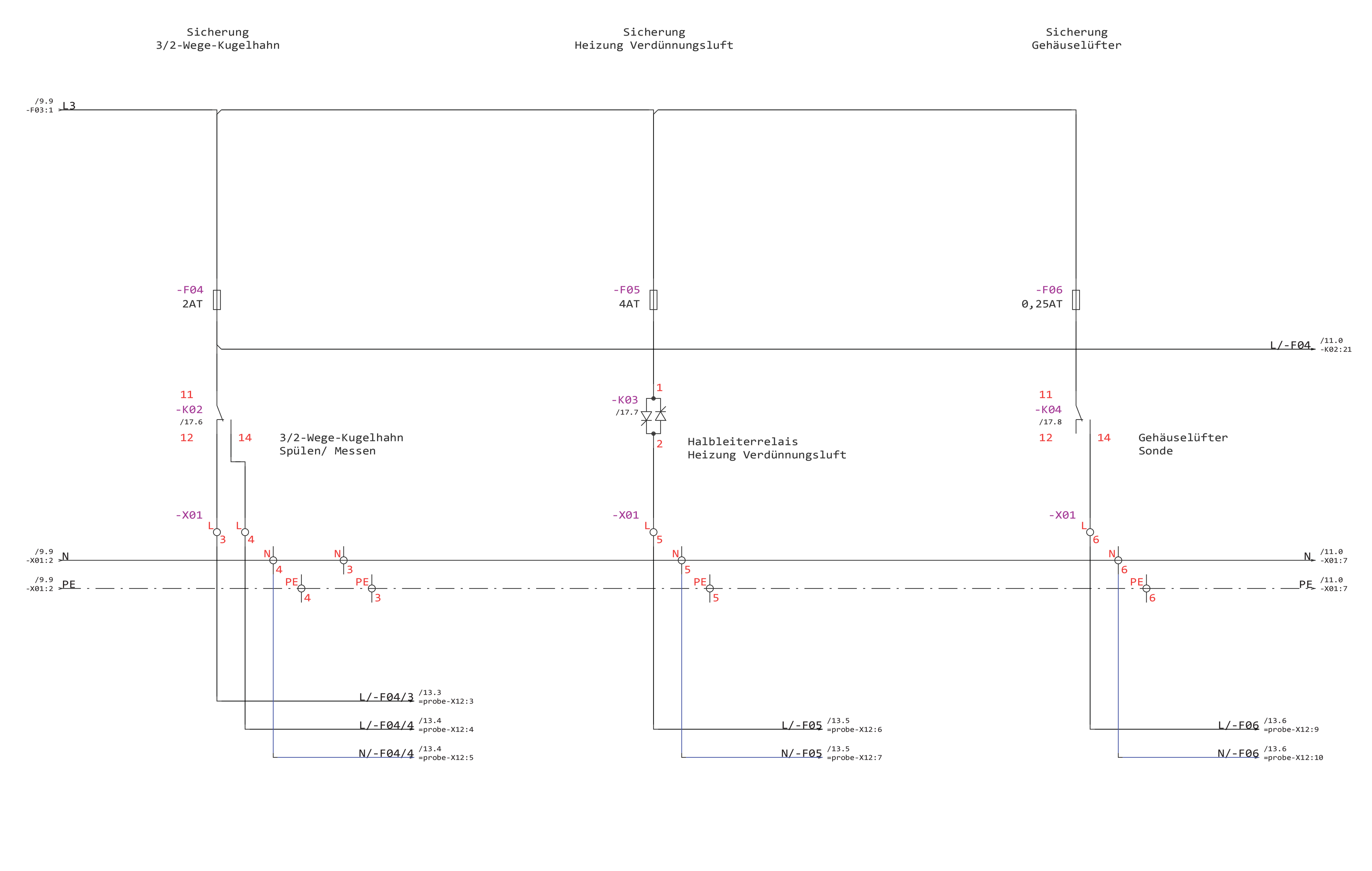


AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			Einspeisung 400VAC	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 8
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d		von: 33

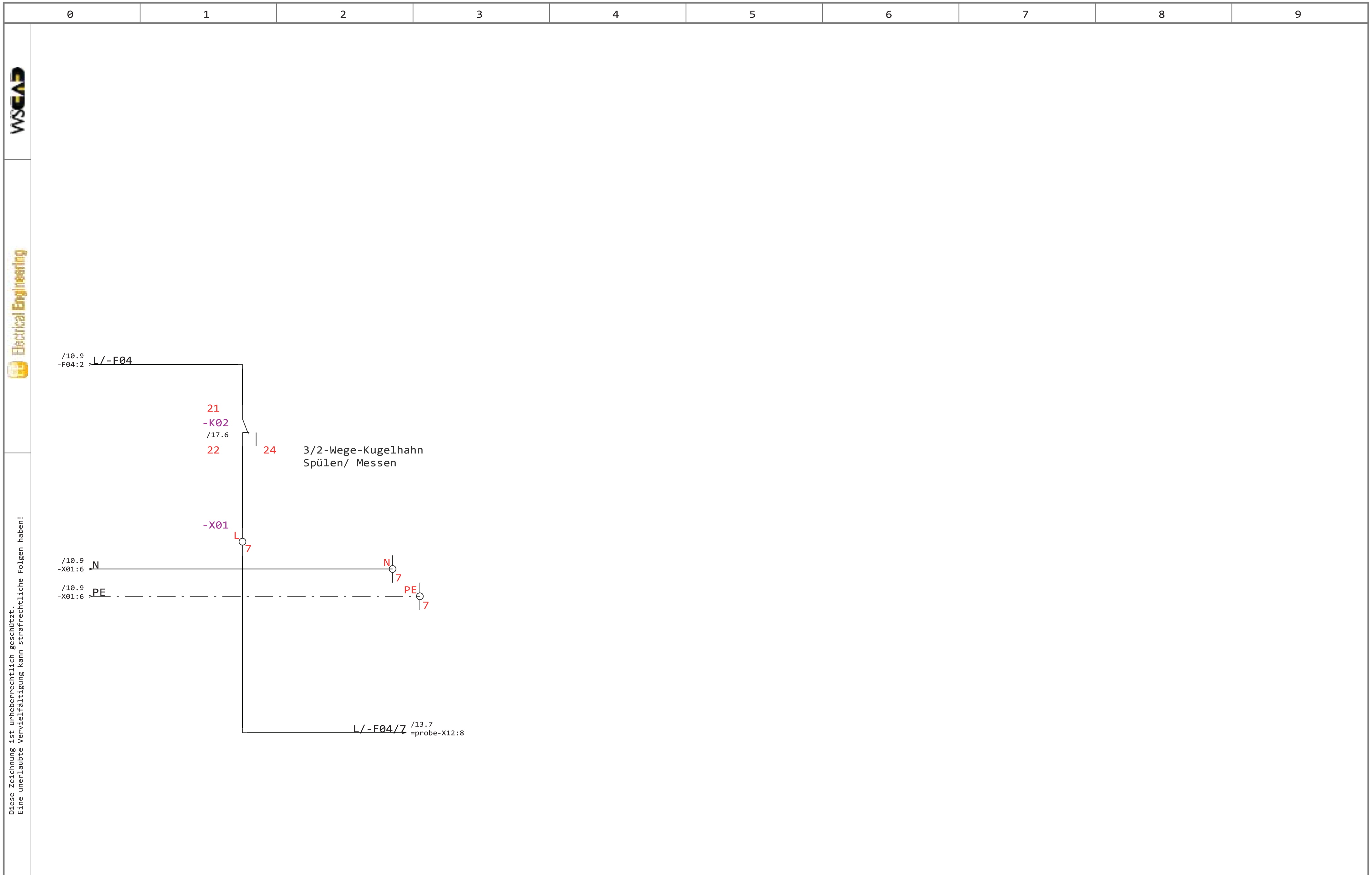


AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude	Verteilung 230VAC	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf		Zeichnungsnummer:	Blatt: 9
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. / Ers.f / Ers.d		von: 33

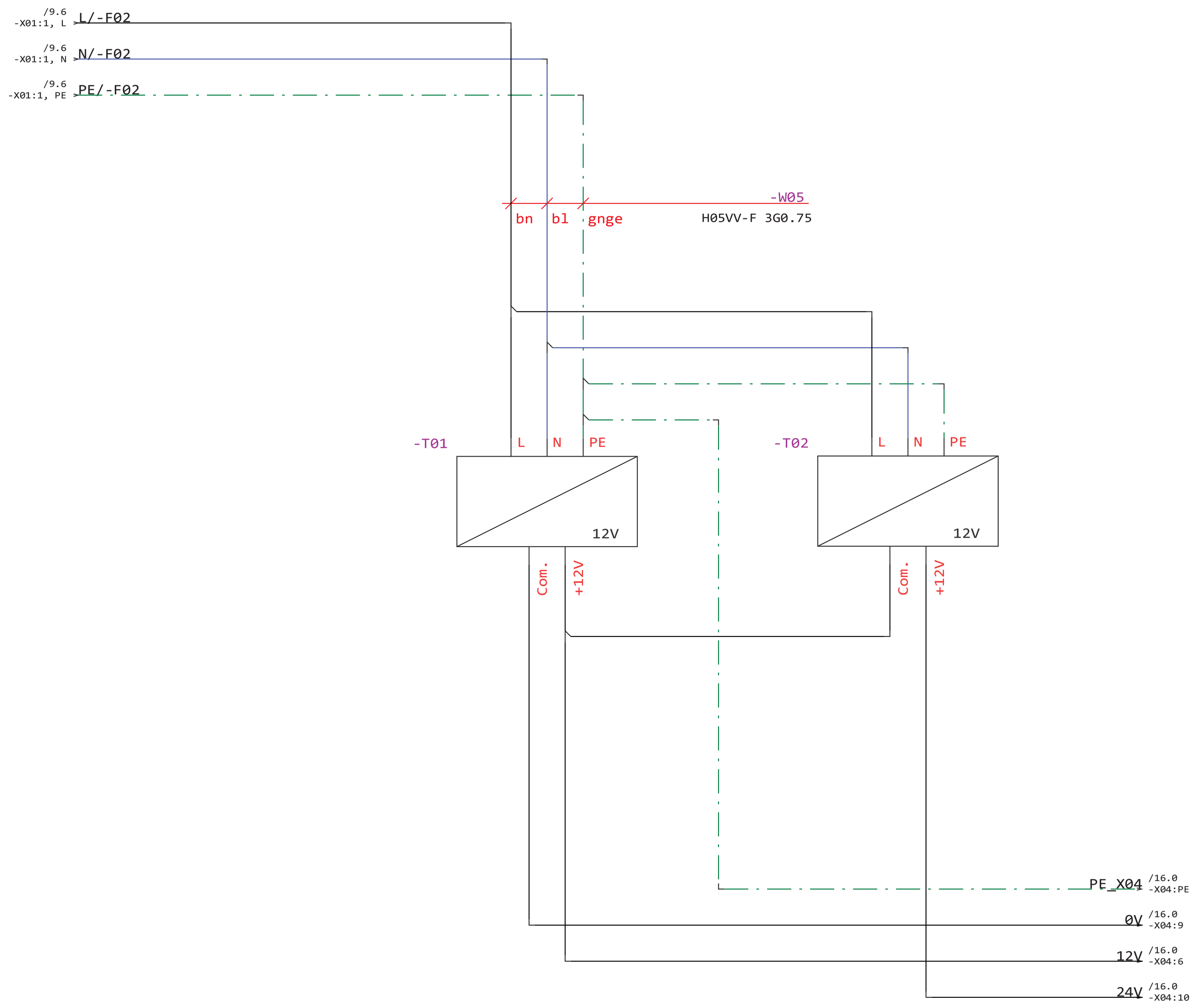
Dr. Födisch Umweltmesstechnik AG



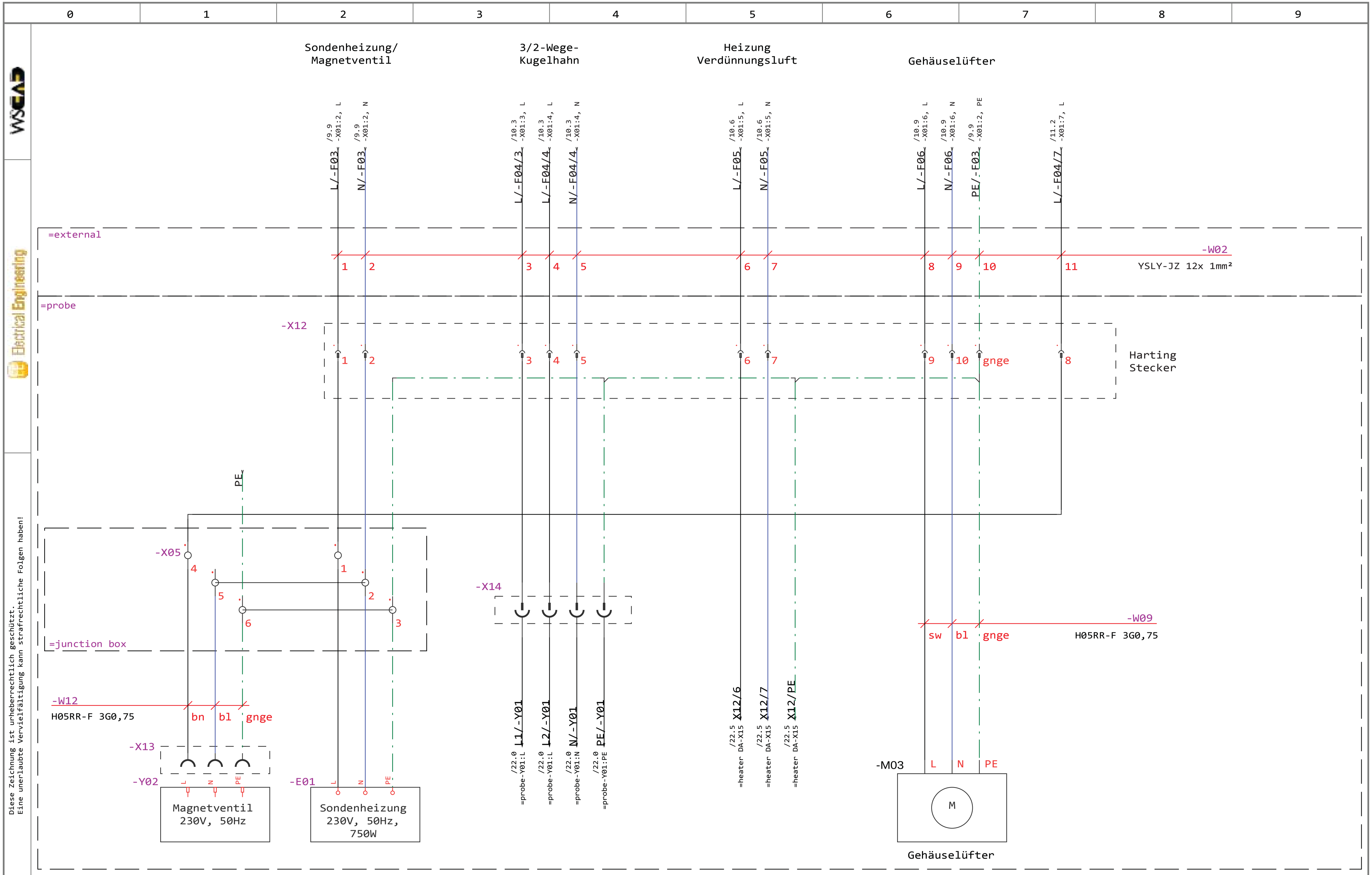
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014		PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude		Verteilung 230VAC	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf			Zeichnungsnummer:	Blatt: 10
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. / Ers.f / Ers.d			von: 33



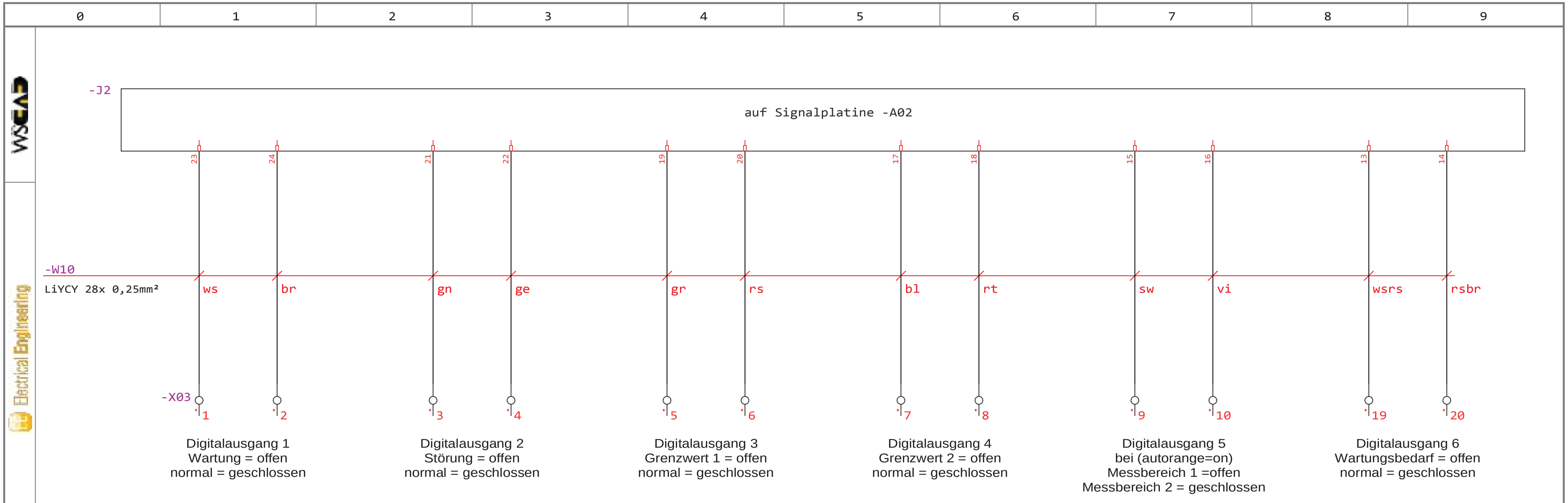
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014			Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude				Verteilung 230VAC	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf					Zeichnungsnummer:	Blatt: 11
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33



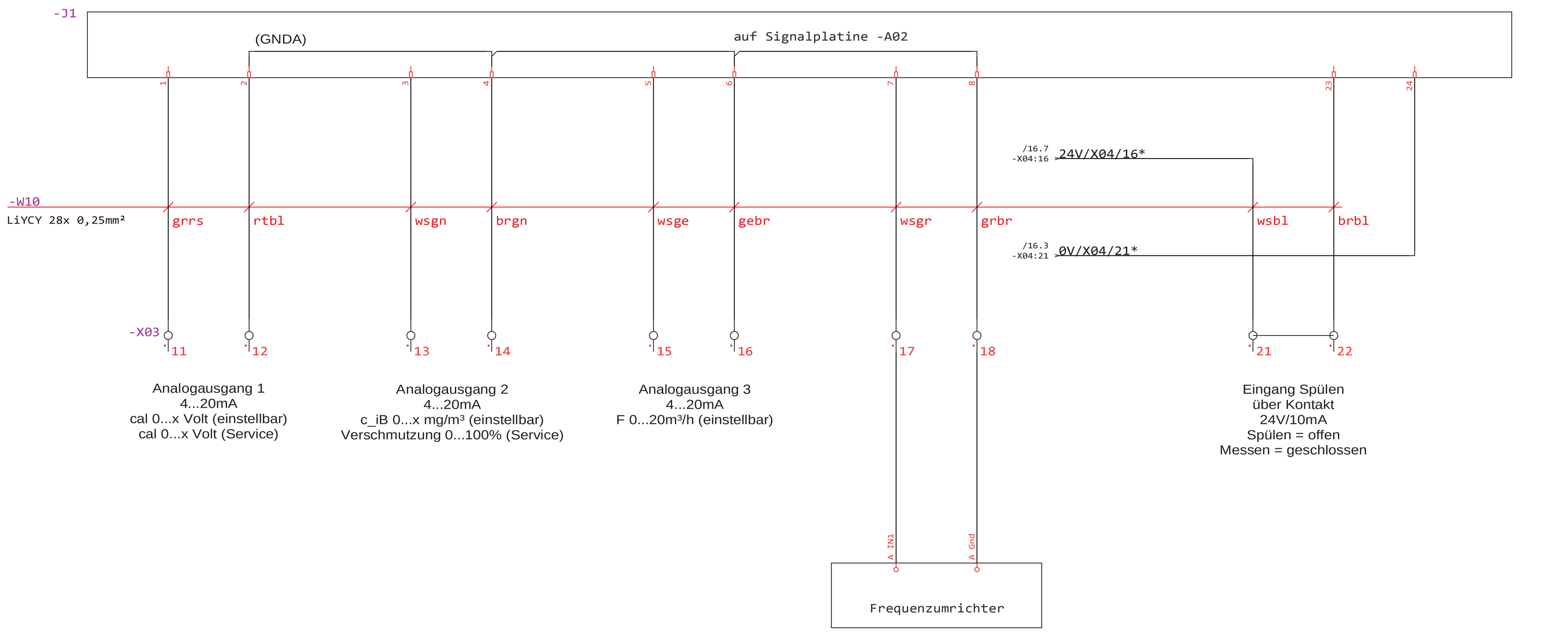
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014			Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude				Netzteile	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf					Zeichnungsnummer:	Blatt: 12
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33



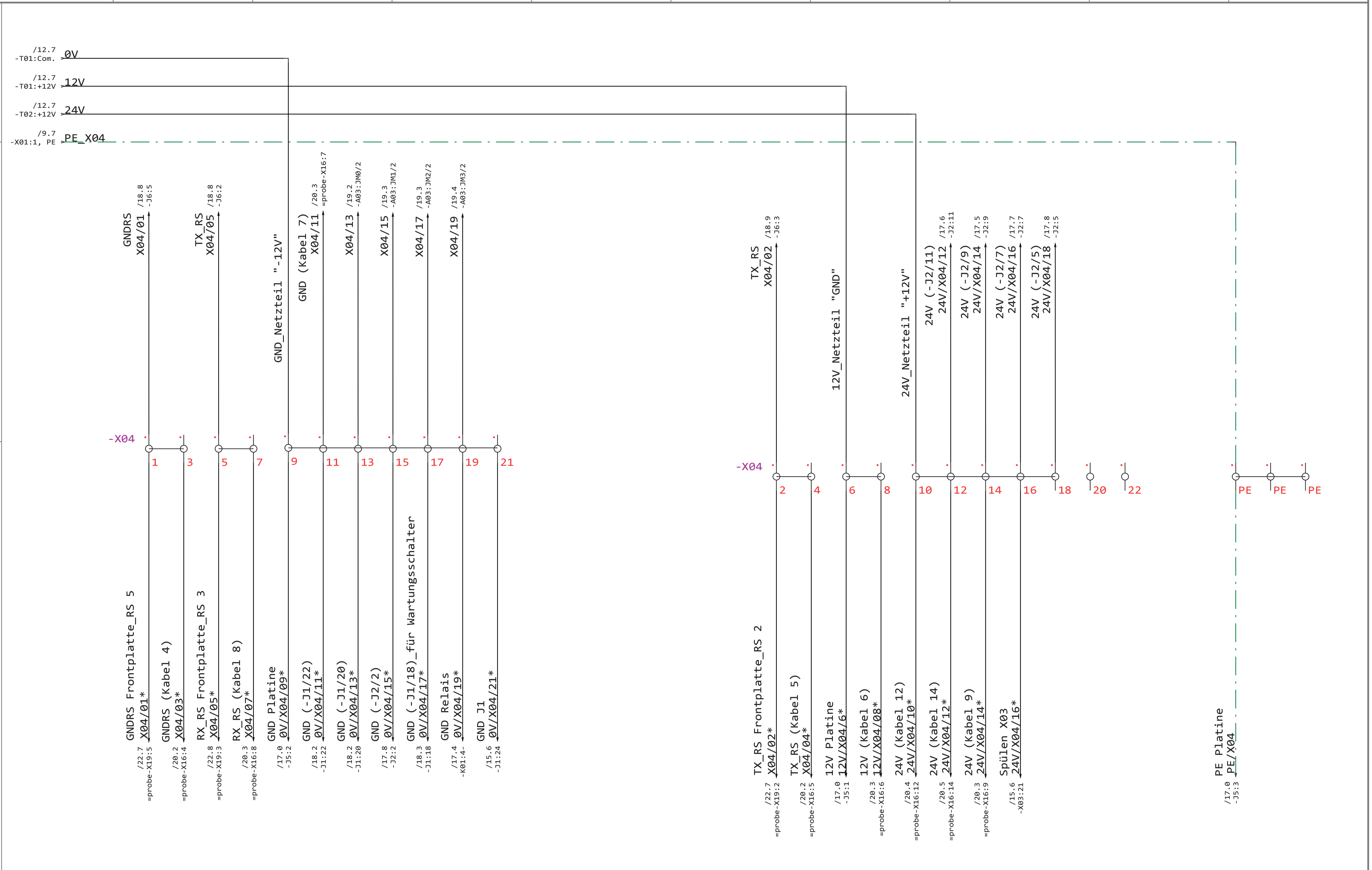
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			Sondenheizung, Gehäuselüfter, Magnetventil	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 13
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f	Ers.d			von: 33



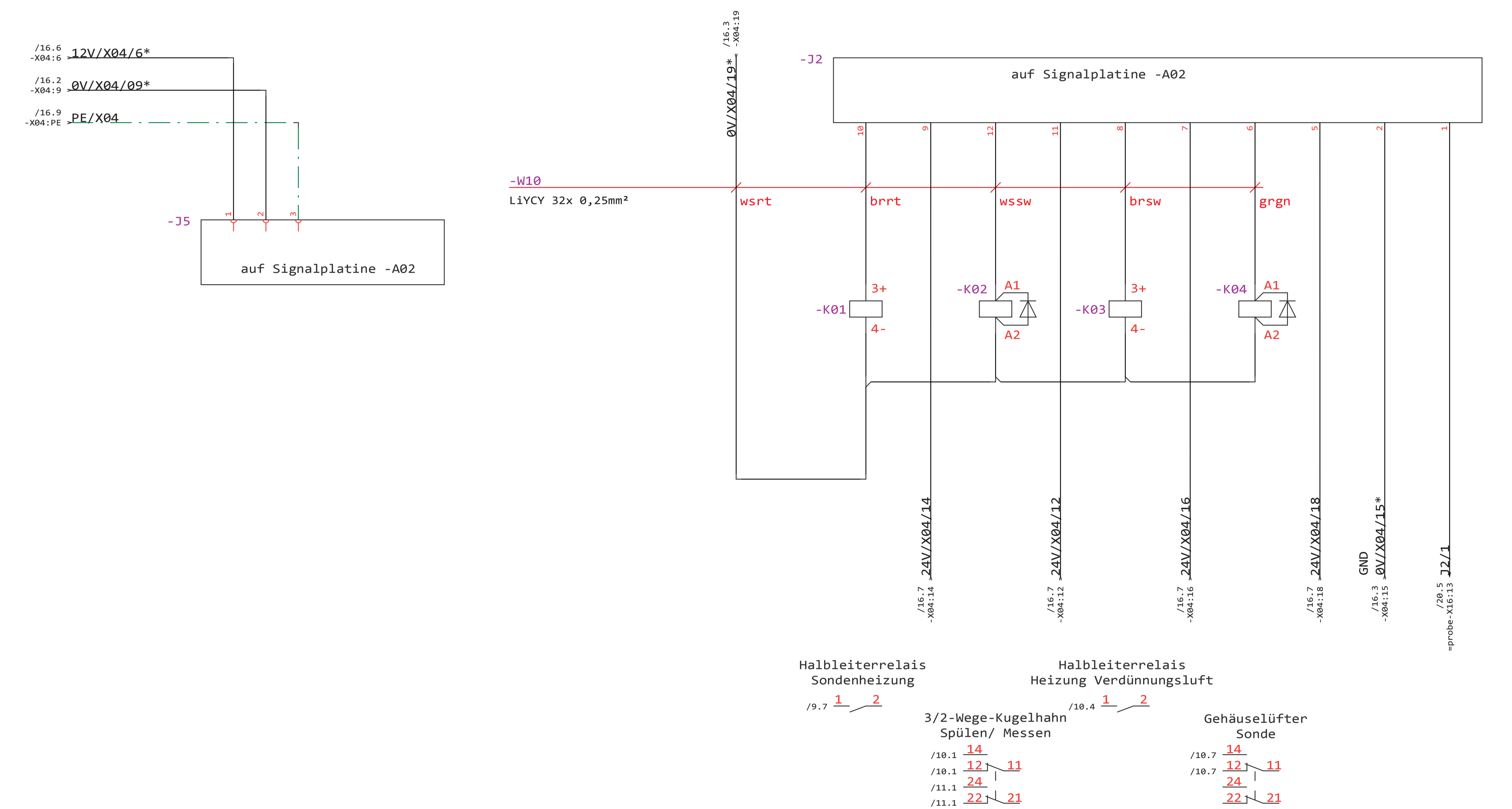
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014			Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude				Digitalsignale	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf					Zeichnungsnummer:	Blatt: 14
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33



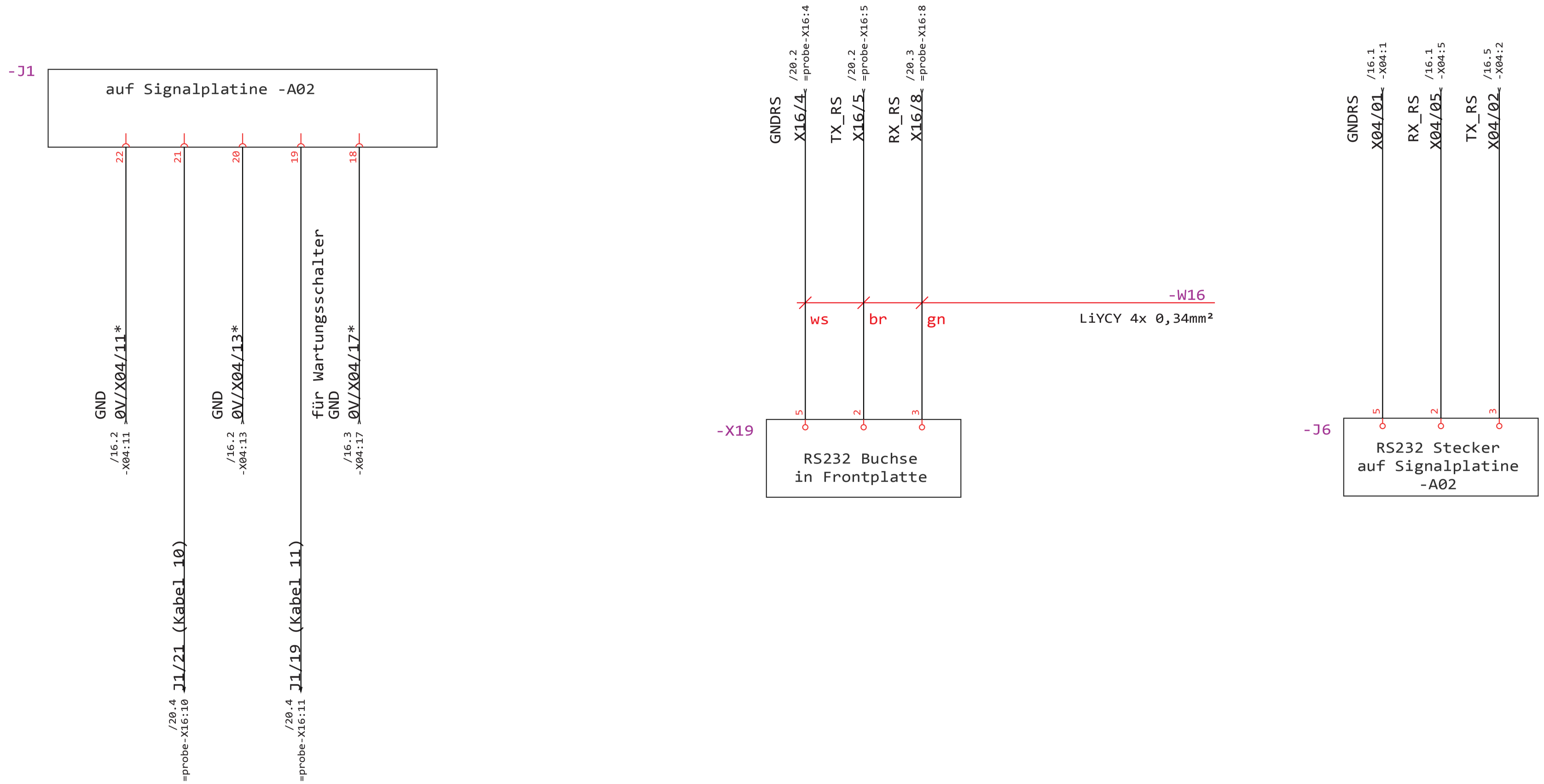
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014	Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude		Analogsignale	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf			Zeichnungsnummer:	Blatt: 15
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. / Ers.f / Ers.d			von: 33



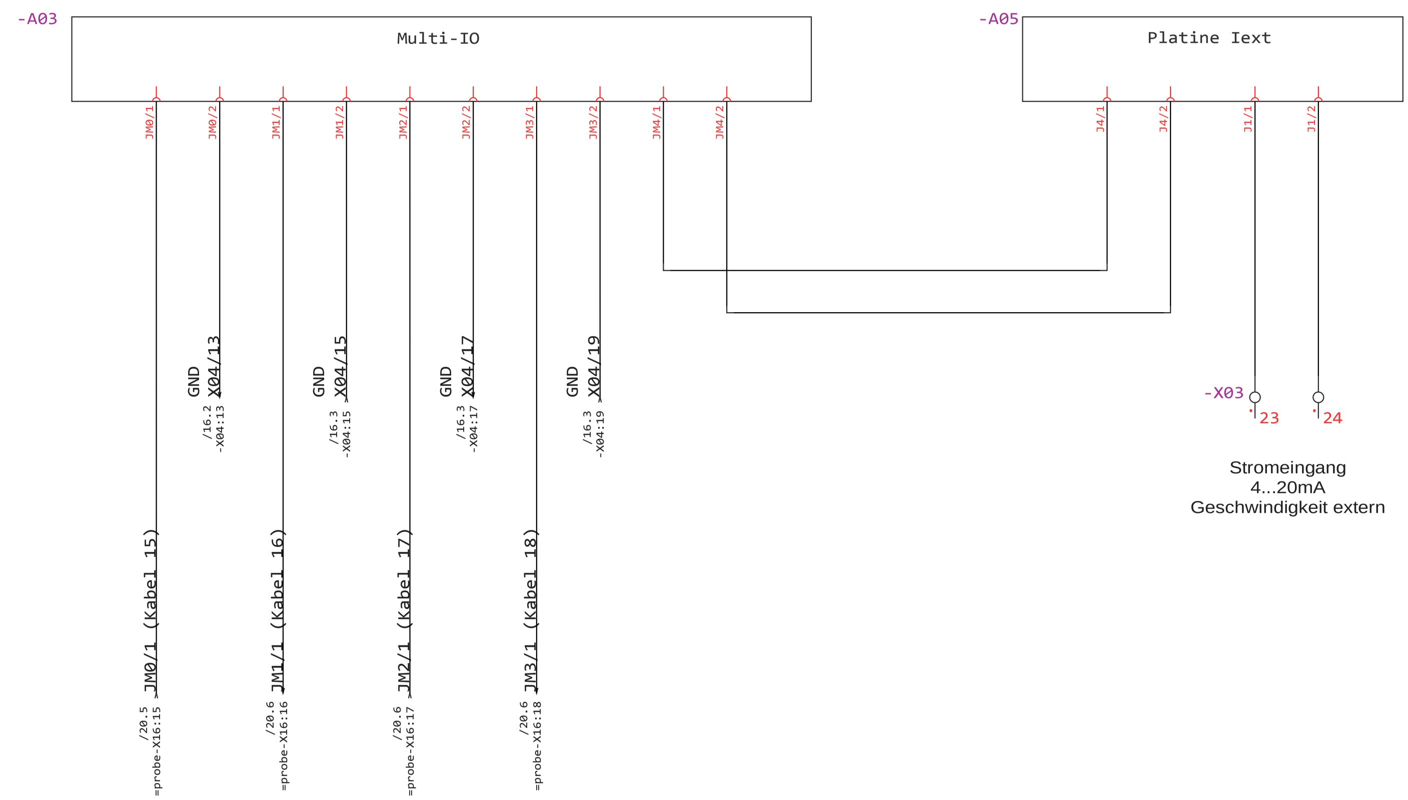
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014	PFM 06 ED	Projektnummer: =PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude	Verteilung 24V & 12V, Steuerspannung	Serie +
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf	Dr. Födisch Umweltmesstechnik AG	Zeichnungsnummer: Blatt: 16
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f Ers.d	von: 33



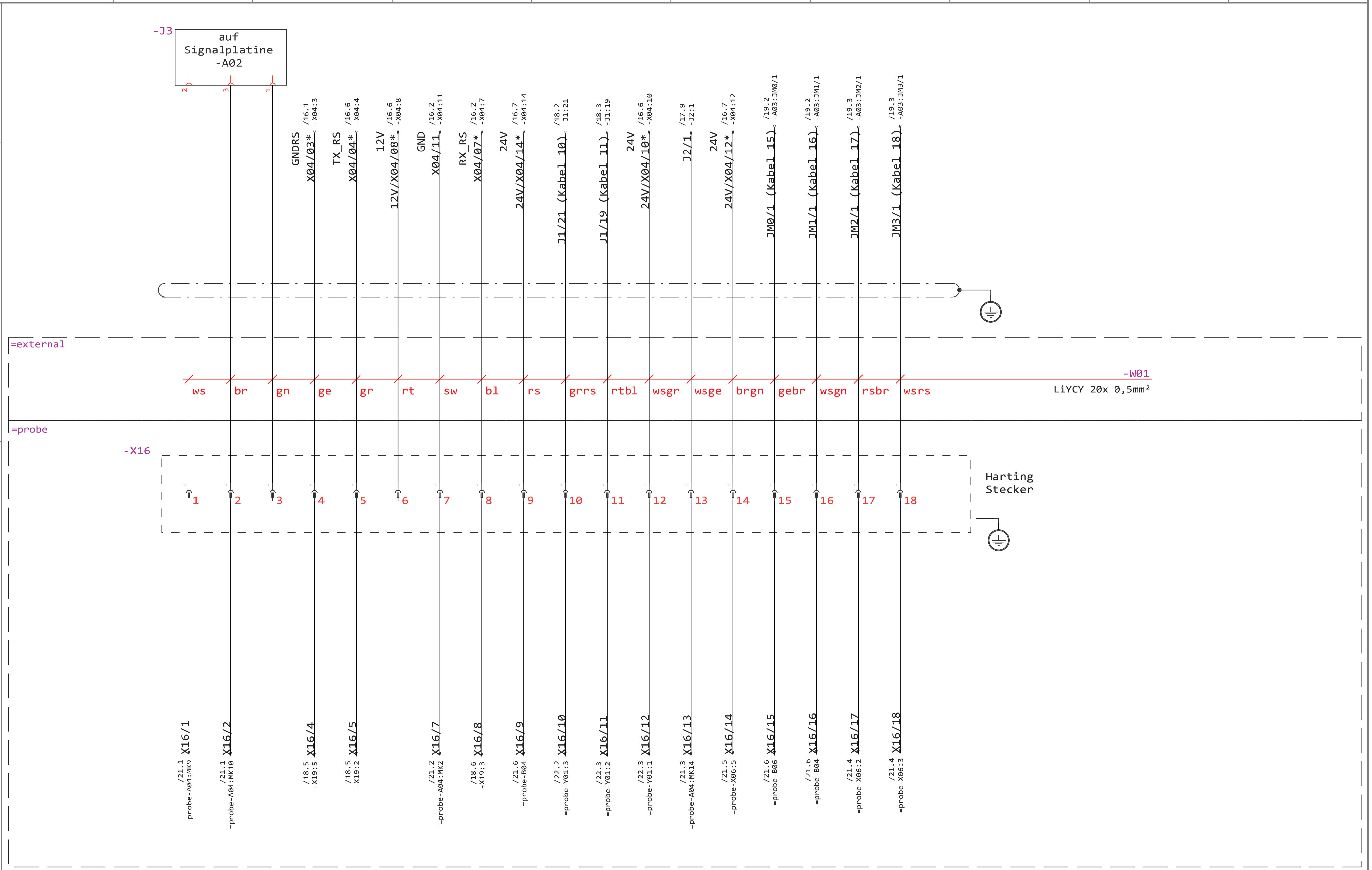
AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			Stromversorgung Signalplatine, Spannung Relais	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 17
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f	Ers.d			von: 33



AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014			Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude				Signalplatine, RS232-Anschlüsse	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf					Zeichnungsnummer:	Blatt: 18
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33

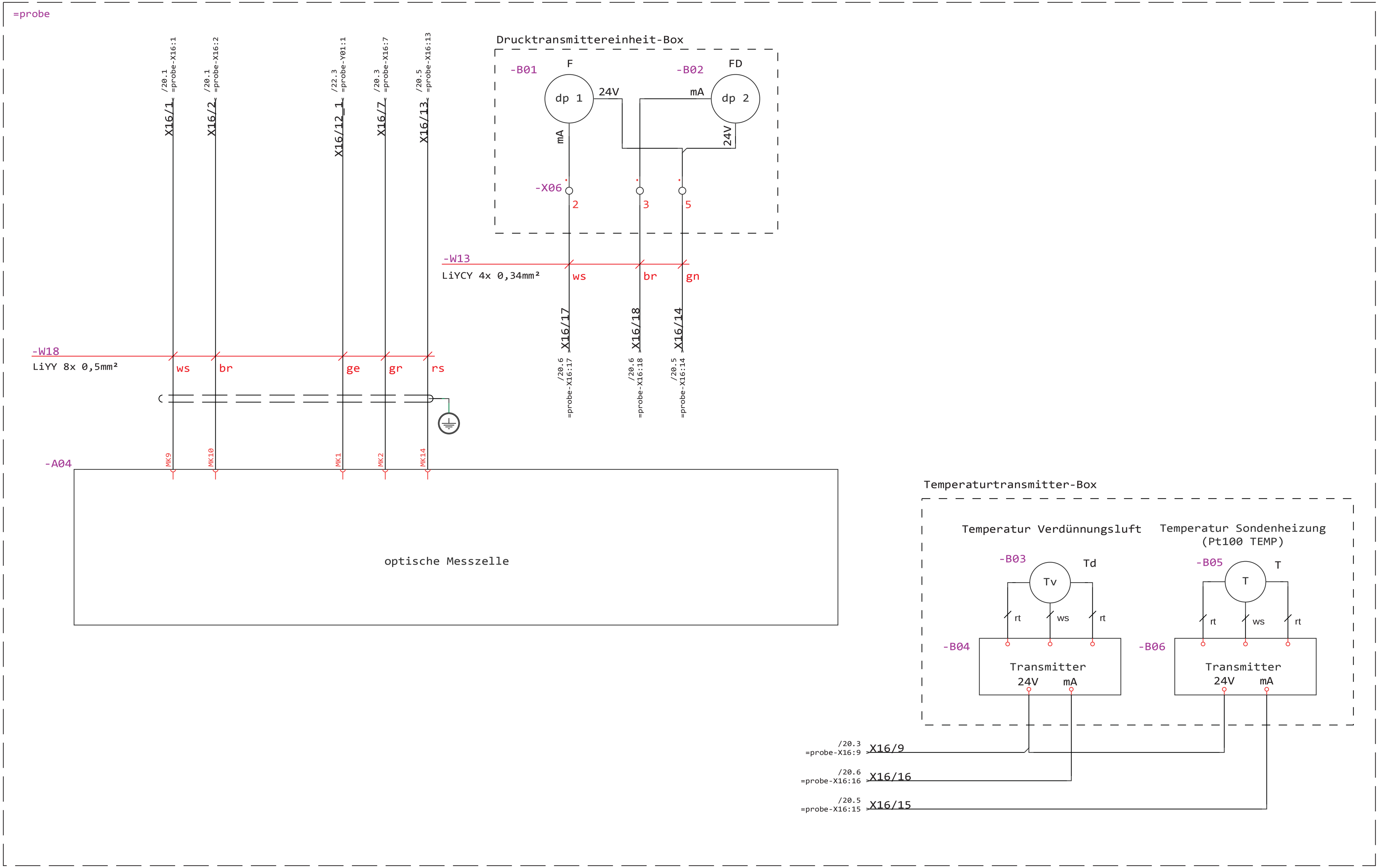


AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014			Dr. Födisch Umweltmesstechnik AG	Multi-IO	PFM 06 ED	Projektnummer: =PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude						Serie +
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf						Zeichnungsnummer: Blatt: 19
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33



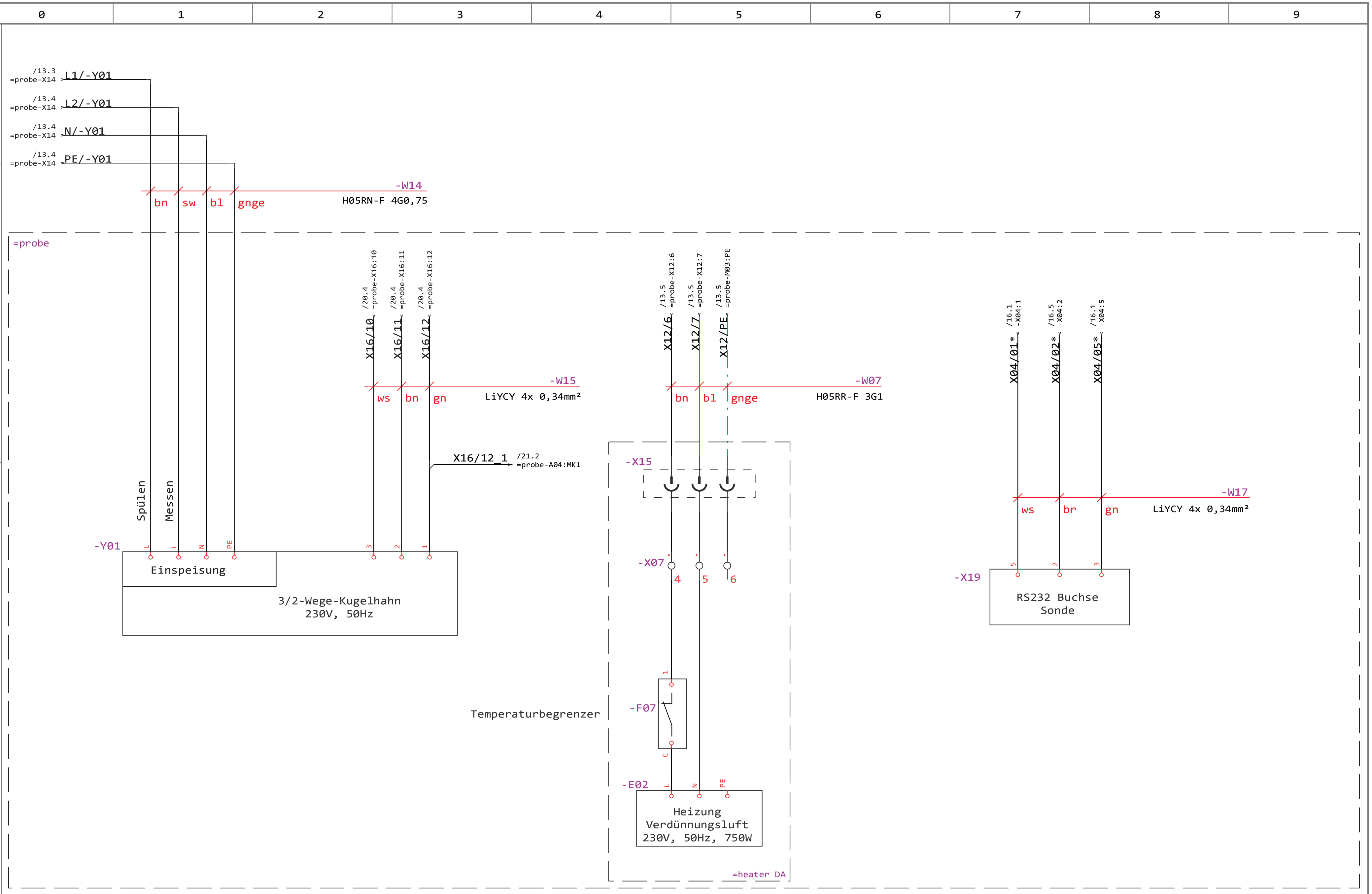
Diese Zeichnung ist urheberrechtlich geschützt.
Eine unerlaubte Vervielfältigung kann strafrechtliche Folgen haben!

AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014	PFM 06 ED	Projektnummer: =PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude	Belegung Stecker -X16	Serie +
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf		Zeichnungsnummer: Blatt: 20
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f Ers.d	von: 33



WSCAD Electrical Engineering

AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			optische Messzelle, Temperaturtransmitter	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 21
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. / Ers.f	Ers.d			von: 33



AC	-F4	14.03.2017	Ro.Kronberg	Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
AB	Rev.02	29.02.2016	Ro.Kronberg	Bearb.	Bude			Kugelhahn, Heizung Verdünnungsluft, R232-Buchse	Serie	+
AA	Phasenverteilung	17.09.2015	Bude	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 22
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d		von: 33

Klemmenplan: PFM 06 ED

Klemmleiste: -X01

Kabel extern		Funktionstext	Ziel extern: Bezeichnung	Ziel extern: Anschluss	Klemmentyp	Klemmennummer	Ziel intern: Bezeichnung	Ziel intern: Anschluss	Kabel intern	Blatt/Pfad
	bn	Netzteil	-T01	L	PIK 4-PE/L/N	1:L	-F02	2		/9.4
	bl		-T01	N	PIK 4-PE/L/N	1:N				/9.5
	gnge		-T01	PE	PIK 4-PE/L/N	1:PE				/9.5
1		Sondenheizung	=probe-X12	1	PIK 4-PE/L/N	2:L	-K01	2		/9.7
2			=probe-X12	2	PIK 4-PE/L/N	2:N				/9.8
10			=probe-X12	gnge	PIK 4-PE/L/N	2:PE				/9.8
3		Kugelhahn_Spülen	=probe-X12	3	PIK 4-PE/L/N	3:L	-K02	12		/10.1
					PIK 4-PE/L/N	3:N				/10.2
					PIK 4-PE/L/N	3:PE				/10.2
4		Kugelhahn_Messen	=probe-X12	4	PIK 4-PE/L/N	4:L	-K02	14		/10.1
5			=probe-X12	5	PIK 4-PE/L/N	4:N				/10.2
					PIK 4-PE/L/N	4:PE				/10.2
6		Heizung Verdünnungsluft	=probe-X12	6	PIK 4-PE/L/N	5:L	-K03	2		/10.4
7			=probe-X12	7	PIK 4-PE/L/N	5:N				/10.5
					PIK 4-PE/L/N	5:PE				/10.5
8		Lüfter Sonde	=probe-X12	9	PIK 4-PE/L/N	6:L	-K04	14		/10.7
9			=probe-X12	10	PIK 4-PE/L/N	6:N				/10.8
					PIK 4-PE/L/N	6:PE				/10.8
11		Magnetventil	=probe-X12	8	PIK 4-PE/L/N	7:L	-K02	22		/11.1
					PIK 4-PE/L/N	7:N				/11.2
					PIK 4-PE/L/N	7:PE				/11.3

Drahtbrücke / Stegbrücke

Zust.	Änderung	Datum	Name		
				Datum	10.09.2014
				Bearb.	Bude
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf
				Norm	DIN 61346

Urspr. Ers.f Ers.d

Dr. Födisch Umweltmesstechnik AG

PFM 06 ED
terminal diagram: -X01

Projektnummer: =PFM06ED
Serie +
Zeichnungsnummer:
Blatt: 23
von: 33

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: -X03

Kabel intern

Farbe	Funktionstext	Ziel extern Bezeichnung	Ziel extern Anschluss	Klemmentyp	Klemmennummer	Stegbrücke	Drahtbrücke	Ziel intern Bezeichnung	Ziel intern Anschluss	Blatt/Pfad
ws	DA 1: Wartung	-J2	23	UKK5	1					/14.1
br		-J2	24	UKK5	2					/14.1
gn	DA 2: Störung	-J2	21	UKK5	3					/14.2
ge		-J2	22	UKK5	4					/14.3
gr	DA 3: Grenzwert 1	-J2	19	UKK5	5					/14.4
rs		-J2	20	UKK5	6					/14.4
bl	DA 4: Grenzwert 2	-J2	17	UKK5	7					/14.5
rt		-J2	18	UKK5	8					/14.6
sw	DA 5: Messbereich 1/2	-J2	15	UKK5	9					/14.7
vi		-J2	16	UKK5	10					/14.7
grrs	AA 1	-J1	1	UKK5	11					/15.1
rtbl		-J1	2	UKK5	12					/15.1
wsgn	AA 2	-J1	3	UKK5	13					/15.2
brgn		-J1	4	UKK5	14					/15.3
wsge	AA 3	-J1	5	UKK5	15					/15.4
gebr		-J1	6	UKK5	16					/15.4
wsgr	AA 4	-J1	7	UKK5	17			-14Y1	A IN1	/15.5
grbr		-J1	8	UKK5	18			-14Y1	A Gnd	/15.6
wsrs	DA 6: Wartungsbedarf	-J2	13	UKK5	19					/14.8
rsbr		-J2	14	UKK5	20					/14.9
wsbl	Eingang Spülen	-X04	16	UKK5	21					/15.7
brbl		-J1	23	UKK5	22					/15.8
		-A05	J1/1	UKK5	23					/19.7
		-A05	J1/2	UKK5	24					/19.7

				Datum	10.09.2014	
				Bearb.	Bude	
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf	
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. / Ers.f / Ers.d

Dr. Födisch Umweltmesstechnik AG

PFM 06 ED

terminal diagram: -X03

Projektnummer: =PFM06ED

Serie +

Zeichnungsnummer:

Blatt: 24

von: 33

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: -X04

Kabel intern

Kabel extern			Funktionstext	Ziel extern Bezeichnung	Ziel extern Anschluss	Klemmentyp	Klemmennummer	Stegbrücke / Drahtbrücke	Ziel intern Bezeichnung	Ziel intern Anschluss	Kabel intern		Blatt/Pfad
		ws	GNDRS	=probe-X19	5	UKKB5	1		-J6	5			/16.1
		br	TX__RS	=probe-X19	2	UKKB5	2		-J6	3			/16.5
	ge			=probe-X16	4	UKKB5	3						/16.1
	gr			=probe-X16	5	UKKB5	4						/16.6
		gn	TX_RX, RX_RS	=probe-X19	3	UKKB5	5		-J6	2			/16.1
			Verteilung 12V	-J5	1	UKKB5	6		-T01	+12V			/16.6
	bl			=probe-X16	8	UKKB5	7						/16.2
	rt			=probe-X16	6	UKKB5	8						/16.6
			Verteilung 0V	-J5	2	UKKB5	9		-T01	Com.			/16.2
	wsgr		Verteilung 24V	=probe-X16	12	UKKB5	10		-T02	+12V			/16.6
				-J1	22	UKKB5	11		=probe-X16	7		sw	/16.2
	brgn			=probe-X16	14	UKKB5	12		-J2	11			/16.7
				-J1	20	UKKB5	13		-A03	JM0/2			/16.2
	rs			=probe-X16	9	UKKB5	14		-J2	9			/16.7
				-J2	2	UKKB5	15		-A03	JM1/2			/16.3
wsbl				-X03	21	UKKB5	16		-J2	7			/16.7
				-J1	18	UKKB5	17		-A03	JM2/2			/16.3
						UKKB5	18		-J2	5			/16.7
wsrt				-K01	4-	UKKB5	19		-A03	JM3/2			/16.3
						UKKB5	20						/16.8
				-J1	24	UKKB5	21						/16.3
						UKKB5	22						/16.8
						UKKB5	PE						/16.9
						UKKB5	PE						/16.9
				-J5	3	UKKB5	PE		-X01	1	gnge		/16.9

				Datum	10.09.2014	Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer: =PFM06ED
				Bearb.	Bude		terminal diagram: -X04	Serie +
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf			Zeichnungsnummer: Blatt: 25
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f Ers.d		von: 33

0 1 2 3 4 5 6 7 8 9

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: =heater DA-X07

Kabel intern

Funktionstext	Ziel extern: Bezeichnung	Ziel extern: Anschluss	Klemmentyp	Klemmennummer	Stegbrücke	Drahtbrücke	Ziel intern: Bezeichnung	Ziel intern: Anschluss	Blatt/Pfad
L: Heizung Verdünnungsluft	-F07	1	UK10N	4			-X15		/22.5
N: Heizung Verdünnungsluft	-E02	N	UK10N	5			-X15		/22.5
PE: Heizung Verdünnungsluft			UK10N	6			-X15		/22.5

				Datum	10.09.2014	PFM 06 ED	Projektnummer: =PFM06ED
				Bearb.	Bude	terminal diagram: =heater DA-X07	Serie +
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf		Zeichnungsnummer: Blatt: 26
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f Ers.d	von: 33

Dr. Födisch Umweltmesstechnik AG

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: =junction box-X05

Kabel intern

Kabel extern	Funktionstext	Ziel extern: Bezeichnung	Ziel extern: Anschluss	Klemmentyp	Klemmennummer	Stegbrücke / Drahtbrücke	Ziel intern: Bezeichnung	Ziel intern: Anschluss	Blatt/Pfad
	L: Sondenheizung	=probe-E01	L	UK10N	1		=probe-X12	1	/13.2
	N: Sondenheizung	=probe-E01	N	UK10N	2		=probe-X12	2	/13.2
	PE: Sondenheizung	=probe-E01	PE	UK10N	3		=probe-X14		/13.2
bn	L: Magnetventil	=probe-X13		UK10N	4		=probe-X12	8	/13.1
bl	N: Magnetventil	=probe-X13		UK10N	5				/13.1
gnge	PE: Magnetventil	=probe-X13		UK10N	6				/13.1

Zust.	Änderung	Datum	Name		
				Datum	10.09.2014
				Bearb.	Bude
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf
				Norm	DIN 61346

Urspr. | Ers.f | Ers.d

Dr. Födisch Umweltmesstechnik AG

PFM 06 ED

terminal diagram: =junction box-X05

Projektnummer: =PFM06ED

Serie +

Zeichnungsnummer:

Blatt: 27

von: 33

0	1	2	3	4	5	6	7	8	9

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: =probe-X06

Kabel intern

Kabel extern	Funktionstext	Ziel extern: Bezeichnung	Ziel extern: Anschluss	Klemmentyp	Klemmennummer	Stegbrücke	Drahtbrücke	Ziel intern: Bezeichnung	Ziel intern: Anschluss	Blatt/Pfad
ws	Drucktransmitter	-X16	17	UK10N	2			-B01		/21.4
br		-X16	18	UK10N	3			-B02		/21.4
gn		-X16	14	UK10N	5			-B01		/21.5

				Datum	10.09.2014				PFM 06 ED	Projektnummer:	=PFM06ED
				Bearb.	Bude			Dr. Födisch Umweltmesstechnik AG	terminal diagram: =probe-X06	Serie	+
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf					Zeichnungsnummer:	Blatt: 28
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d			von: 33

0	1	2	3	4	5	6	7	8	9

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: =probe-X12

Kabel intern

Kabel extern		Funktionstext	Ziel extern: Bezeichnung	Ziel extern: Anschluss	Klemmentyp	Klemmennummer	Stegbrücke	Drahtbrücke	Ziel intern: Bezeichnung	Ziel intern: Anschluss	Kabel intern		Blatt/Pfad
		Harting Stecker	=junction box-X05	1		1			-X01	2		bn	/13.2
			=junction box-X05	2		2			-X01	2		bl	/13.2
			-X14			3			-X01	3	3		/13.3
			-X14			4			-X01	4	4		/13.4
			-X14			5			-X01	4	5		/13.4
	bn		=heater DA-X15			6			-X01	5	6		/13.5
	bl		=heater DA-X15			7			-X01	5	7		/13.5
			=junction box-X05	4		8			-X01	7	11		/13.7
sw			-M03	L		9			-X01	6	8		/13.6
bl			-M03	N		10			-X01	6	9		/13.6
gnge			-M03	PE		gnge			-X01	2		gnge	/13.7

				Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer: =PFM06ED
				Bearb.	Bude			terminal diagram: =probe-X12	Serie +
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf				Zeichnungsnummer: Blatt: 29
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f	Ers.d		von: 33

0	1	2	3	4	5	6	7	8	9

Klemmenplan: PFM 06 ED

Kabel extern

Klemmleiste: =probe-X16

Kabel intern

									Ziel extern						Ziel intern				
								Funktionstext	Bezeichnung	Anschluss	Klemmentyp	Klemmennummer	Stegbrücke	Drahtbrücke	Bezeichnung	Anschluss			Blatt/Pfad
							ws	Harting Stecker	-A04	MK9		1			-J3	2		ws	/20.1
							br		-A04	MK10		2			-J3	3		br	/20.1
												3			-J3	1		gn	/20.2
						ws			-X19	5		4			-X04	3		ge	/20.2
						br			-X19	2		5			-X04	4		gr	/20.2
												6			-X04	8		rt	/20.3
							gr		-A04	MK2		7			-X04	11		sw	/20.3
						gn			-X19	3		8			-X04	7		bl	/20.3
									-B04			9			-X04	14		rs	/20.3
					ws				-Y01	3		10			-J1	21		grrs	/20.4
					bn				-Y01	2		11			-J1	19		rtbl	/20.4
					gn				-Y01	1		12			-X04	10		wsgr	/20.4
							rs		-A04	MK14		13			-J2	1		wsge	/20.5
				gn					-X06	5		14			-X04	12		brgn	/20.5
									-B06			15			-A03	JM0/1		gebr	/20.5
									-B04			16			-A03	JM1/1		wsgn	/20.6
				ws					-X06	2		17			-A03	JM2/1		rsbr	/20.6
				br					-X06	3		18			-A03	JM3/1		wsrs	/20.6

				Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
				Bearb.	Bude			terminal diagram: =probe-X16	Serie	+
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 30
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d		von: 33

0	1	2	3	4	5	6	7	8	9

Kabelliste: PFM 06 ED

Nr.	Leitungsbezeichnung	Leitungstyp	Anzahl Adern	Ziel1	Funktionstext	Blatt/ Pfad
1	-W03	H07RN-F 4G1,5	4	=control unit-M01	Anschlusskabel für Seitenkanalverdichter SD42	/8.7
2	-W04	H07RN-F 4G1,5	4	=control unit	Anschlusskabel für Seitenkanalverdichter SD2n/M	/9.1
3	-W05	H05VV-F 3G0.75	3	-T01	Anschlusskabel Netzteile	/12.4
4	-W06	H05VV-F 3G0.75	3	-X11	Anschlusskabel Servicesteckdose	/8.4
5	-W10	LiYCY 28x 0,25mm²	27	-J2	Signalkabel Grundplatte	/14.1
6	-W11	H07RN-F 4G1	3	-X01	Anschlusskabel Sondenheizung/ Magnetventil	/9.7
7	-W14	H05RN-F 4G0,75	4	=probe-X14	Antrieb Kugelhahn Sonde	/22.1
8	-W16	LiYCY 4x 0,34mm²	3	-X19	Kommunikationskabel Sonde	/18.5
9	=external-W01	LiYCY 20x 0,5mm²	18	-J3	Sondenanschlusskabel 24V	/20.1
10	=external-W02	YSLY-JZ 12x 1mm²	11	-X01	Sondenanschlusskabel 230V	/13.2
11	=probe-W07	H05RR-F 3G1	3	=heater DA-X15	Anschluss Verdünnungsluftheizung	/22.5
12	=probe-W09	H05RR-F 3G0,75	3	=probe-M03	Gehäuselüfter	/13.6
13	=probe-W12	H05RR-F 3G0,75	3	=probe-X13	Magnetventil Sonde	/13.1
14	=probe-W13	LiYCY 4x 0,34mm²	3	=probe-X06	Differenzdrucktransmitter	/21.4
15	=probe-W15	LiYCY 4x 0,34mm²	3	=probe-Y01	Endschalter Kugelhahn	/22.2
16	=probe-W17	LiYCY 4x 0,34mm²	3	=probe-X19	RS232 Buchse_Sonde	/22.7
17	=probe-W18	LiYY 8x 0,5mm²	5	=probe-A04	Anschlusskabel optischer Sensor	/21.1

				Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer:	=PFM06ED
				Bearb.	Bude			Kabelliste	Serie	+
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf				Zeichnungsnummer:	Blatt: 31
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. / Ers.f	Ers.d			von: 33

0	1	2	3	4	5	6	7	8	9

Materialliste: PFM 06 ED

Nr.	Anz.	Name	Artikelnummer	Artikel intern	Bezugnamen	Hersteller	Blatt/ Pfad
1	1	Steuerelektronik PFM97ED	06005 Rev.3.23 4L		-A02	Dr. Födisch AG	/5.33
2	1	Multi-IO			-A03	Dr. Födisch AG	/6.9
3	1	Platine Iext			-A05	Dr. Födisch AG	/19.6
4	1	LS-Schalter, 3-pol, C-Charakteristik, 10A	PXL-C10/3		-F01	Eaton	/8.0
5	1	Fehlerstromschutzschalter, 2-pol, 25A /0,03A	PXF-25/2/003-A		-F01.1	Eaton	/8.5
6	5	UK 5-HESI (5x20) Sicherungsreihenklemme	3004472		-F02,-F03,-F04,-F05	Phoenix	/9.4
					-F06		
7	2	Halbleiterrelais	G3NA-210B	ETL 730	-K01,-K03	Omron	/17.5
8	2	Koppelrelais 24VDC, 2 Wechsler	48.52.7.024.0050	ETL 20701	-K02,-K04	Finder	/17.6
9	1	Display 4x20 Zeichen	EW20400YLY		-P01	EDT	/5.9
10	1	Displaykabel 28AWG, 16os., IDC	IDMS-16-D-0790-T-R			Samtec	/5.9
11	1	Tastatur			-S01		/5.20
12	1	Motorschutzschalter PKZM0 1,5kW	PKZM0-4		-S01	Moeller	/8.7
13	1	Motorschutzschalter PKZM0, 0,75kW	PKZM0-2,5		-S02	Eaton	/9.1
14	2	PFM97ED_Netzteil	TXL 025-12S		-T01,-T02	Traco Power	/12.3
15	1	Steckdose	HVES 2751829		-X11	Hagemeyer	/8.4
16	1	Wandgehäuse EL, 3-teilig mit Montageplatte und Profilschienen	EL 2246.605		-Z100	Rittal	/4.10
17	1	Seitenkanalverdichter	SD42		=control unit-M01	Elektor airsystems Gmb	/8.7
18	1	Seitenkanalverdichter	SD2n/M		=control unit-M02	Elektor airsystems Gmb	/9.1
19	1	Frequenzumrichter, 0,75kW, 400VAC	Kostal INVEOR			Elektror airsystems Gm	/9.1
20	2	Sondenheizung,Verdünnungsluftheizung, 750W, 230V	Hotcoil Heizelemente		=heater DA-E02	KSG Gerätetechnik	/22.4
					=probe-E01		
21	1	Temperaturbegrenzer	TS320-SR		=heater DA-F07	Rainbow	/22.4
22	1	Steckverbinder Stecker + Dose 2pol+PE			=heater DA-X15		/22.5
23	1	Differenzdrucktransmitter	432LP10D-PCB	ETL 7651	=probe-B01	First Sensor AG	/21.4
24	1	Differenzdrucktransmitter Fd	SQ231- 13130	ETL 7602	=probe-B02	Sensor Technics	/21.5
25	2	Temperaturfühler Pt100	52-40101199-0040.V3	ETL 7604	=probe-B03,=probe-B05	Malek Günther GmbH	/21.7

				Datum	10.09.2014	Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer: =PFM06ED
				Bearb.	Bude		Materialliste	Serie +
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf			Zeichnungsnummer: Blatt: 32
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr. Ers.f Ers.d		von: 33

0	1	2	3	4	5	6	7	8	9

Materialliste: PFM 06 ED

Nr.	Anz.	Name	Artikelnummer	Artikel intern	Bezugnamen	Hersteller	Blatt/ Pfad
26	2	Temperaturtransmitter Td	ETL 7639		=probe-B04,=probe-B06	Dr.Födisch UMT	/21.7
27	1	Gehäuselüfter	DV4650-470		=probe-M03	EBM Pabst	/13.6
28	1	HAN 10E-F	09330102701		=probe-X12	Harting	/13.2
29	1	Gehäuse	09300101441			Harting	/13.2
30	1	Dichtung	0930000990300			Harting	/13.2
31	1	Gehäuse	09300100901			Harting	/13.2
32	1	Stecker	09330102601			Harting	/13.2
33	1	Gerätesteckdose	008369		=probe-X13	Bürkert	/13.1
34	1	Steckverbinder Stecker + Dose 3pol+PE			=probe-X14		/13.4
35	1	HAN 18EE-F	09320183101		=probe-X16	Harting	/20.1
36	1	Gehäuse	09300101441			Harting	/20.1
37	18	Goldkontakt	0933006222			Harting	/20.1
38	1	3-Wege-Kugelhahn	399H20-363G12-T	ETL 7617	=probe-Y01	Schwer	/22.1
39	1	Magnetventil	56178		=probe-Y02	Bürkert	/13.1

				Datum	10.09.2014		Dr. Födisch Umweltmesstechnik AG	PFM 06 ED	Projektnummer: =PFM06ED
				Bearb.	Bude			Materialliste	Serie +
AC	-F4	14.03.2017	Ro.Kronberg	Gepr.	Krummsdorf				Zeichnungsnummer:
Zust.	Änderung	Datum	Name	Norm	DIN 61346	Urspr.	Ers.f	Ers.d	Blatt: 33 von: 33